

Session Four

Course Workbook

Table of Contents

About this Workbook

[Slide 3](#)

Pre-Lab: Workshop Pre-requisites

[Slide 4](#)

Lab 4: TMR MicroBlaze

[Slide 7](#)

About this Workbook

The contents of this workbook are created by Aduvo Engineering & Training, Ltd.

If you have any questions about the contents, or need assistance, please contact Adam Taylor at adam@aduvoengineering.com.

Pre-Lab

Workshop Pre-requisites

Required Hardware

The hands-on labs in this course use the following hardware:

- Digilent Arty S7-50 development board
- USB-A to Micro-B cable

Downloads and Installations

Step 1 – Download and install the following at least one day prior to the workshop. This may take a significant amount of time and drive space.

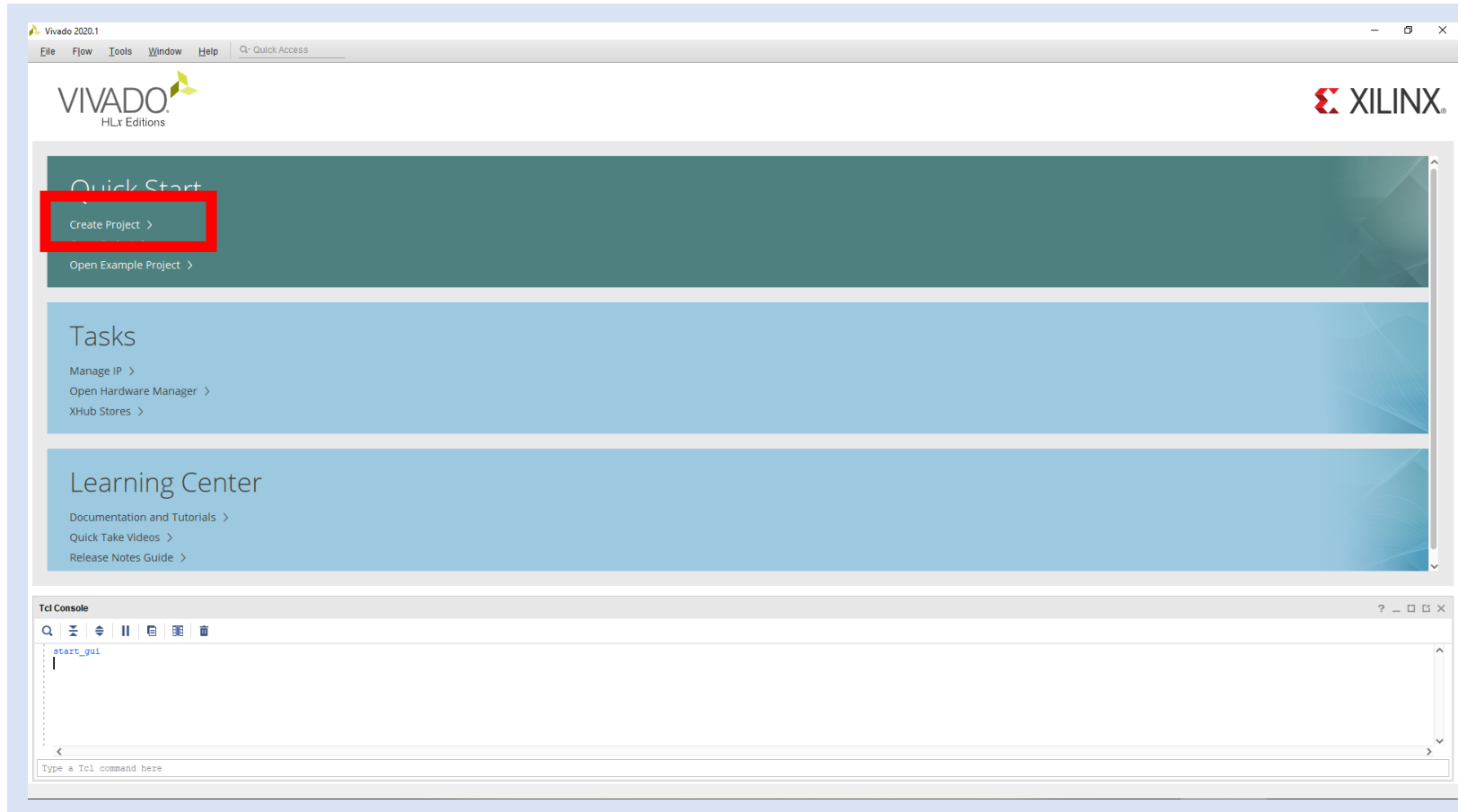
Vivado 2020.1	Download
---------------	--------------------------

Lab 4

TMR MicroBlaze

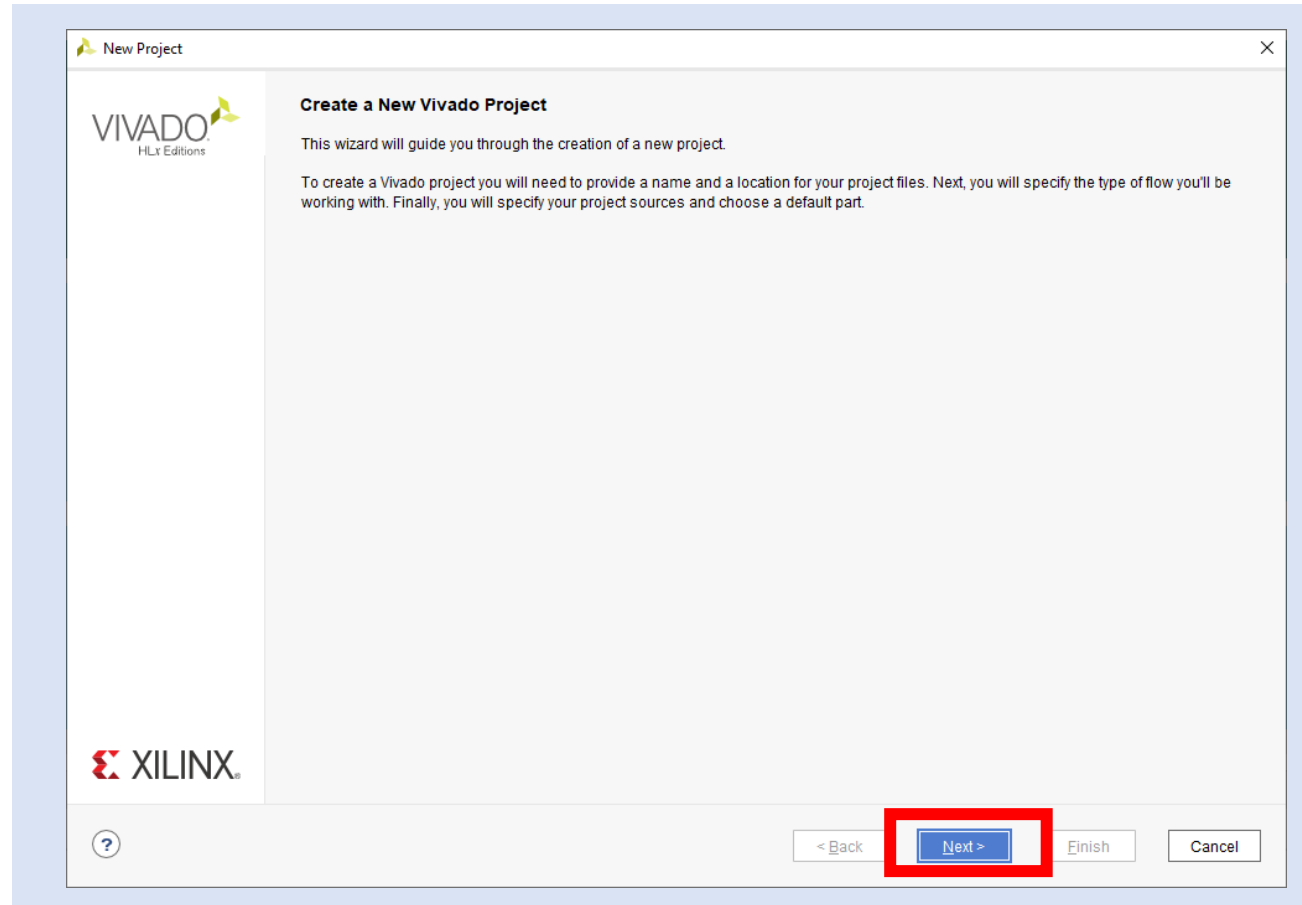
Lab 4: Overview and Introduction to Vivado

Step 1 – Create a new Vivado project



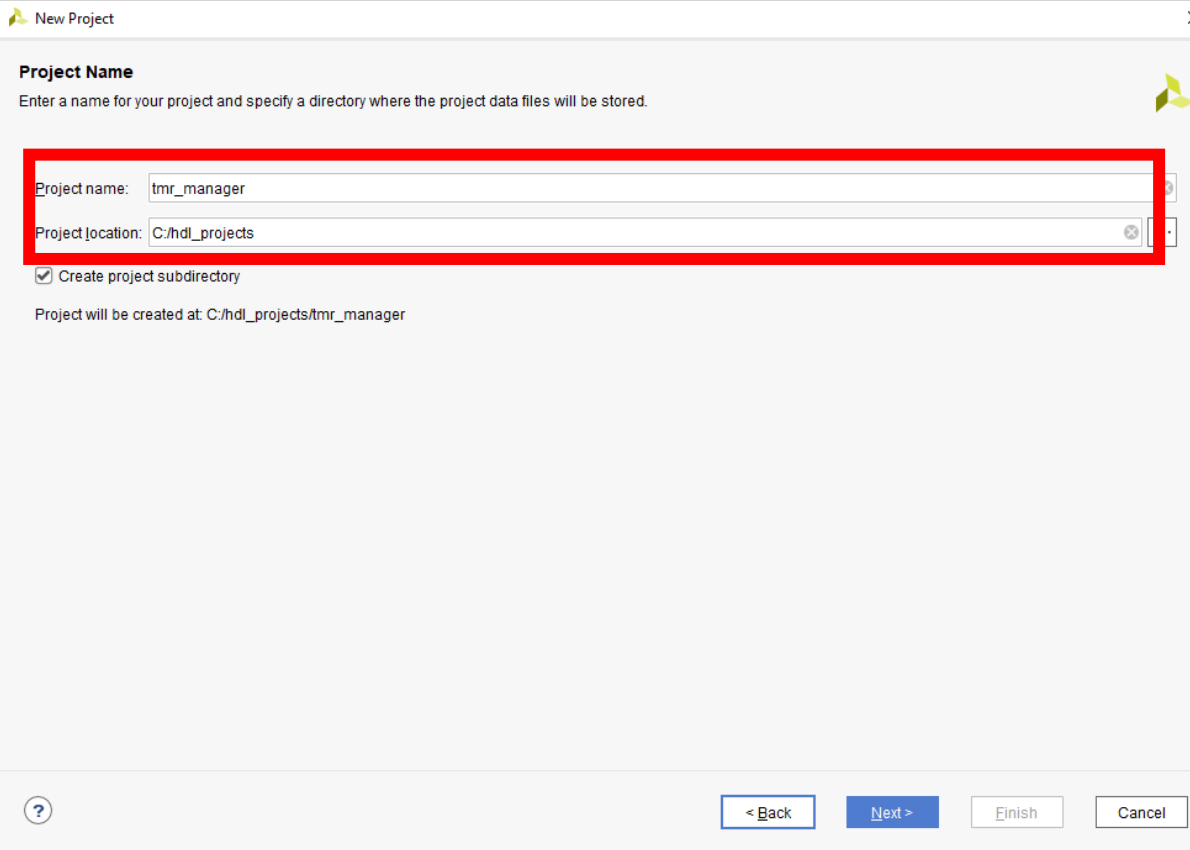
Lab 4: Overview and Introduction to Vivado

Step 2 – Create a new Vivado project



Lab 4: Overview and Introduction to Vivado

Step 3 – Select the project name and storage location, click next



New Project

Project Name
Enter a name for your project and specify a directory where the project data files will be stored.

Project name:

Project location:

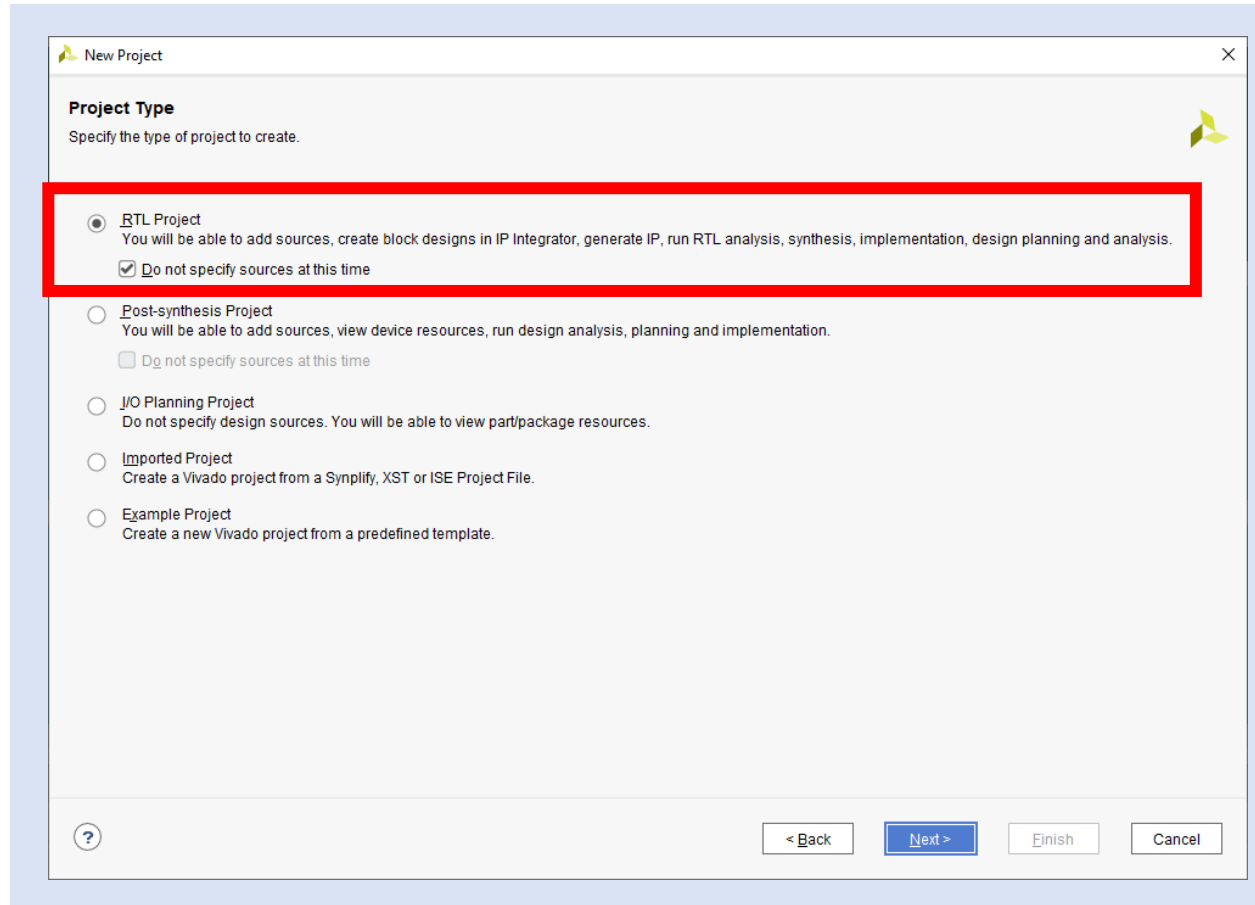
☒ Create project subdirectory

Project will be created at: C:/hdl_projects/tmr_manager

[?](#) [< Back](#) [Next >](#) [Finish](#) [Cancel](#)

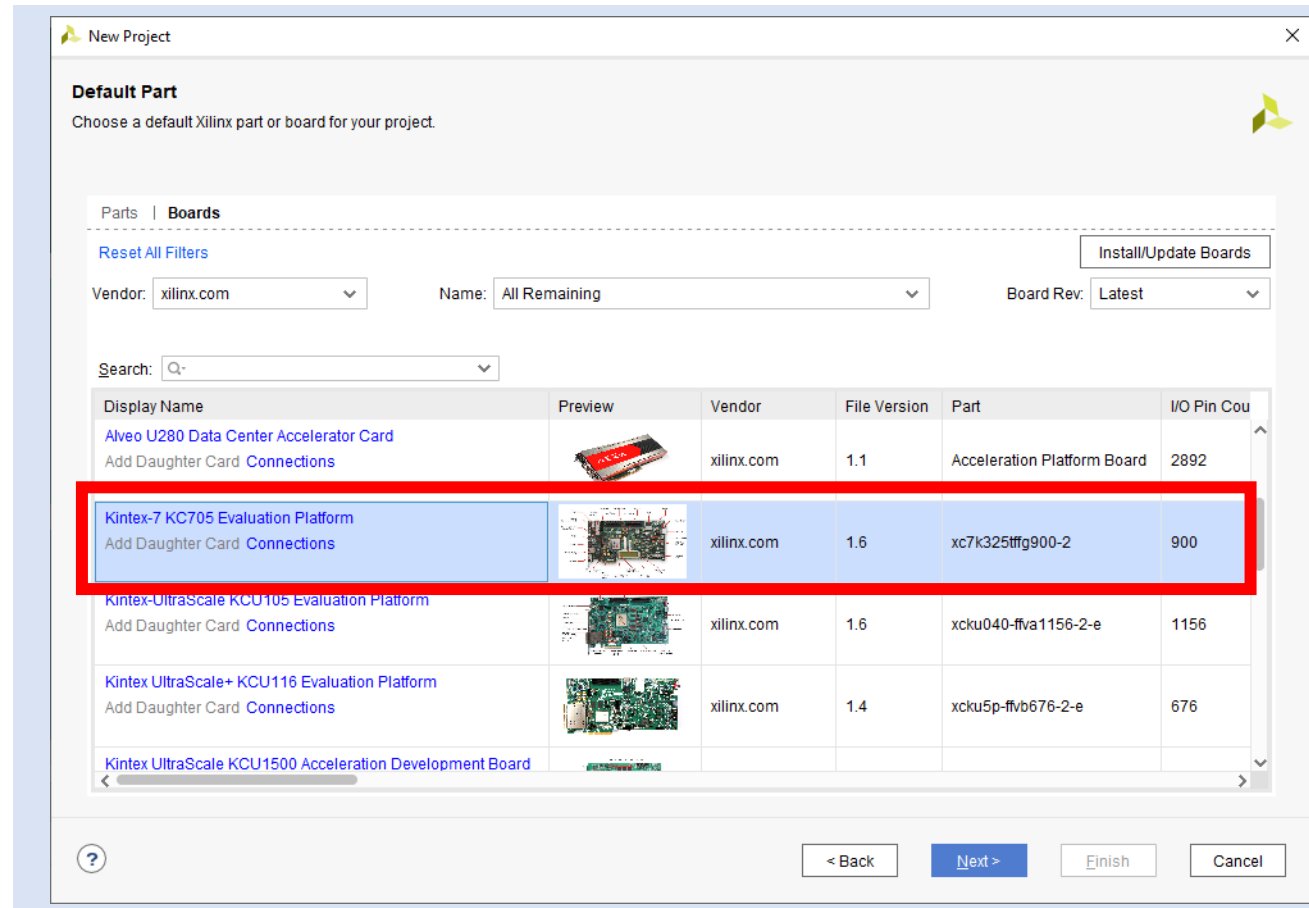
Lab 4: Overview and Introduction to Vivado

Step 4 – Select RTL Project and check the Do not specify sources at this time



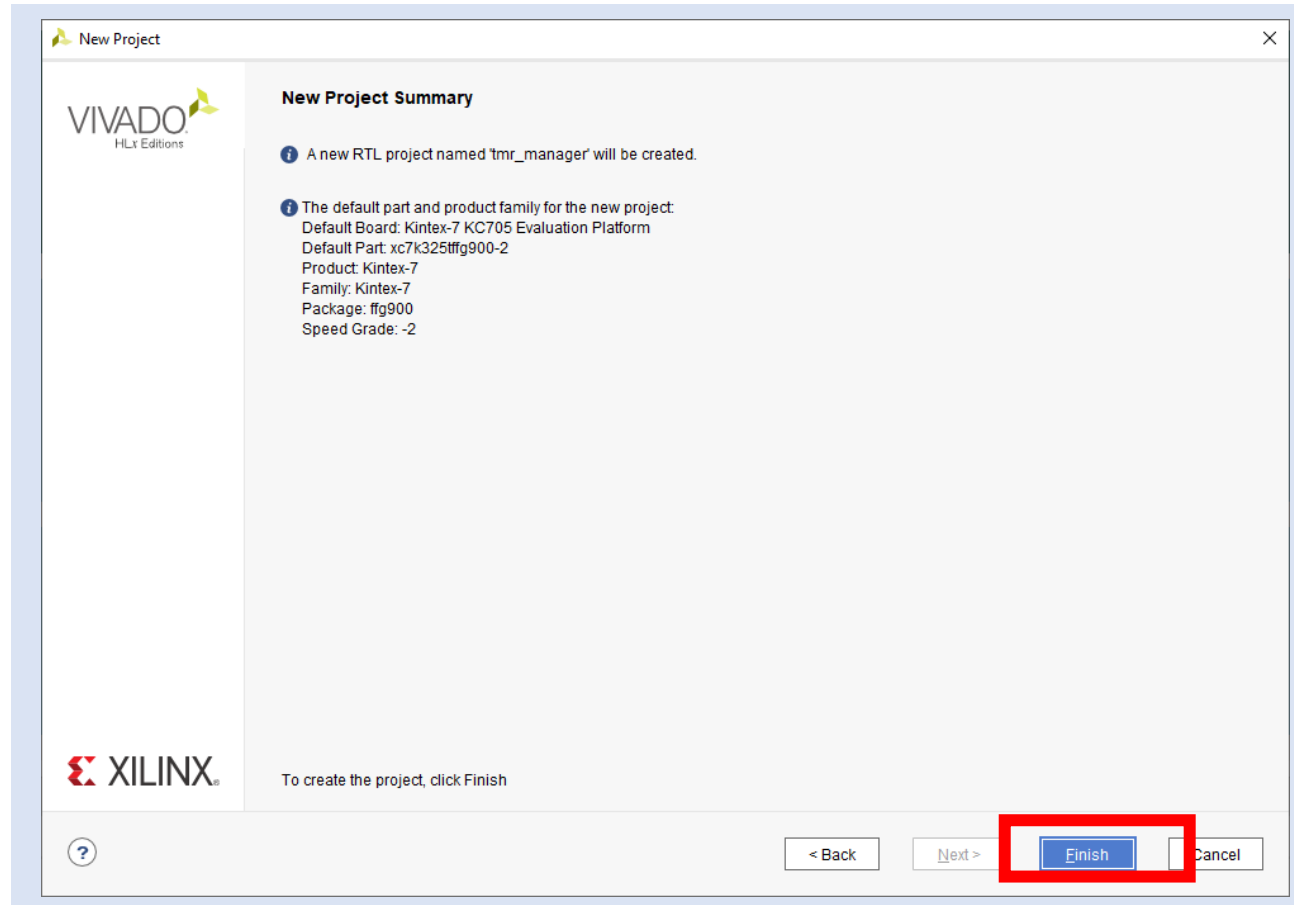
Lab 4: Overview and Introduction to Vivado

Step 5 – Select the Kintex KC705 Eval Board (NOT Arty S7)



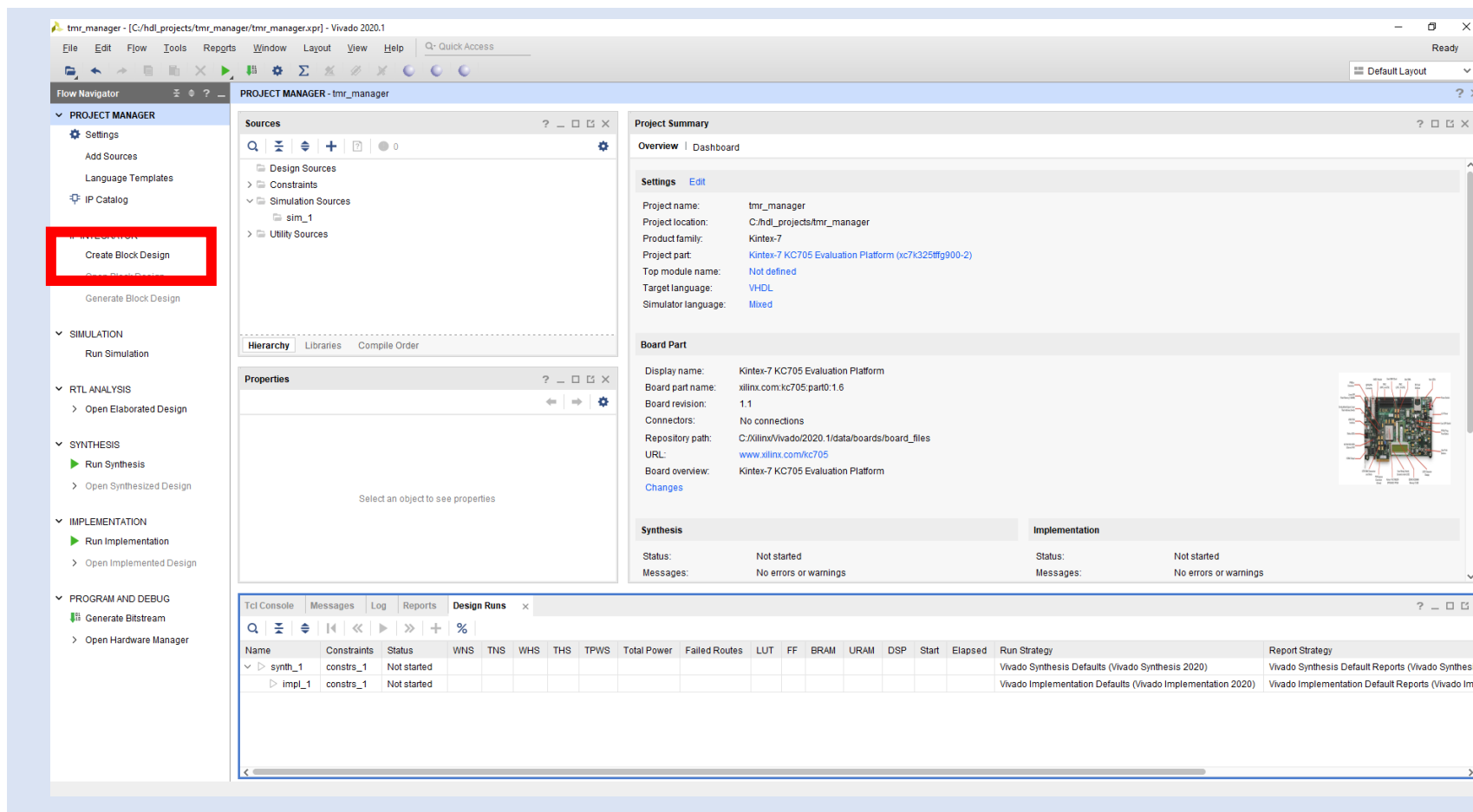
Lab 4: Overview and Introduction to Vivado

Step 6 – Click Finish



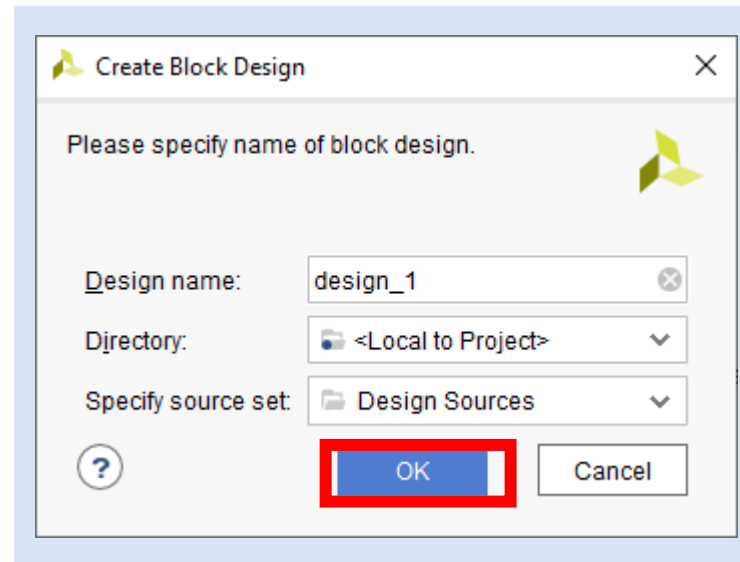
Lab 4: Overview and Introduction to Vivado

Step 7 – Create Block Diagram



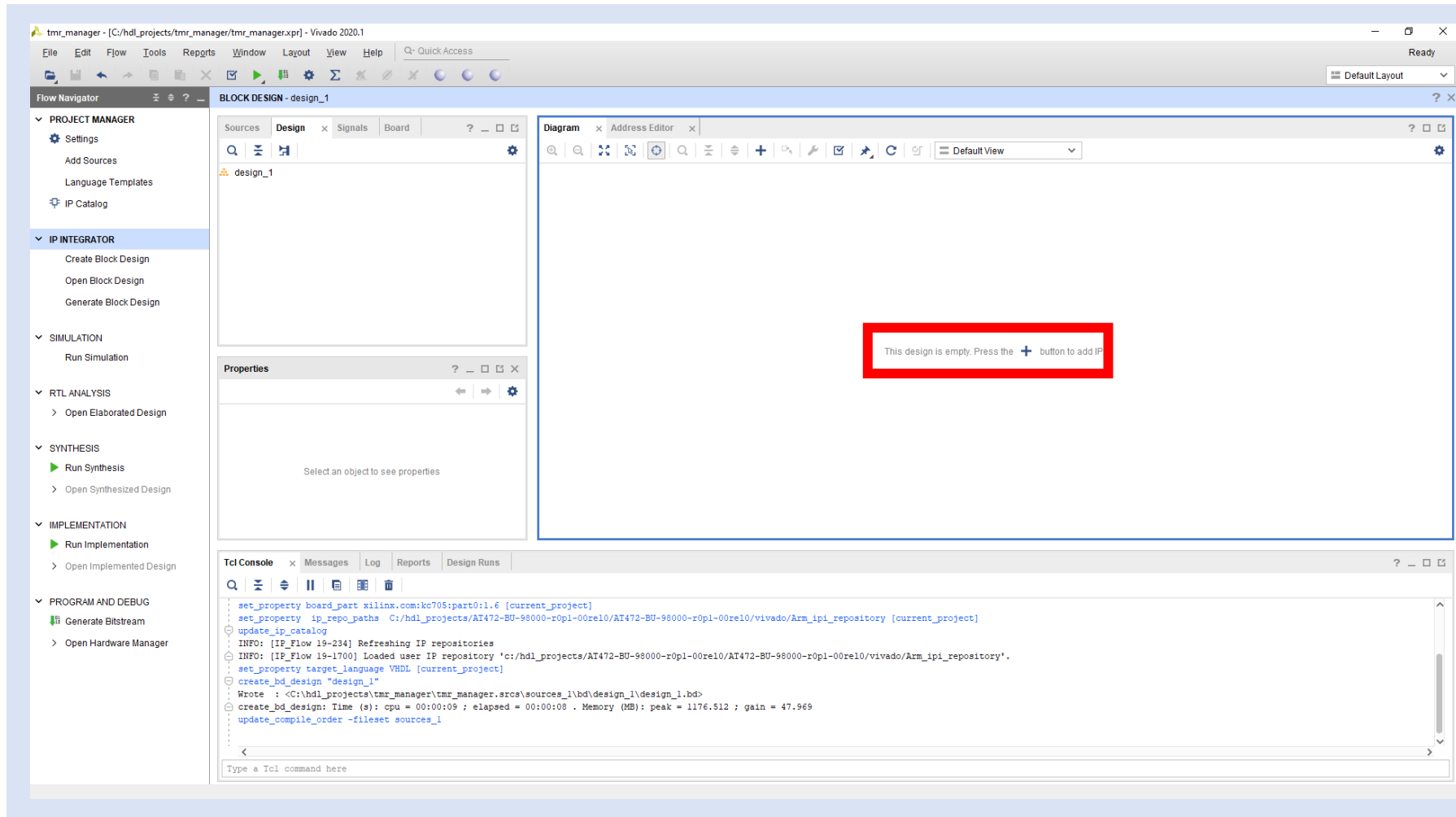
Lab 4: Overview and Introduction to Vivado

Step 8 – Leave the settings unchanged and click OK



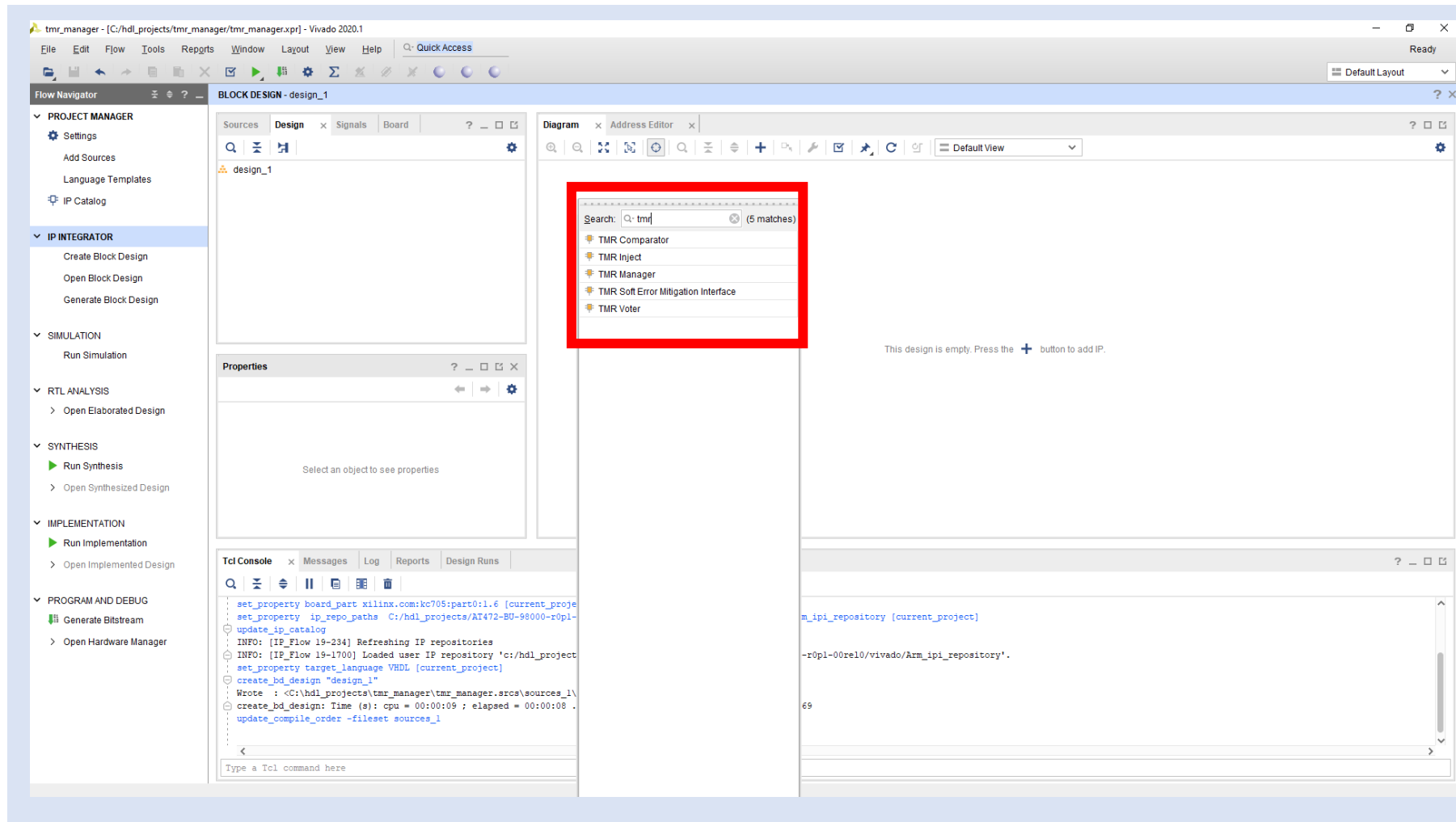
Lab 4: Overview and Introduction to Vivado

Step 9 – Click on the + signal to open the IP library



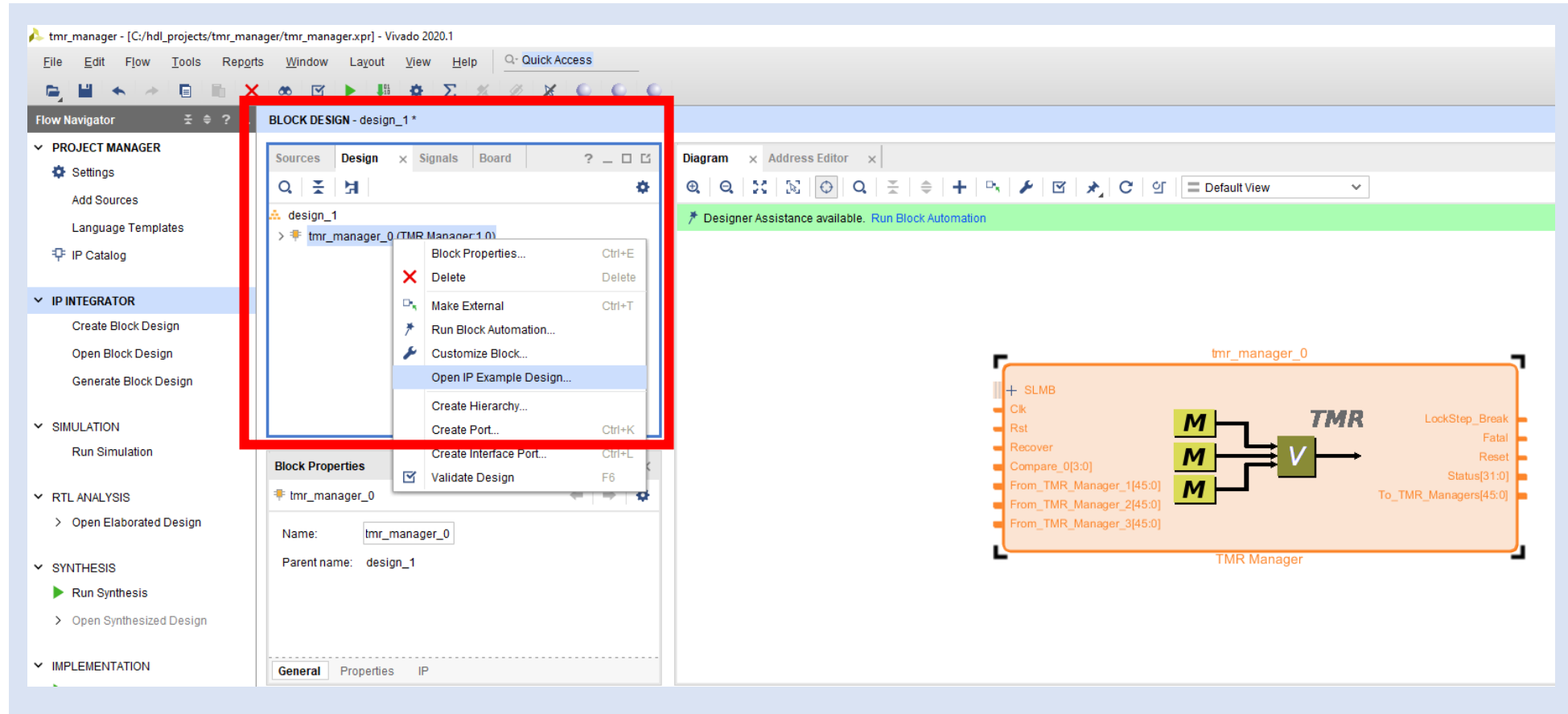
Lab 4: Overview and Introduction to Vivado

Step 10 – Select the TMR Manager and add it into the block diagram



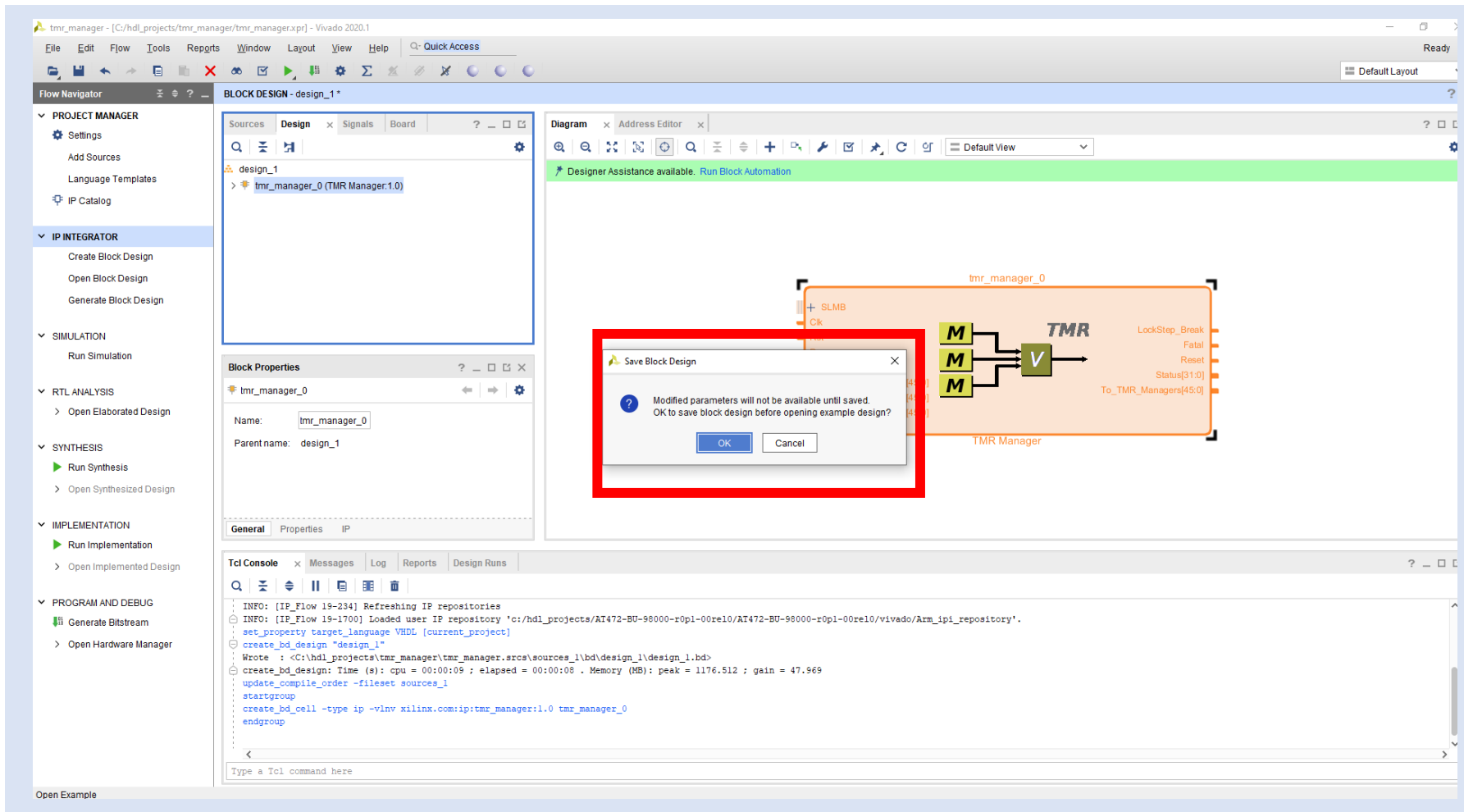
Lab 4: Overview and Introduction to Vivado

Step 11 – In the design view, select the TMR Manager, right click and select Open IP Example Design



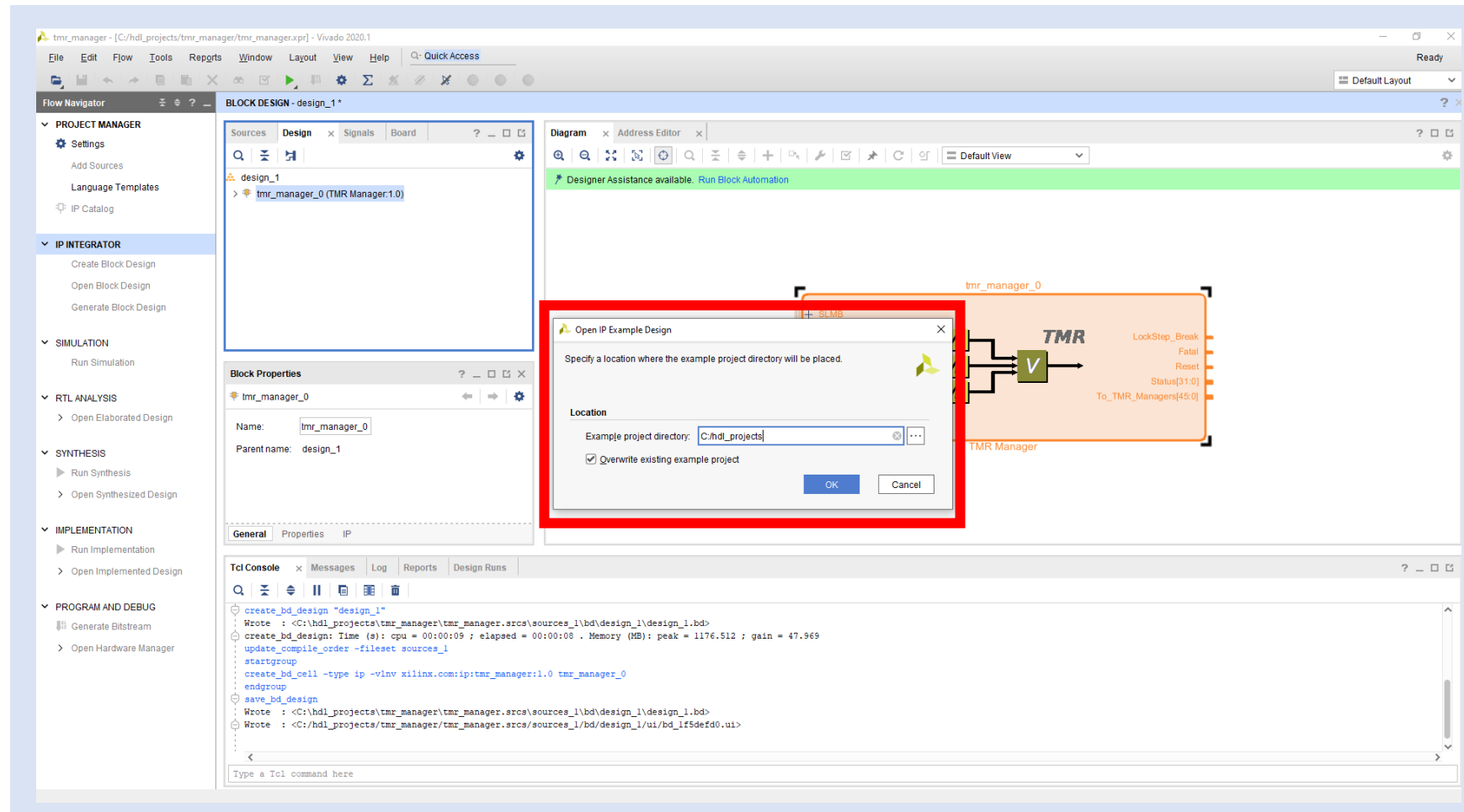
Lab 4: Overview and Introduction to Vivado

Step 12 – Click on OK on the dialog which appears



Lab 4: Overview and Introduction to Vivado

Step 13 – Select the location to save the example project



Lab 4: Overview and Introduction to Vivado

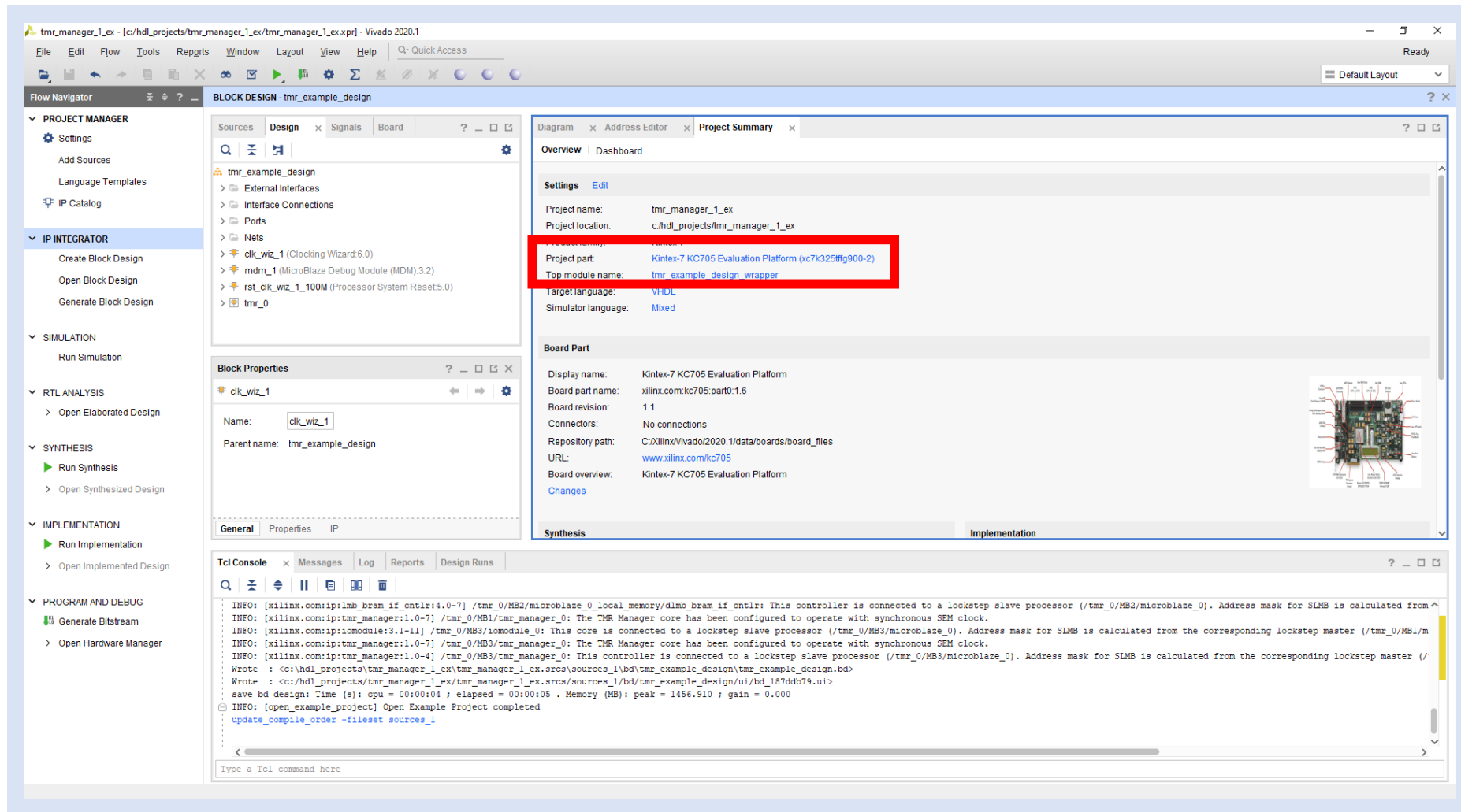
Step 14 – A new project will open with a TMR MicroBlaze Design – Targeting the Kintex-7 device.
Click on the summary

The screenshot displays the Vivado 2020.1 IDE interface for a project named 'tmr_manager_1_ex'. The top toolbar features a red box around the 'Summary' icon (a document with a checkmark). The left sidebar shows the 'PROJECT MANAGER' and 'IP INTEGRATOR' sections. The central 'Diagram' window shows a block design with components: 'clk_wiz_1' (Clocking Wizard), 'rst_clk_wiz_1_100M' (Processor System Reset), 'mdm_1' (MicroBlaze Debug Module (MDM)), and 'tmr_0' (TMR Manager). The 'tmr_0' block is connected to an 'rs232_uart' block. The bottom 'Tcl Console' window shows the following log messages:

```
INFO: [xilinx.com:ip:lmb_bram_if_cntlr:4.0-7] /tmr_0/MB2/microblaze_0_local_memory/dlmb_bram_if_cntlr: This controller is connected to a lockstep slave processor (/tmr_0/MB2/microblaze_0). Address mask for SLMB is calculated from...
INFO: [xilinx.com:ip:tmr_manager:1.0-7] /tmr_0/MB1/tmr_manager_0: The TMR Manager core has been configured to operate with synchronous SEM clock.
INFO: [xilinx.com:ip:iomodule:3.1-11] /tmr_0/MB3/iomodule_0: This core is connected to a lockstep slave processor (/tmr_0/MB3/microblaze_0). Address mask for SLMB is calculated from the corresponding lockstep master (/tmr_0/MB1/m...
INFO: [xilinx.com:ip:tmr_manager:1.0-7] /tmr_0/MB3/tmr_manager_0: The TMR Manager core has been configured to operate with synchronous SEM clock.
INFO: [xilinx.com:ip:tmr_manager:1.0-4] /tmr_0/MB3/tmr_manager_0: This controller is connected to a lockstep slave processor (/tmr_0/MB3/microblaze_0). Address mask for SLMB is calculated from the corresponding lockstep master (/...
Wrote : <c:/hdl_projects/tmr_manager_1_ex/tmr_manager_1_ex.srcs/sources_1/bd/tmr_example_design/tmr_example_design.bd>
Wrote : <c:/hdl_projects/tmr_manager_1_ex/tmr_manager_1_ex.srcs/sources_1/bd/tmr_example_design/ui/bd187ddb79.ui>
save_bd_design: Time (s): cpu = 00:00:04 ; elapsed = 00:00:05 . Memory (MB): peak = 1456.910 ; gain = 0.000
INFO: [open_example_project] Open Example Project completed
update_compile_order -fileset sources_1
```

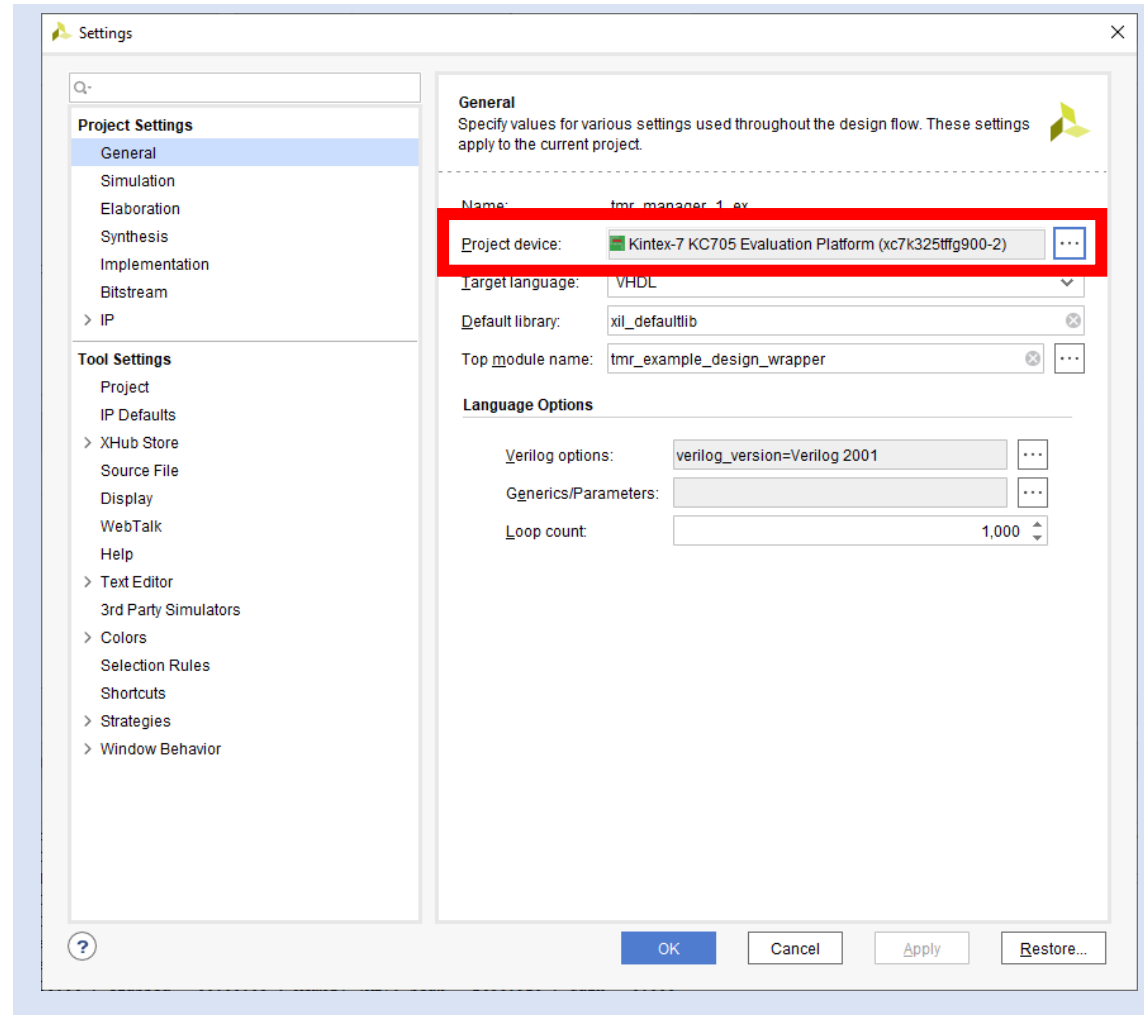
Lab 4: Overview and Introduction to Vivado

Step 15 – When the summary opens click on the project part



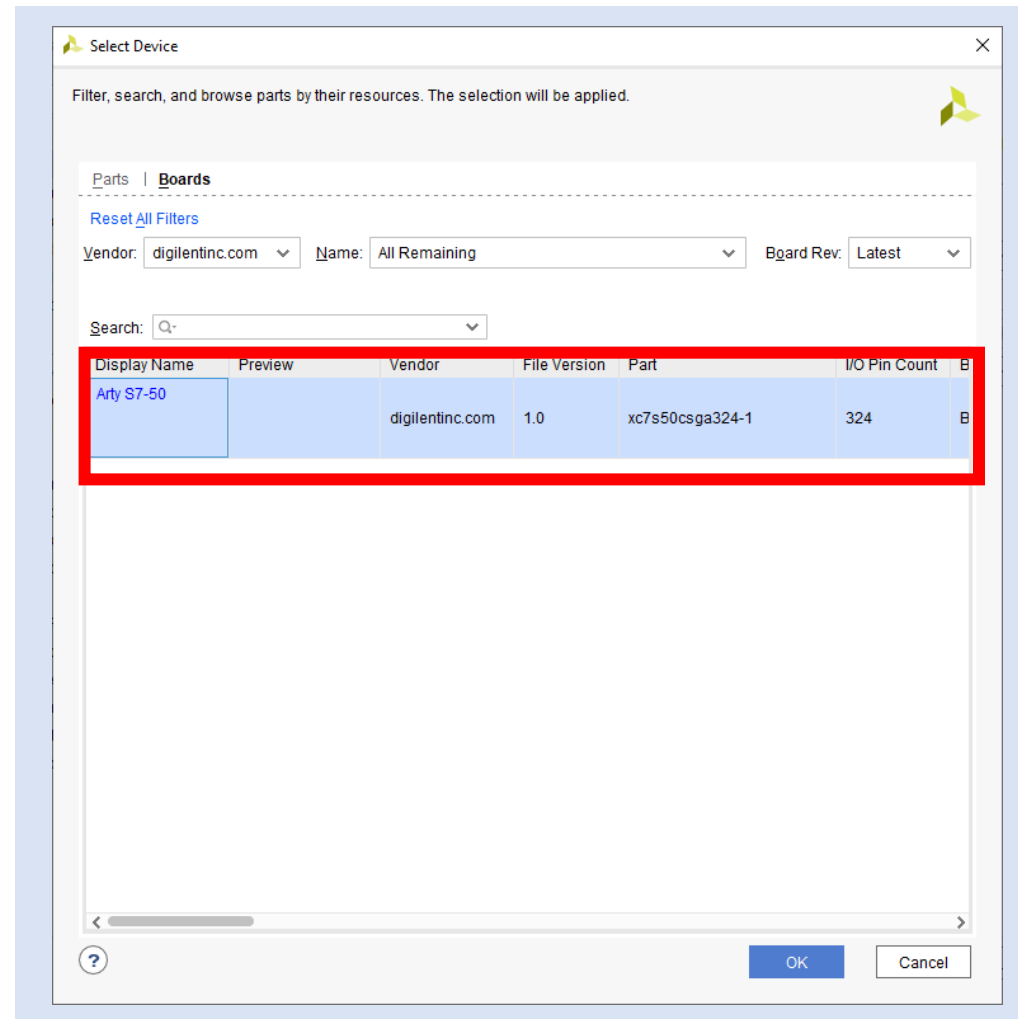
Lab 4: Overview and Introduction to Vivado

Step 16 – Click to change the project device



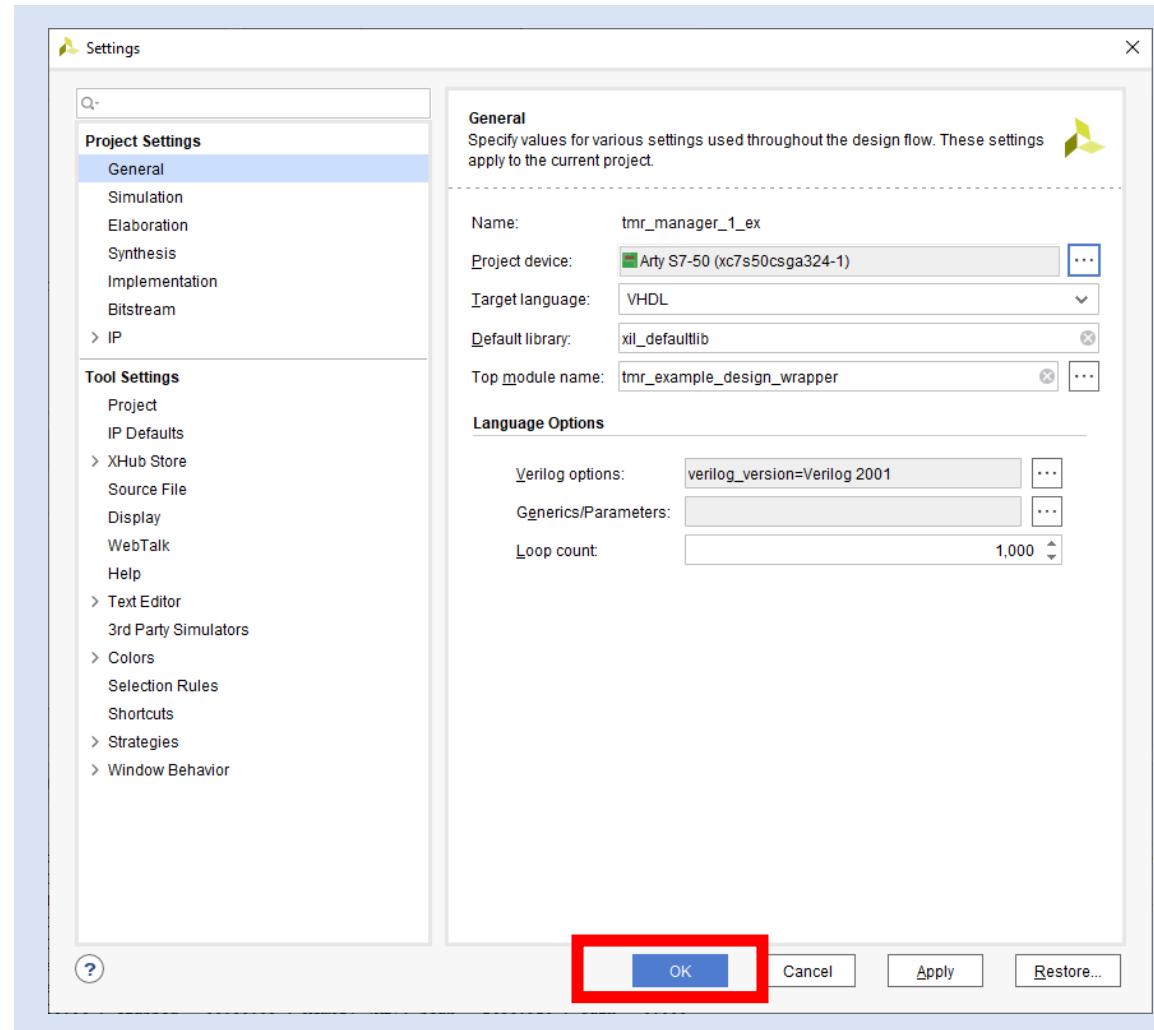
Lab 4: Overview and Introduction to Vivado

Step 17 – Select the Arty S7-50



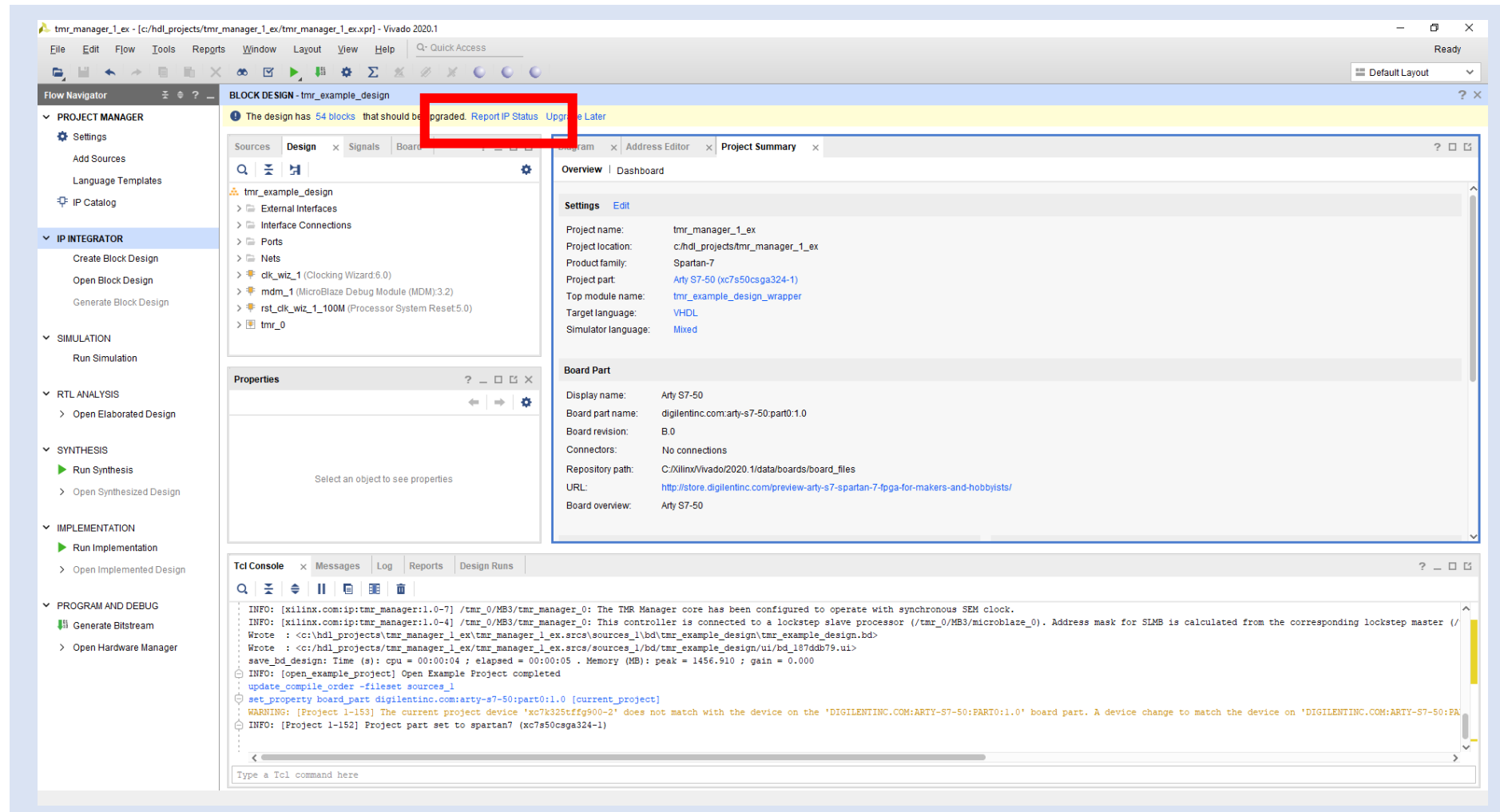
Lab 4: Overview and Introduction to Vivado

Step 18 – Click OK



Lab 4: Overview and Introduction to Vivado

Step 19 – Click on report IP Status



Lab 4: Overview and Introduction to Vivado

Step 20 – Click on upgrade selected

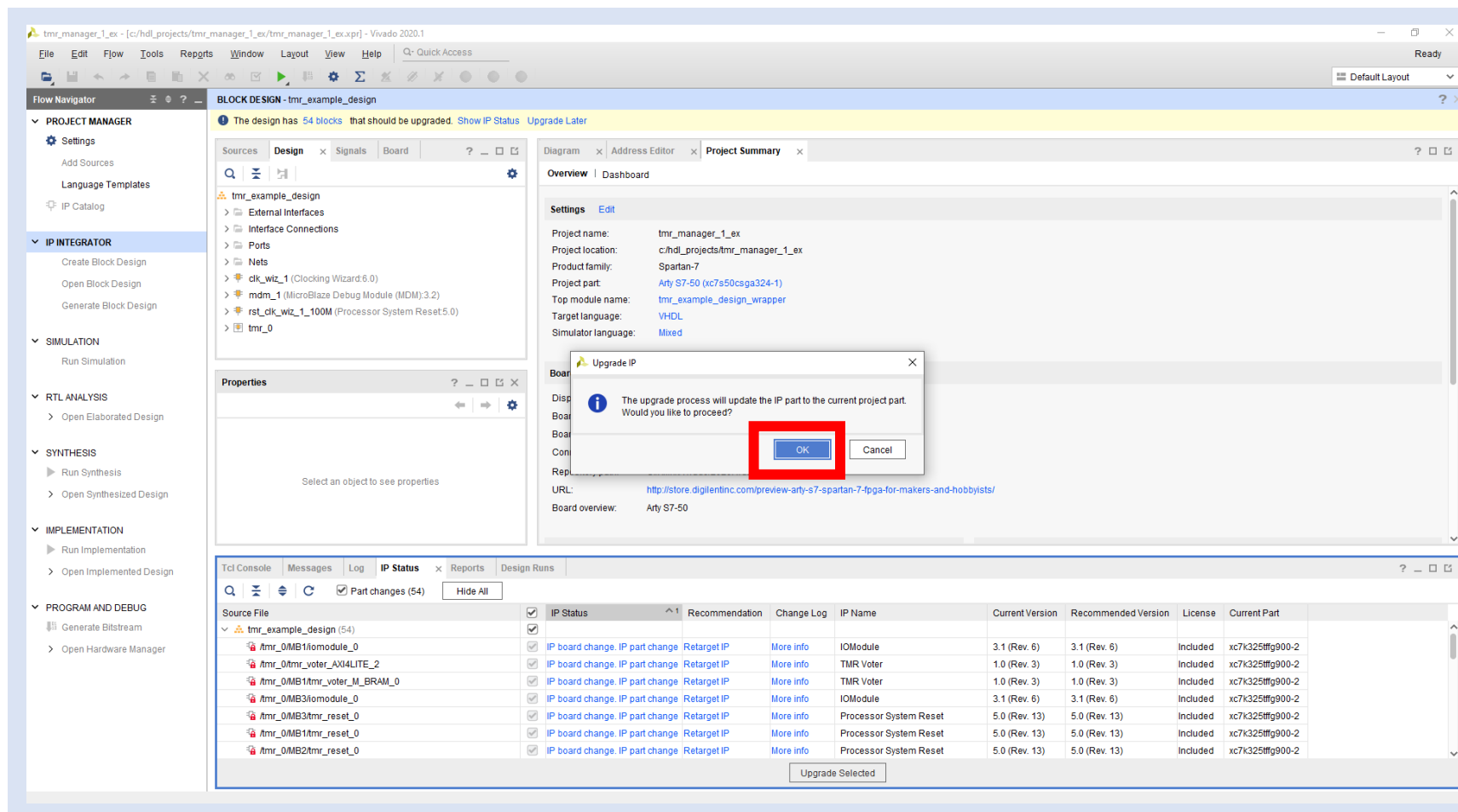
The screenshot shows the Vivado 2020.1 interface. The left sidebar contains the Project Manager, IP Integrator, Simulation, RTL Analysis, Synthesis, Implementation, and Program and Debug sections. The main workspace is divided into several panes. The top pane shows a message: "The design has 54 blocks that should be upgraded. Show IP Status Upgrade Later". Below this, the Design pane shows a tree view of the design components. The Properties pane is empty. The IP Status pane is active, displaying a table of IP components. The table has columns for Source File, IP Status, Recommendation, Change Log, IP Name, Current Version, Recommended Version, License, and Current Part. The table lists 54 IP components, all of which are marked as "IP board change. IP part change" and "Retarget IP". The "Upgrade Selected" button is highlighted with a red box.

Source File	IP Status	Recommendation	Change Log	IP Name	Current Version	Recommended Version	License	Current Part
tmr_example_design (54)	✓							
tmr_0mb1/iomodule_0	✓	IP board change. IP part change	Retarget IP	More info	3.1 (Rev. 6)	3.1 (Rev. 6)	Included	xc7k325ffg900-2
tmr_0mb1/tmr_voter_AXI4LITE_2	✓	IP board change. IP part change	Retarget IP	More info	1.0 (Rev. 3)	1.0 (Rev. 3)	Included	xc7k325ffg900-2
tmr_0mb1/tmr_voter_M_BRAM_0	✓	IP board change. IP part change	Retarget IP	More info	1.0 (Rev. 3)	1.0 (Rev. 3)	Included	xc7k325ffg900-2
tmr_0mb3/iomodule_0	✓	IP board change. IP part change	Retarget IP	More info	3.1 (Rev. 6)	3.1 (Rev. 6)	Included	xc7k325ffg900-2
tmr_0mb3/tmr_reset_0	✓	IP board change. IP part change	Retarget IP	More info	5.0 (Rev. 13)	5.0 (Rev. 13)	Included	xc7k325ffg900-2
tmr_0mb1/tmr_reset_0	✓	IP board change. IP part change	Retarget IP	More info	5.0 (Rev. 13)	5.0 (Rev. 13)	Included	xc7k325ffg900-2
tmr_0mb2/tmr_reset_0	✓	IP board change. IP part change	Retarget IP	More info	5.0 (Rev. 13)	5.0 (Rev. 13)	Included	xc7k325ffg900-2

Upgrade Selected

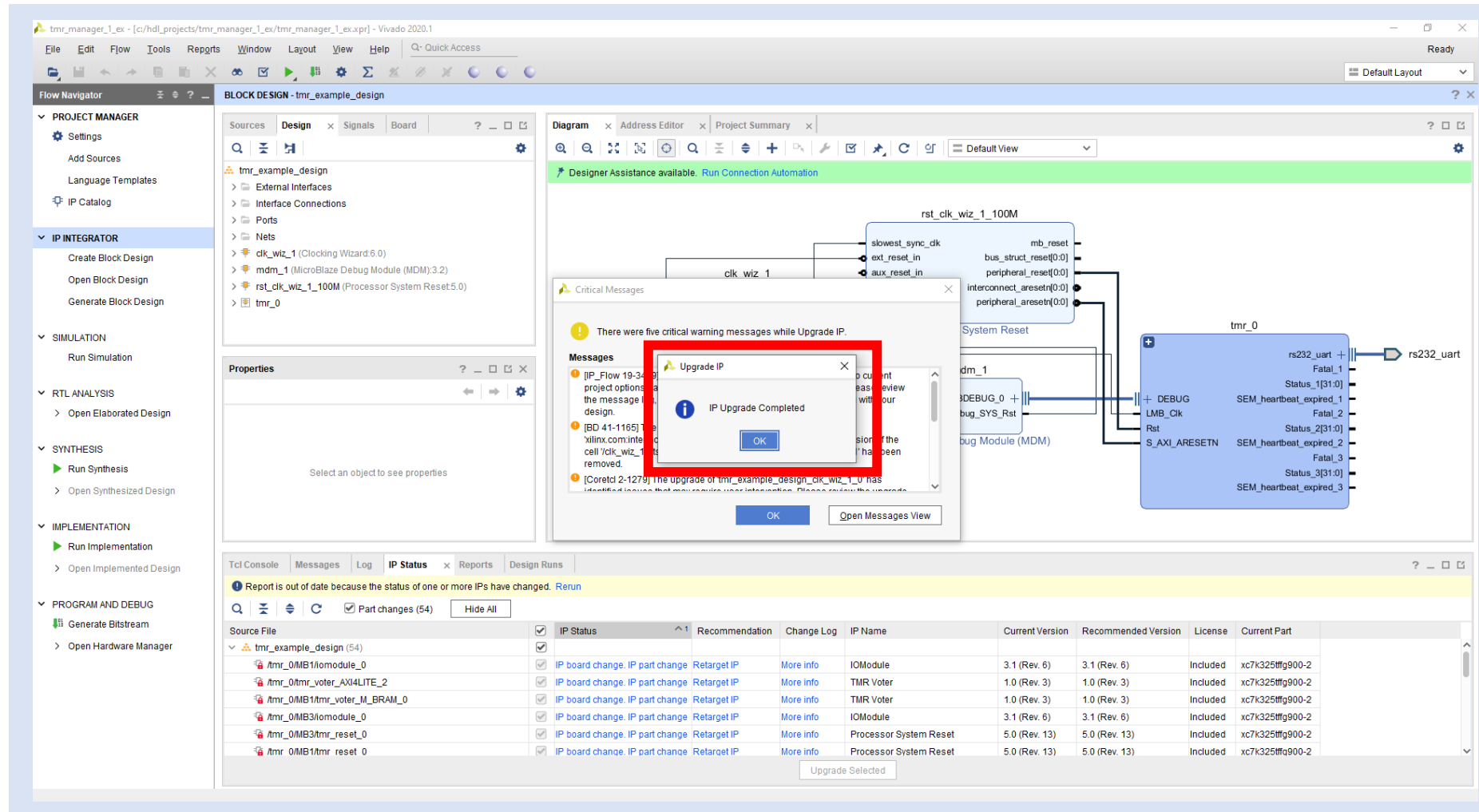
Lab 4: Overview and Introduction to Vivado

Step 21 – Click OK



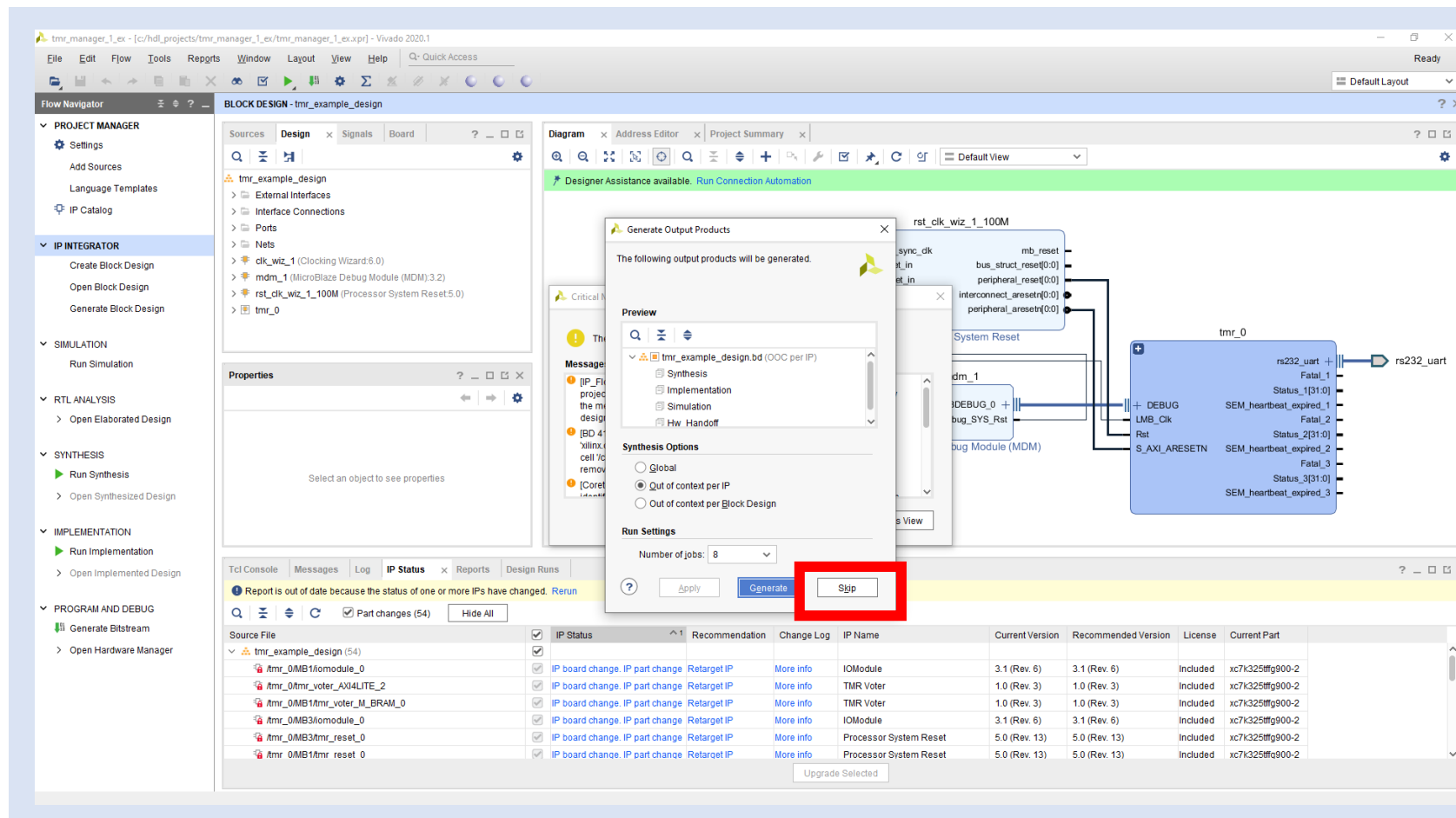
Lab 4: Overview and Introduction to Vivado

Step 22 – Click OK once the IP Upgrade completes



Lab 4: Overview and Introduction to Vivado

Step 23 – Click On Skip



Lab 4: Overview and Introduction to Vivado

Step 24 – Read the messages and then click on OK

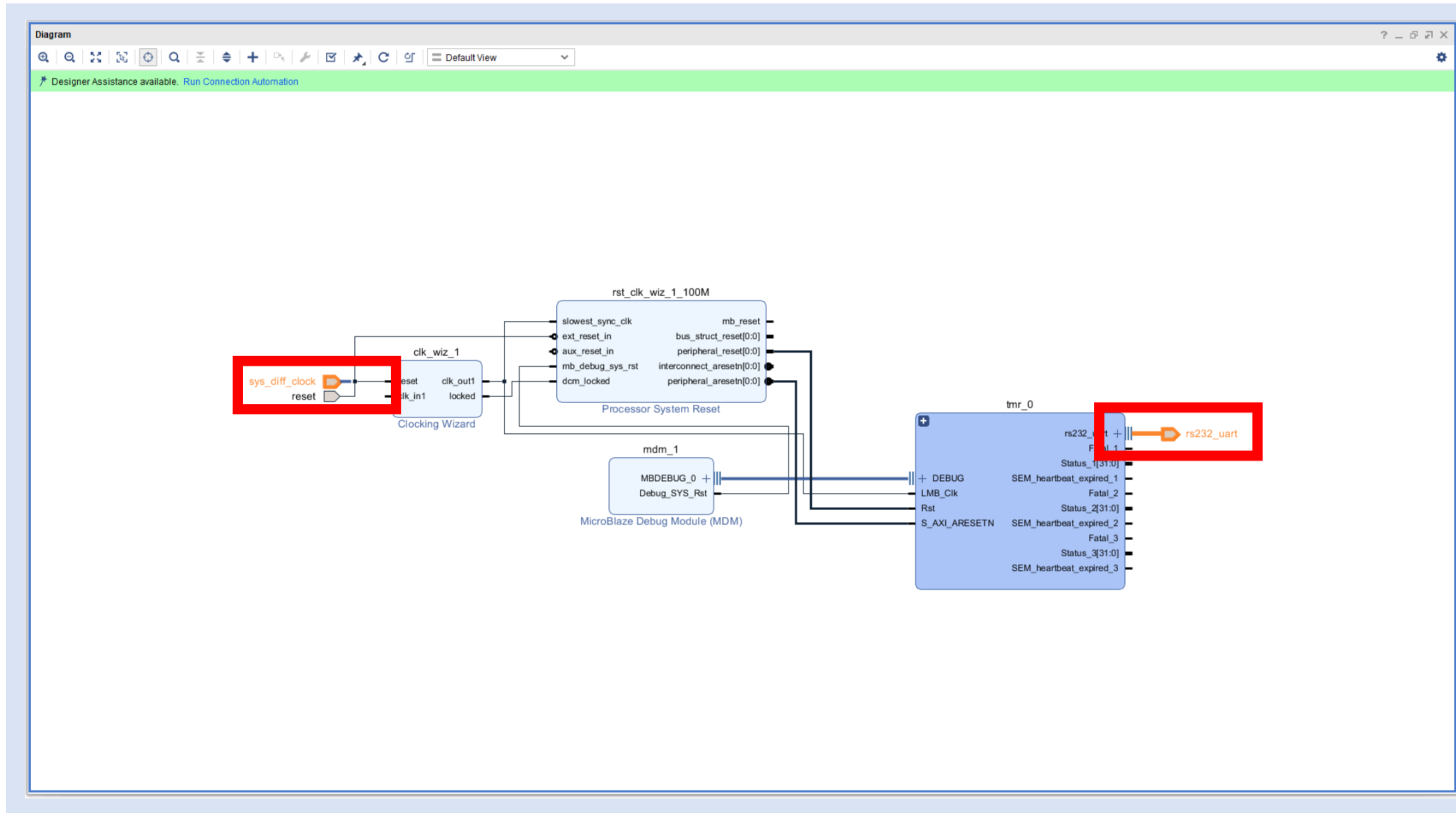
The screenshot displays the Vivado IDE interface during an IP upgrade process. A 'Critical Messages' dialog box is prominently shown in the center, highlighting two critical messages. The first message indicates that five critical warning messages were generated during the IP upgrade. The second message, from Coreltd 2-1279, warns that the upgrade of the 'tmr_example_design_rst_clk_wiz_1_100M_0' IP block has identified issues requiring user intervention, advising the user to review the upgrade log and verify the configuration.

The background shows the 'BLOCK DESIGN - tmr_example_design' with various IP blocks connected. The 'IP Status' tab at the bottom provides a detailed table of the components and their upgrade status.

Source File	IP Status	Recommendation	Change Log	IP Name	Current Version	Recommended Version	License	Current Part
tmr_example_design (54)	✓							
tmr_0/MB1/omodule_0	✓	IP board change. IP part change	Retarget IP	More info	3.1 (Rev. 6)	3.1 (Rev. 6)	Included	xc7k325ffg900-2
tmr_0/MB1/voter_AXI4LITE_2	✓	IP board change. IP part change	Retarget IP	More info	1.0 (Rev. 3)	1.0 (Rev. 3)	Included	xc7k325ffg900-2
tmr_0/MB1/voter_M_BRAM_0	✓	IP board change. IP part change	Retarget IP	More info	1.0 (Rev. 3)	1.0 (Rev. 3)	Included	xc7k325ffg900-2
tmr_0/MB3/omodule_0	✓	IP board change. IP part change	Retarget IP	More info	3.1 (Rev. 6)	3.1 (Rev. 6)	Included	xc7k325ffg900-2
tmr_0/MB3/rst_clk_wiz_1_100M_0	✓	IP board change. IP part change	Retarget IP	More info	5.0 (Rev. 13)	5.0 (Rev. 13)	Included	xc7k325ffg900-2
tmr_0/MB3/rst_clk_wiz_1_100M_0	✓	IP board change. IP part change	Retarget IP	More info	5.0 (Rev. 13)	5.0 (Rev. 13)	Included	xc7k325ffg900-2

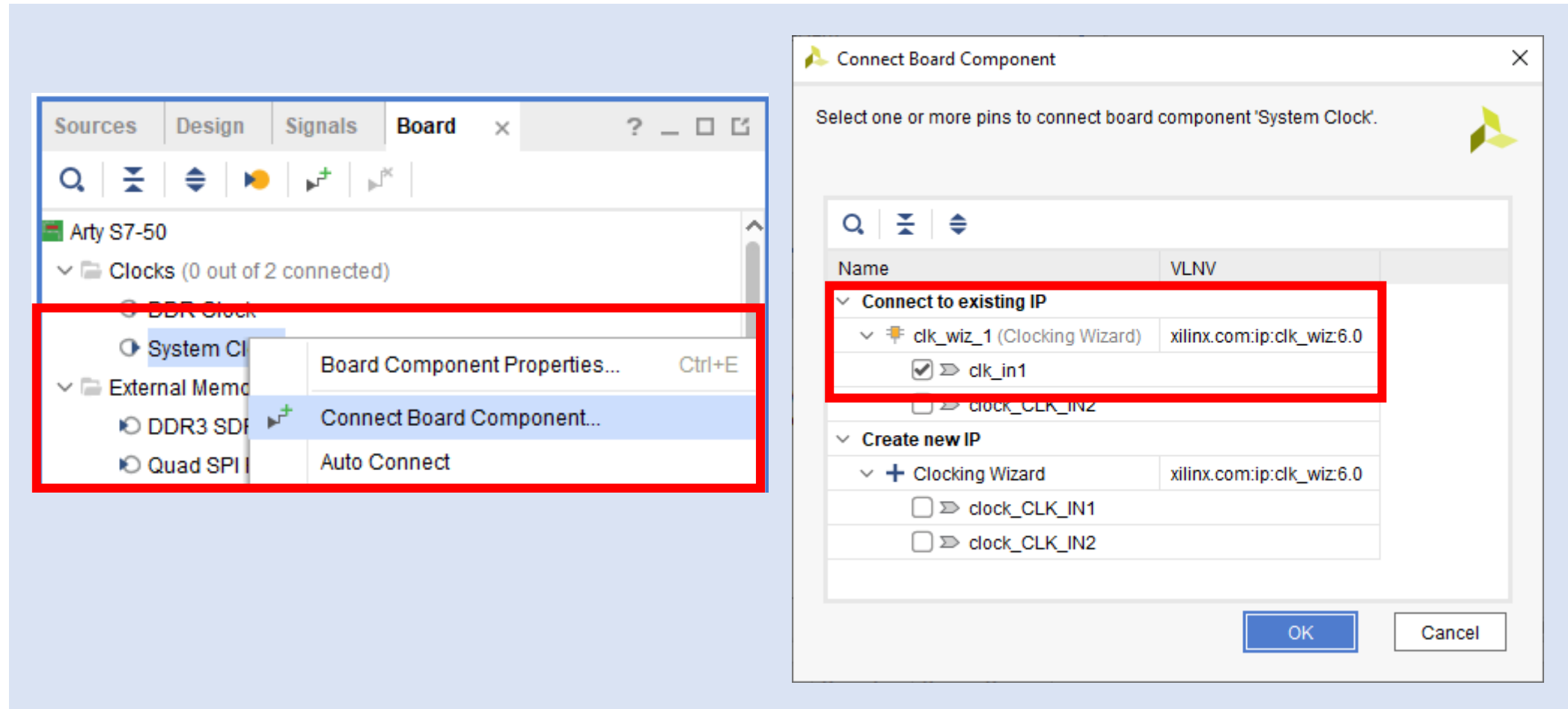
Lab 4: Overview and Introduction to Vivado

Step 25 – Open the block diagram select the diff clock and rs232 uart signals and delete them



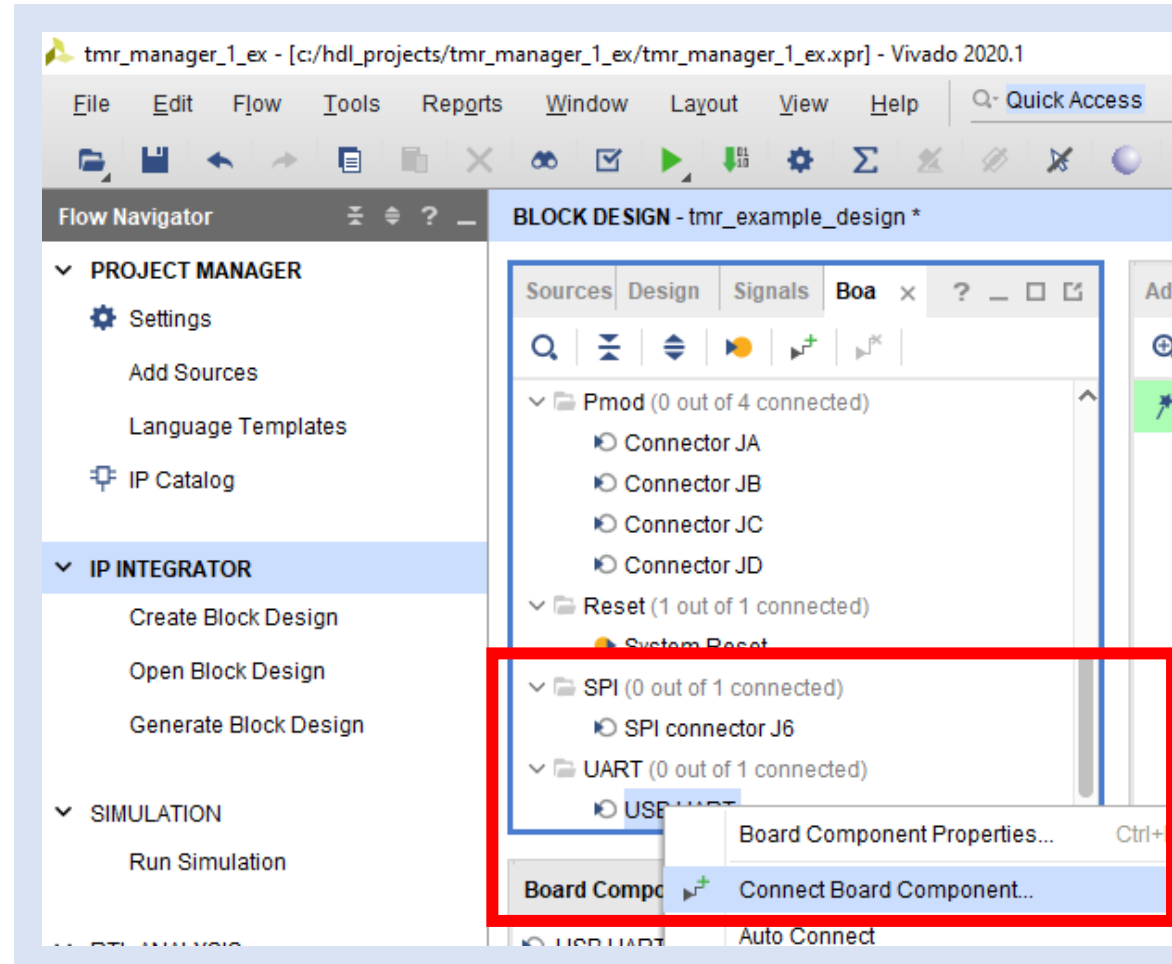
Lab 4: Overview and Introduction to Vivado

Step 26 – Select the boards tab, right click on the Sys Clock and select Connect Board Component



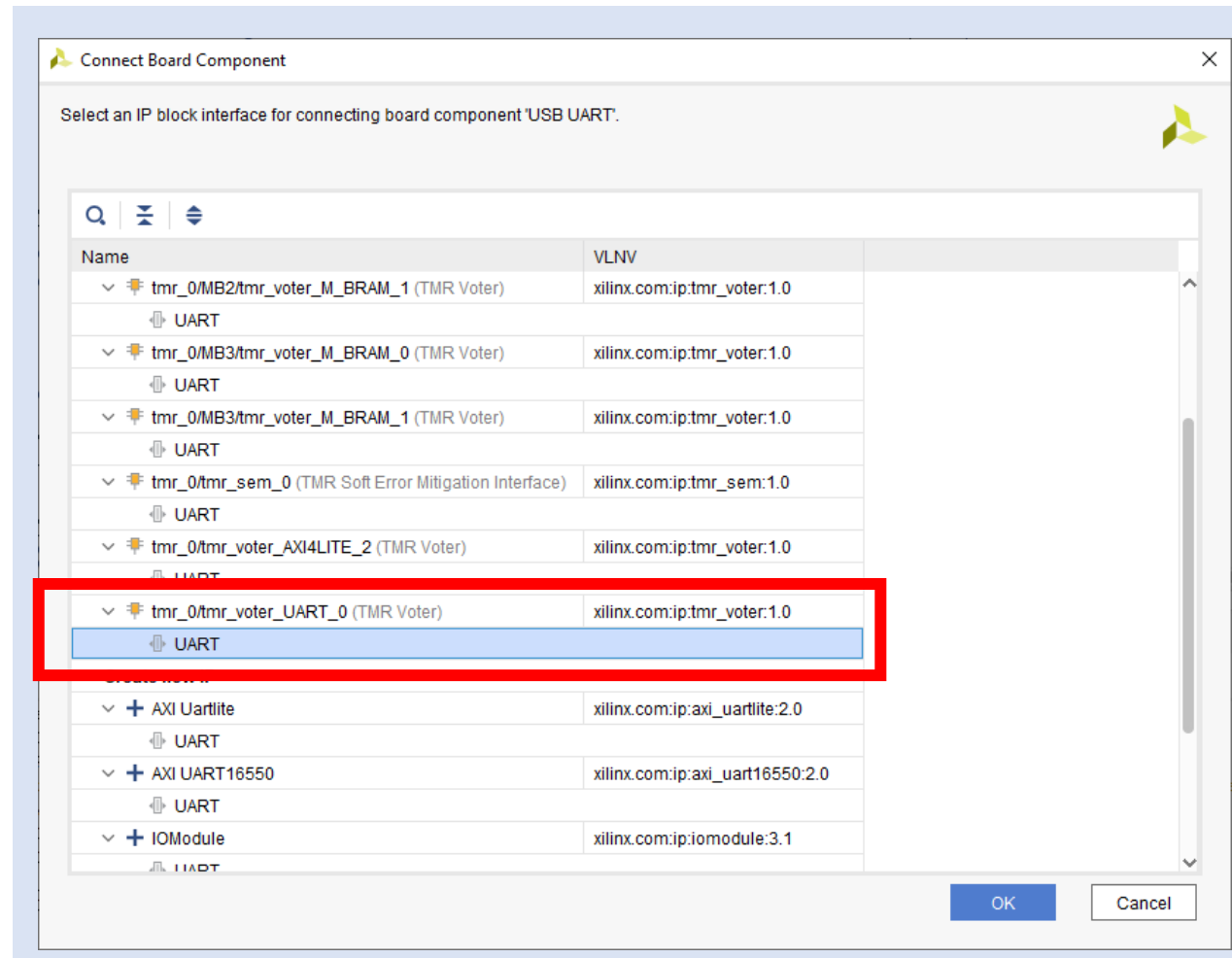
Lab 4: Overview and Introduction to Vivado

Step 27 – Do the same for the UART



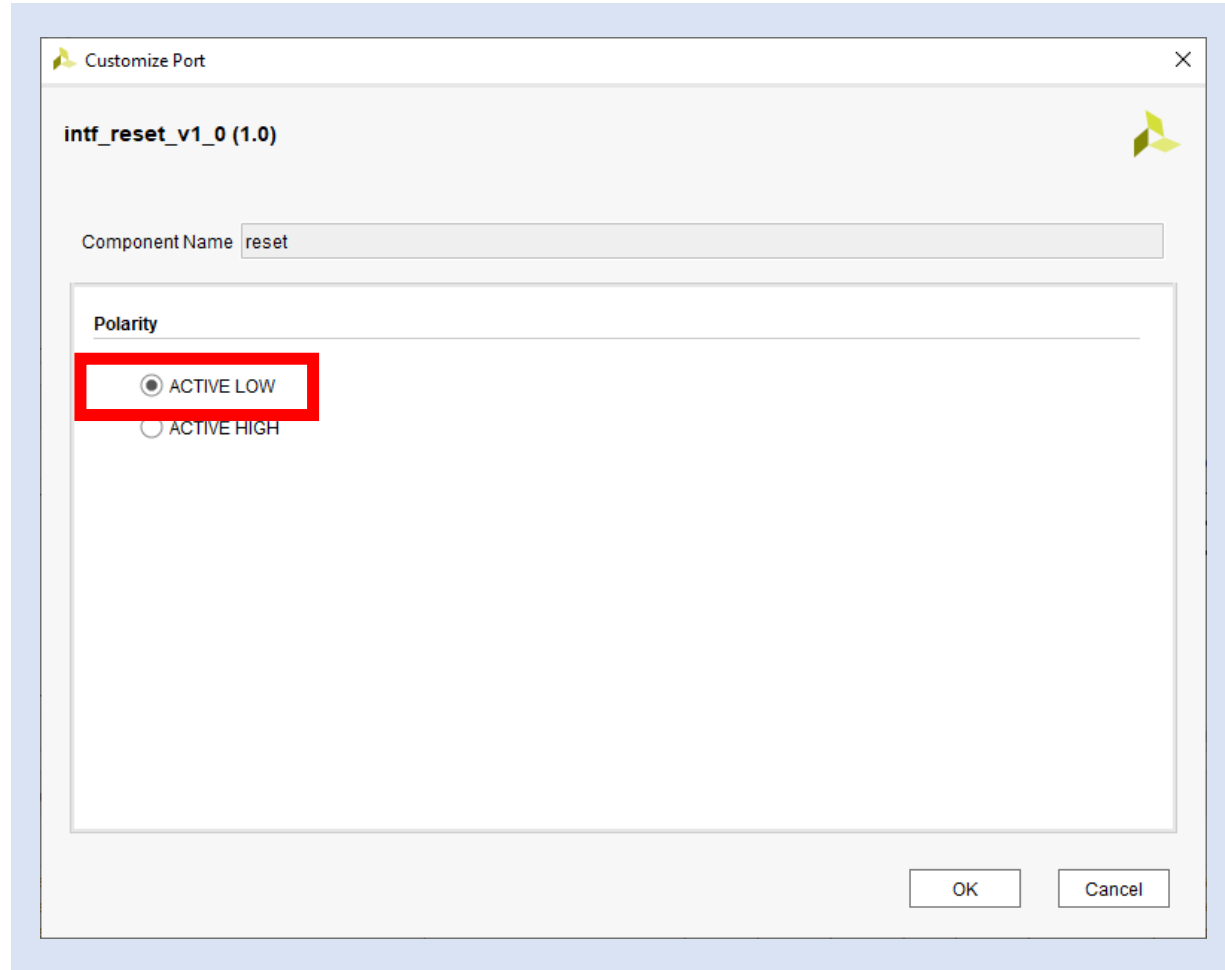
Lab 4: Overview and Introduction to Vivado

Step 28 – Select tmr_voter_UART_0



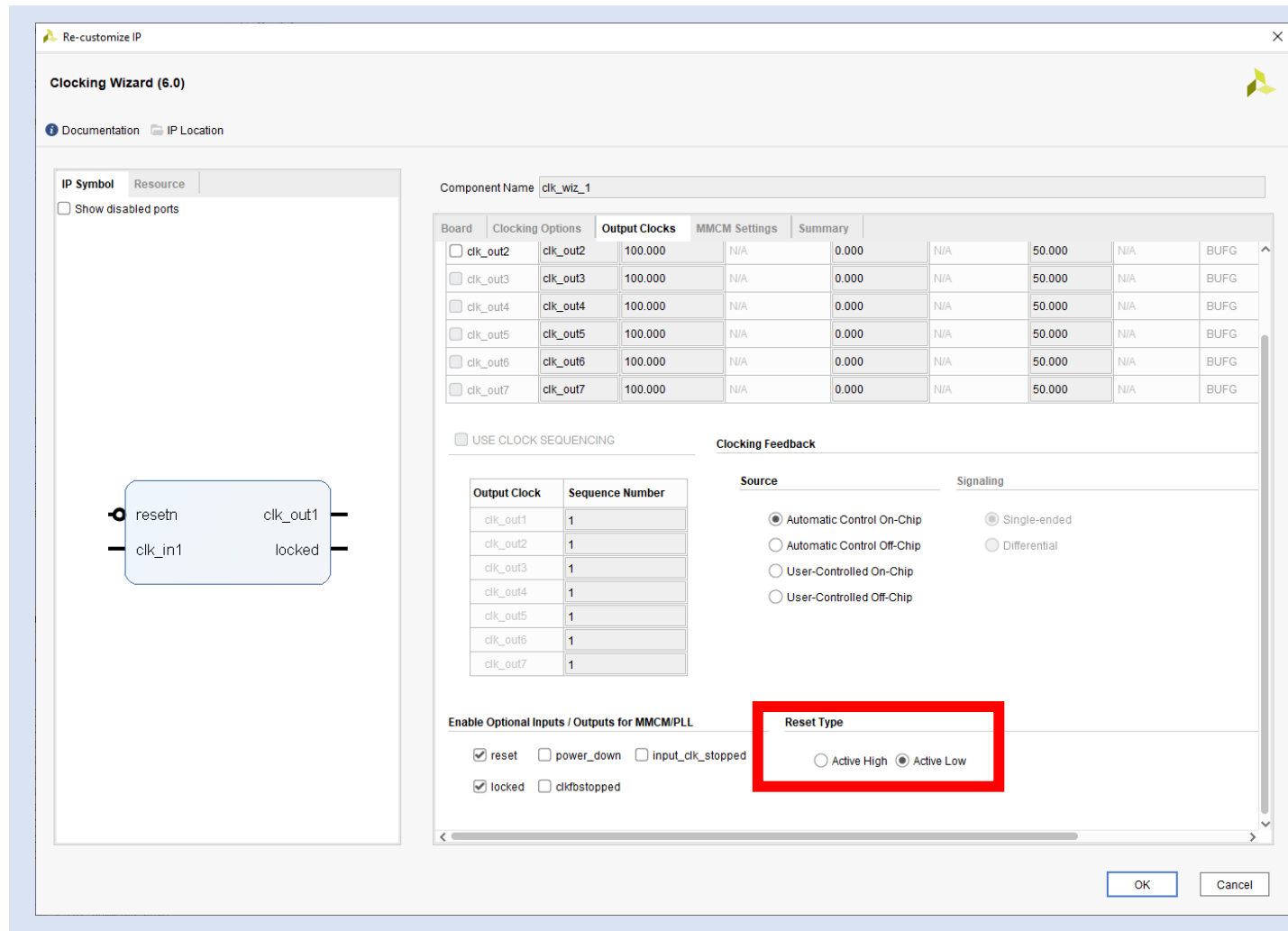
Lab 4: Overview and Introduction to Vivado

Step 29 – Click on the reset port and change the polarity to Active Low



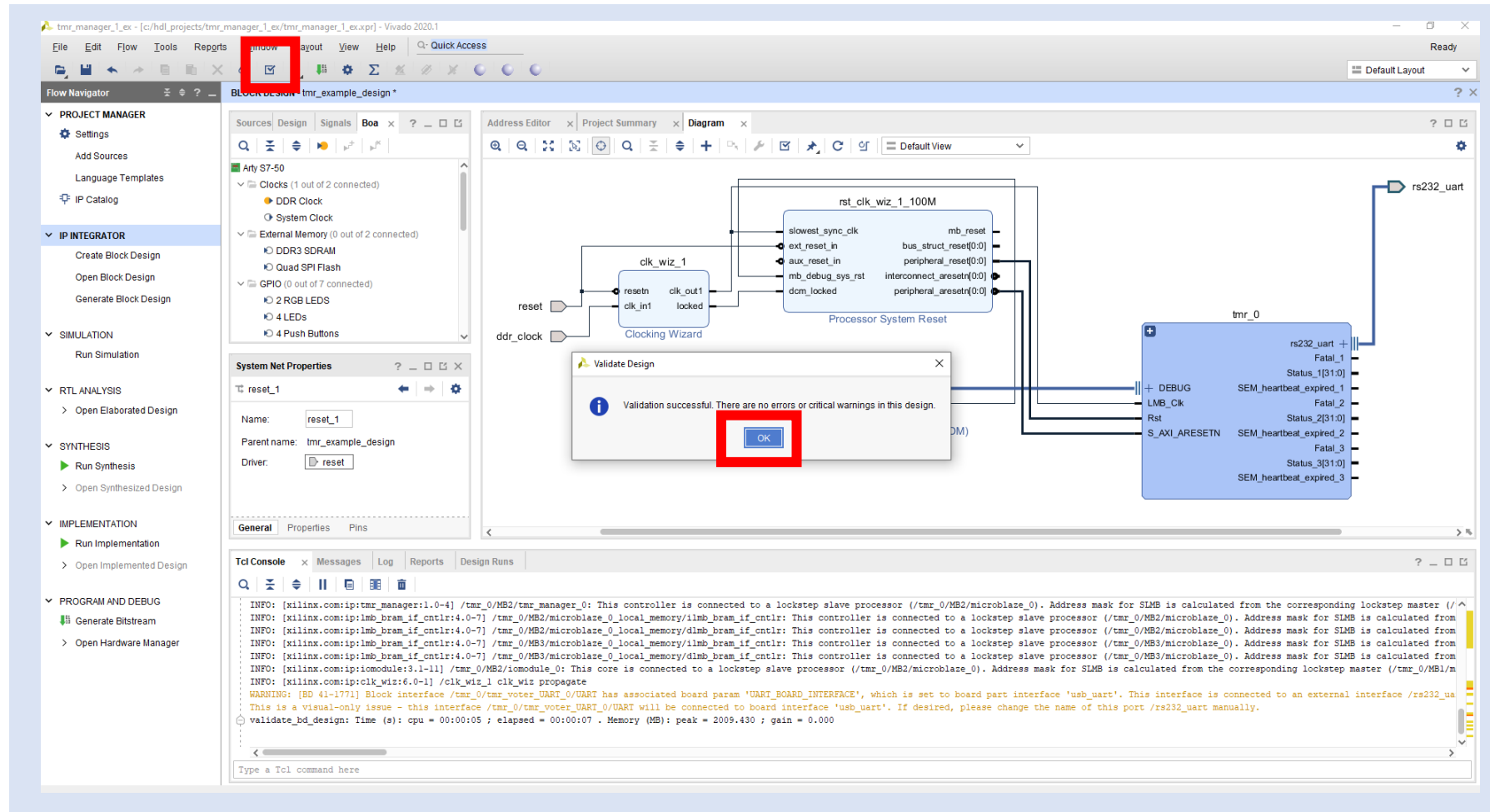
Lab 4: Overview and Introduction to Vivado

Step 30 – Open the clock wizard and change the reset to active low



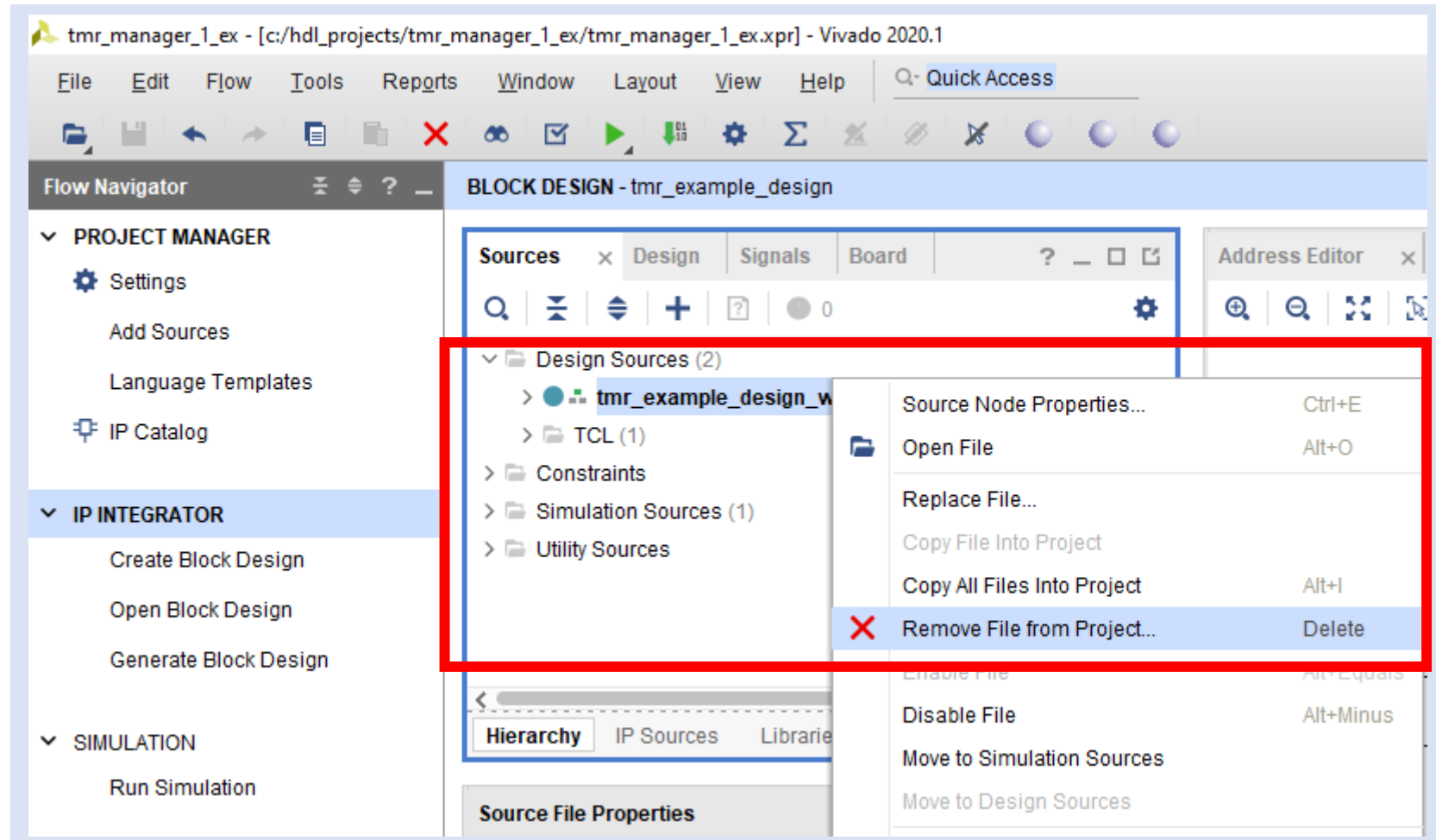
Lab 4: Overview and Introduction to Vivado

Step 31 – Validate the project, when completed click OK



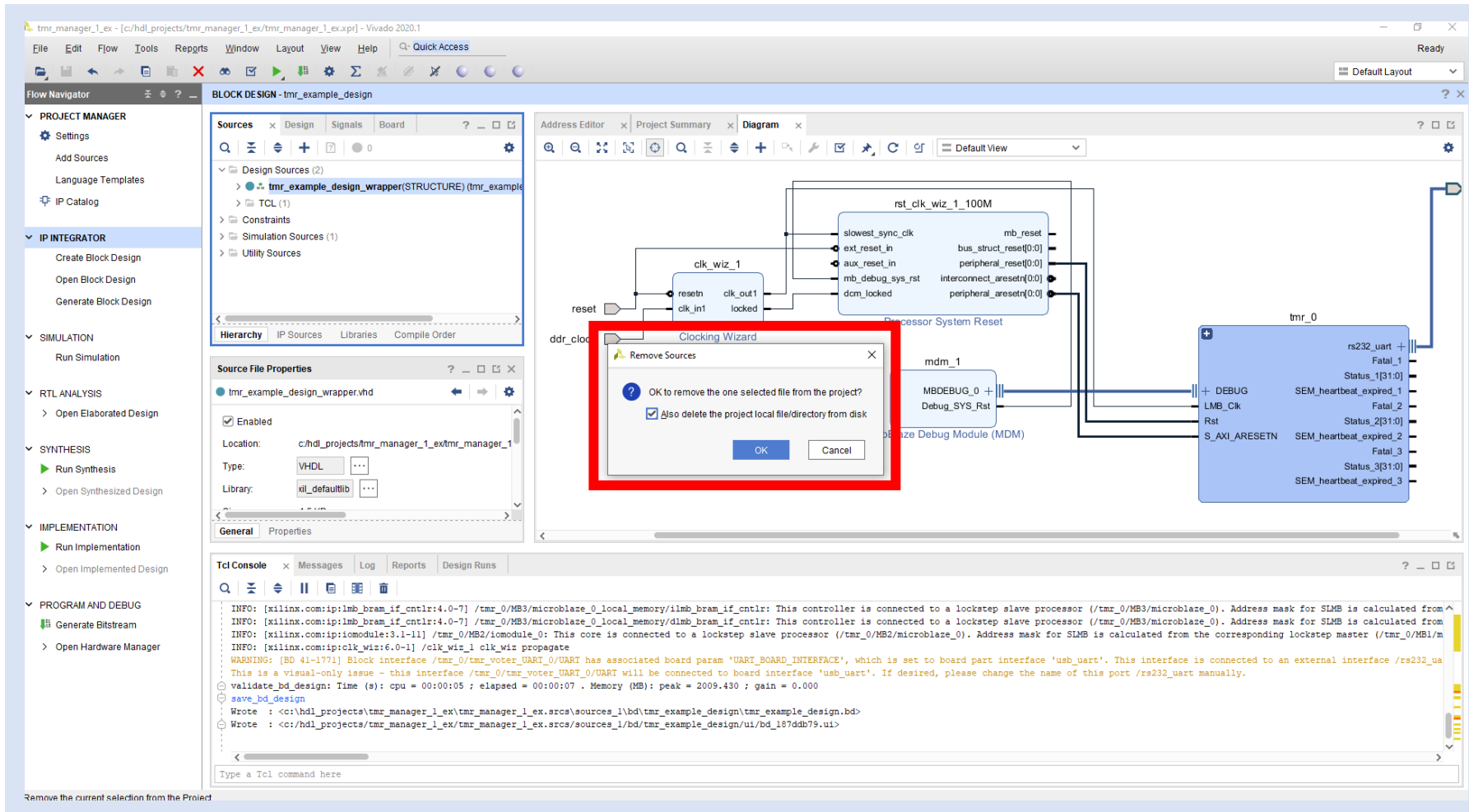
Lab 4: Overview and Introduction to Vivado

Step 32 – In sources select the top-level file and remove from project



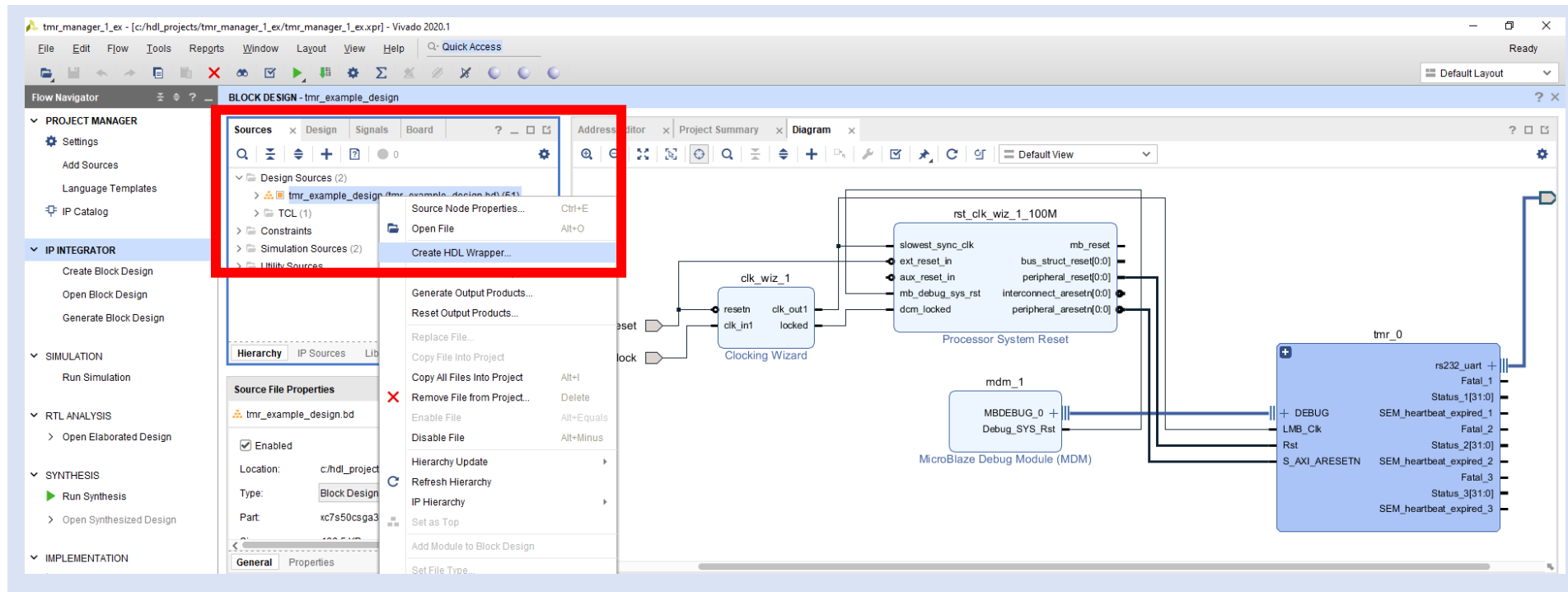
Lab 4: Overview and Introduction to Vivado

Step 33 – When prompted check delete file from disk and click OK



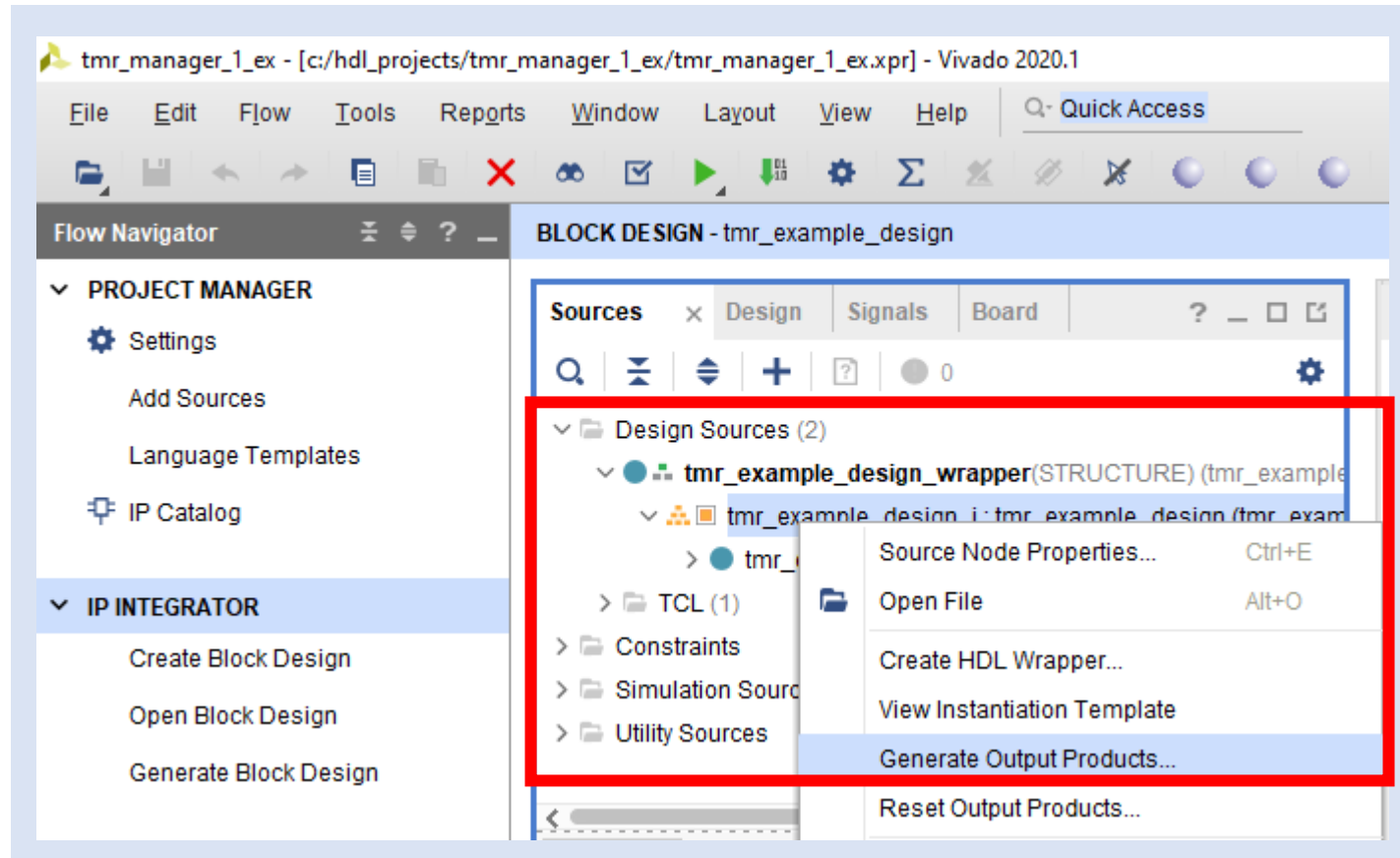
Lab 4: Overview and Introduction to Vivado

Step 34 – In sources select the top level of the block diagram and select create HDL Wrapper – when prompted let Vivado manage it



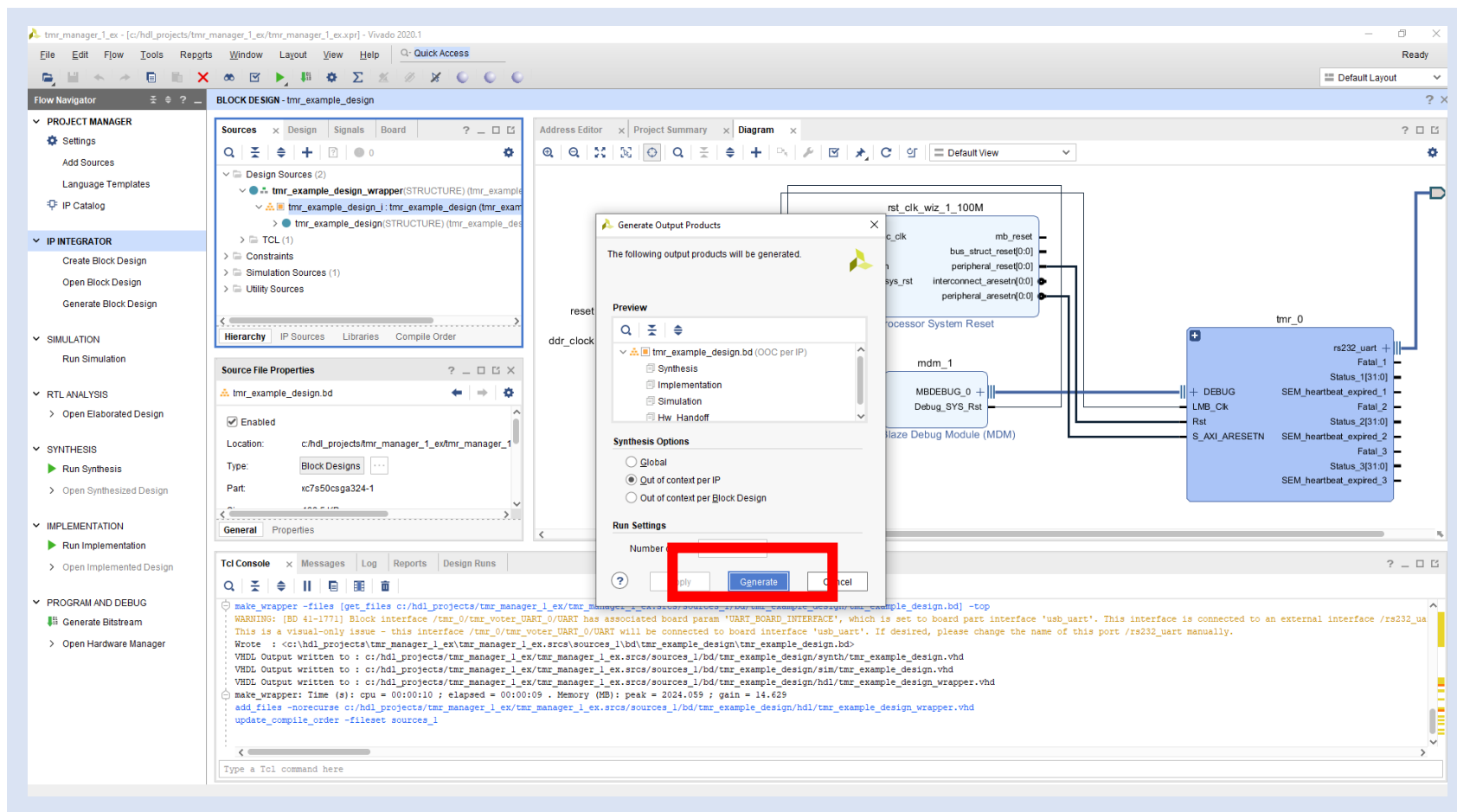
Lab 4: Overview and Introduction to Vivado

Step 35 – Generate the output products



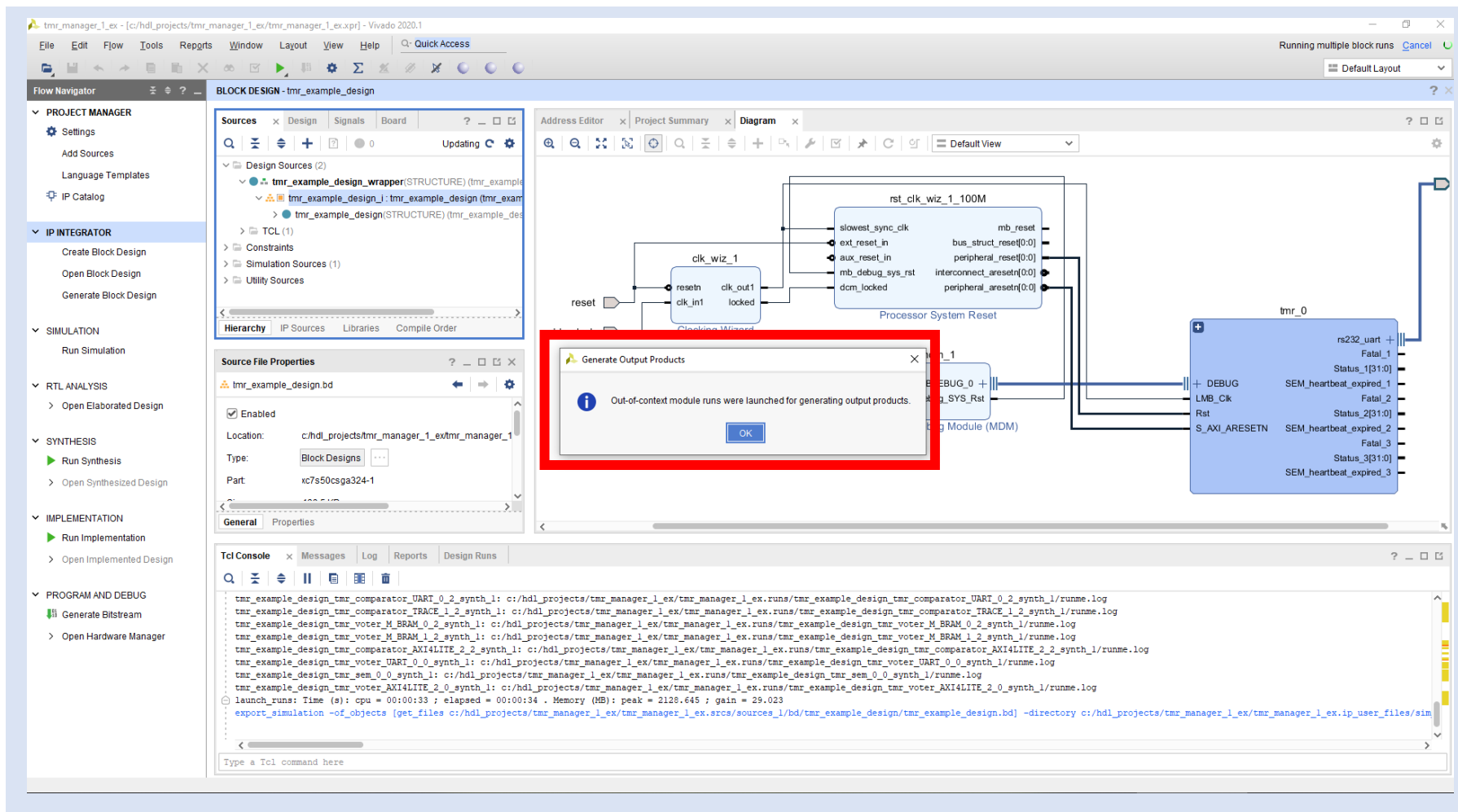
Lab 4: Overview and Introduction to Vivado

Step 36 – Select Generate



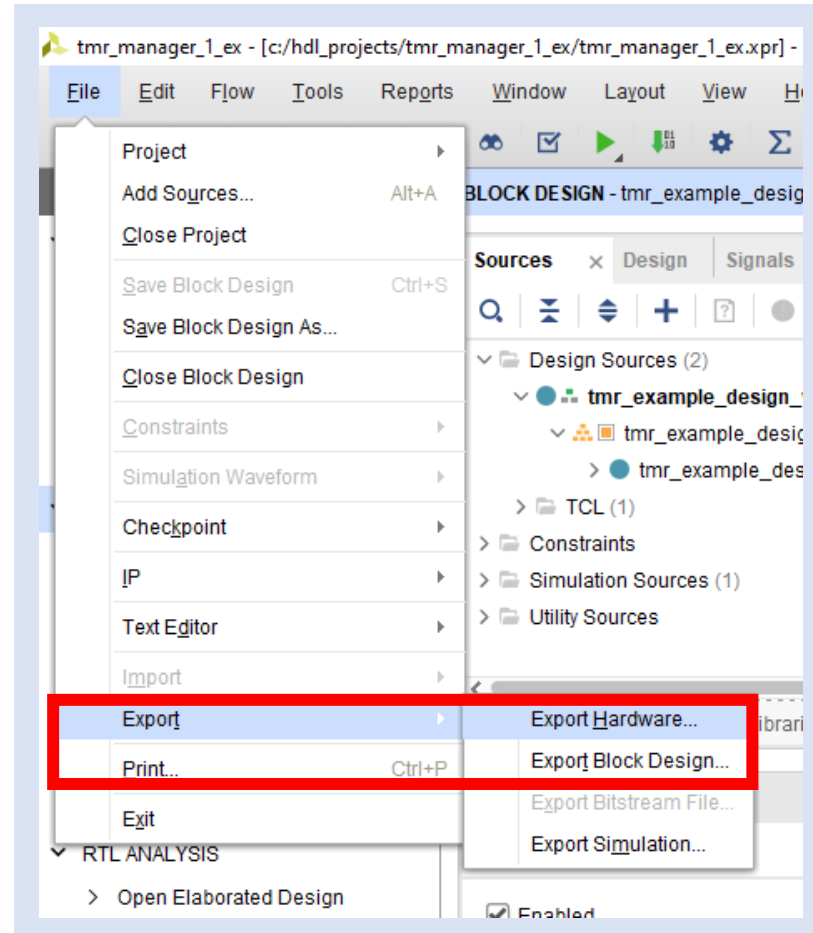
Lab 4: Overview and Introduction to Vivado

Step 37 – Click OK



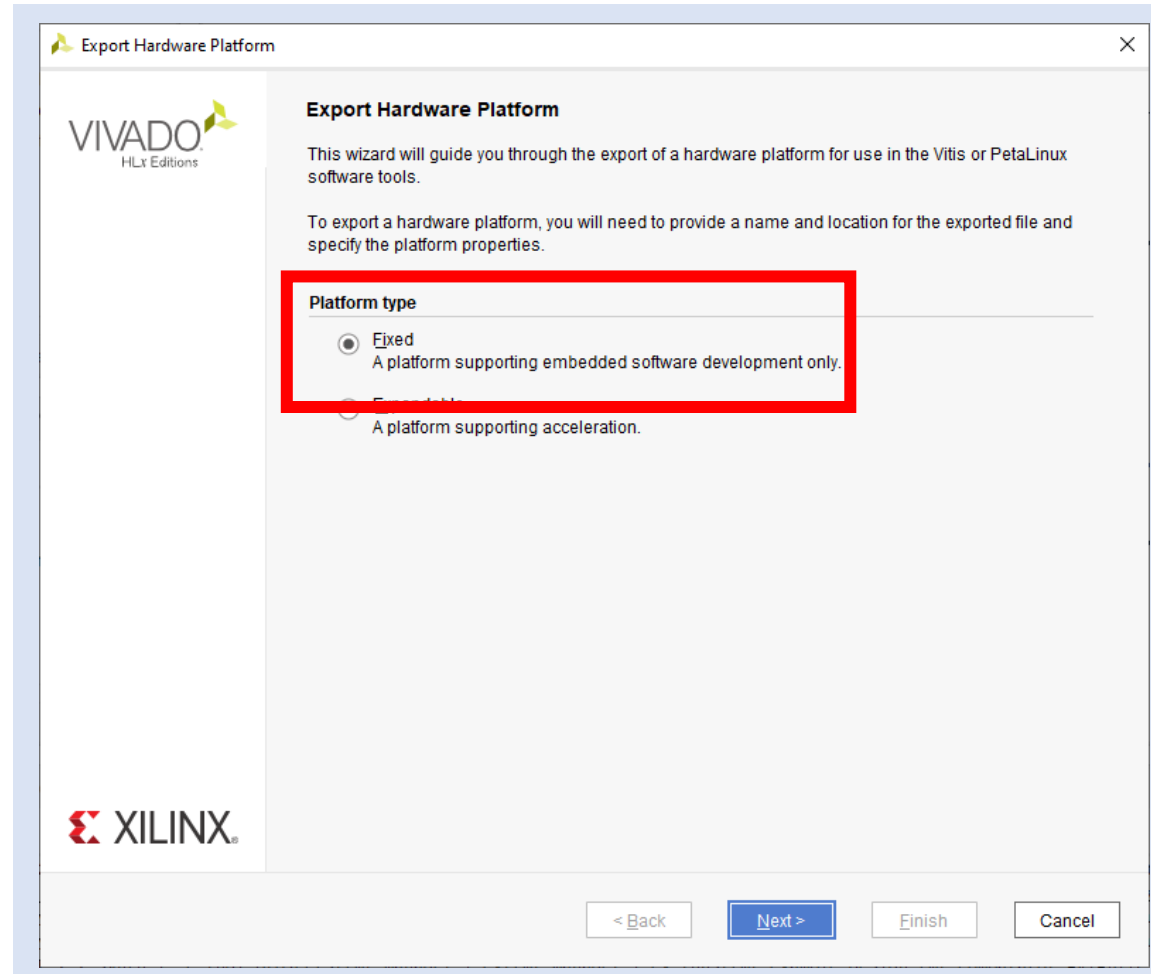
Lab 4: Overview and Introduction to Vivado

Step 38 – Export the hardware definition to create the software application



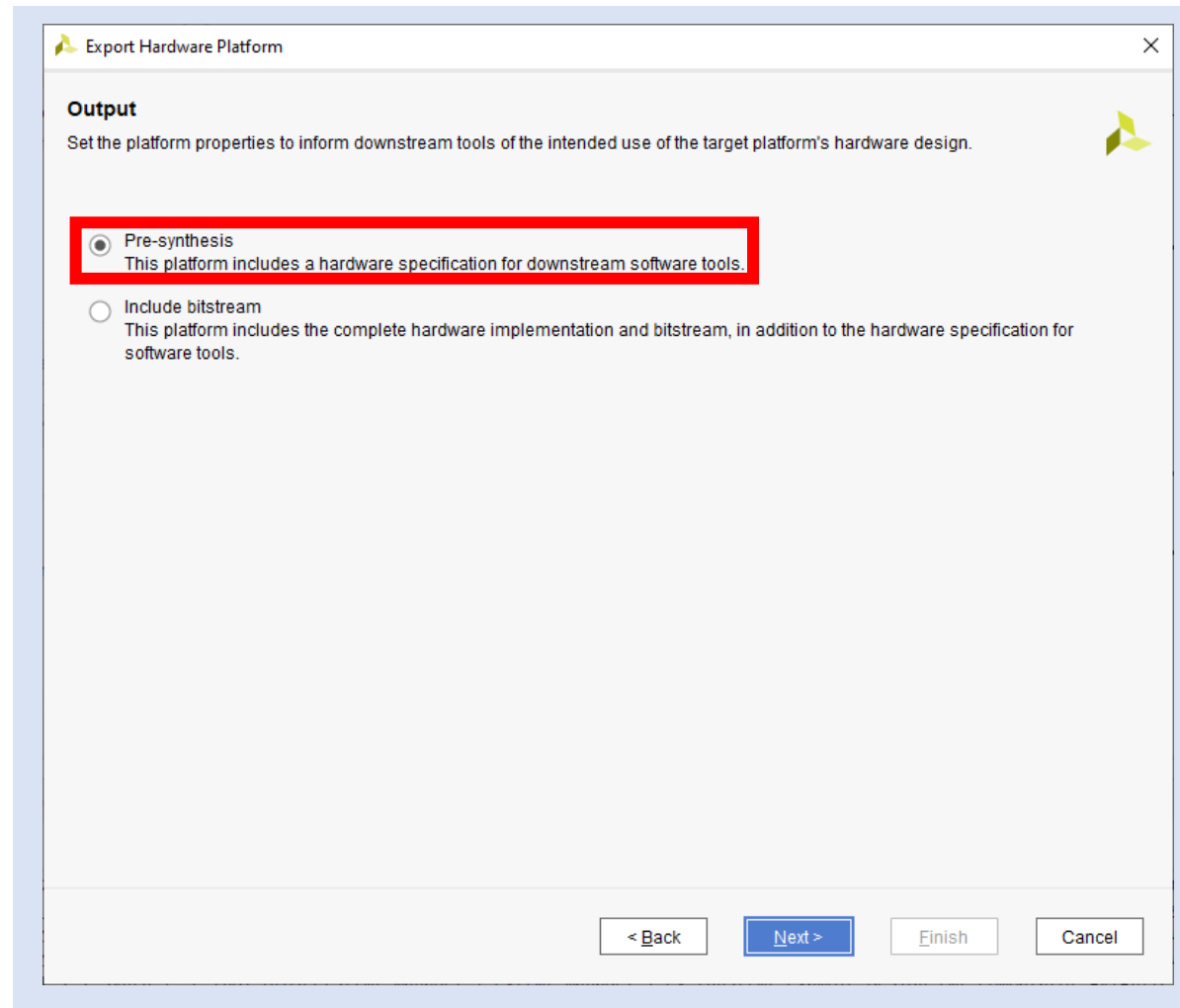
Lab 4: Overview and Introduction to Vivado

Step 39 – Select the fixed Platform Type



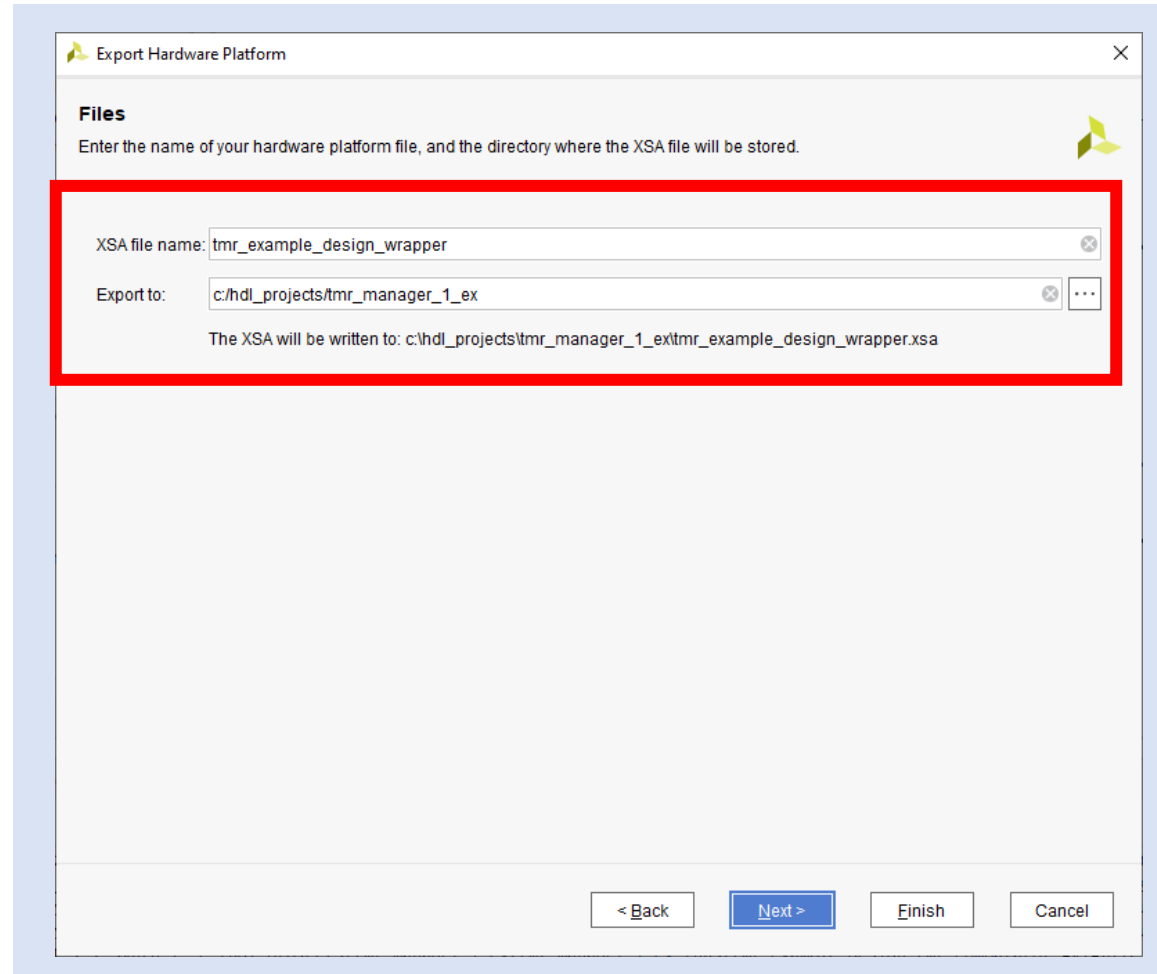
Lab 4: Overview and Introduction to Vivado

Step 40 – Select Pre-Synthesis



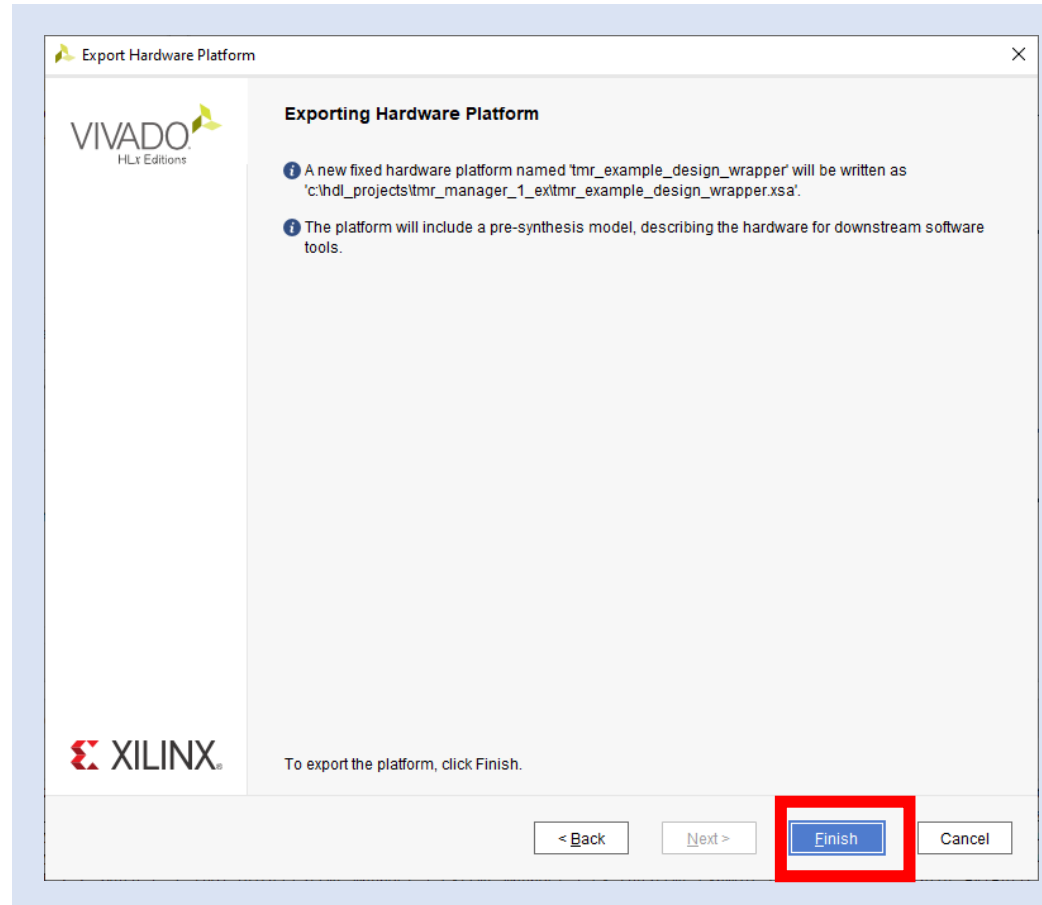
Lab 4: Overview and Introduction to Vivado

Step 41 – Enter a name and location – I recommend leaving defaults



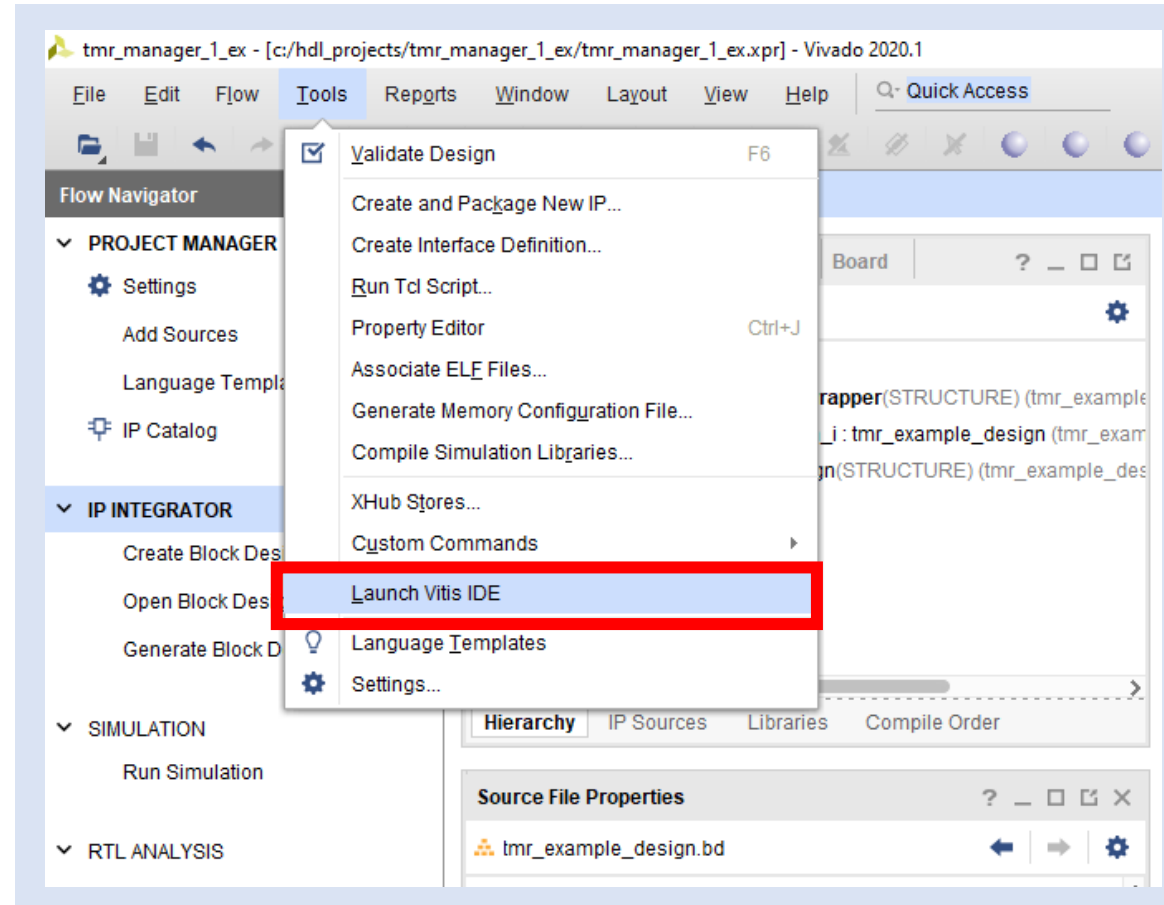
Lab 4: Overview and Introduction to Vivado

Step 42 – Click Finish



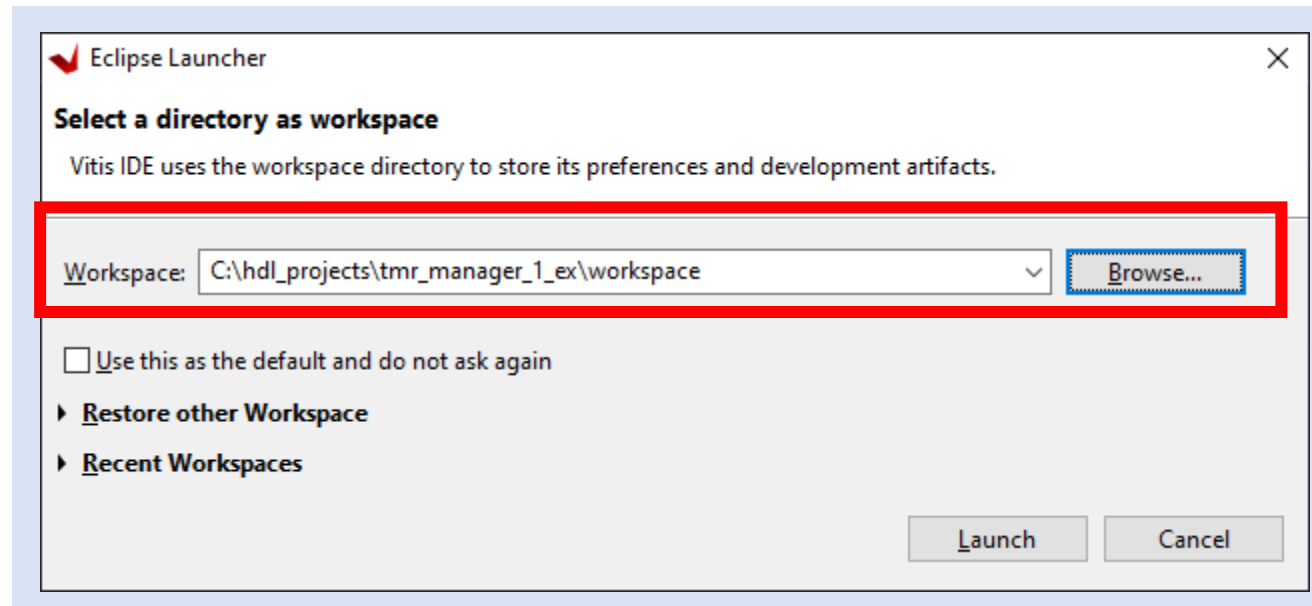
Lab 4: Overview and Introduction to Vivado

Step 43 – Open Vitis from the Tools Menu



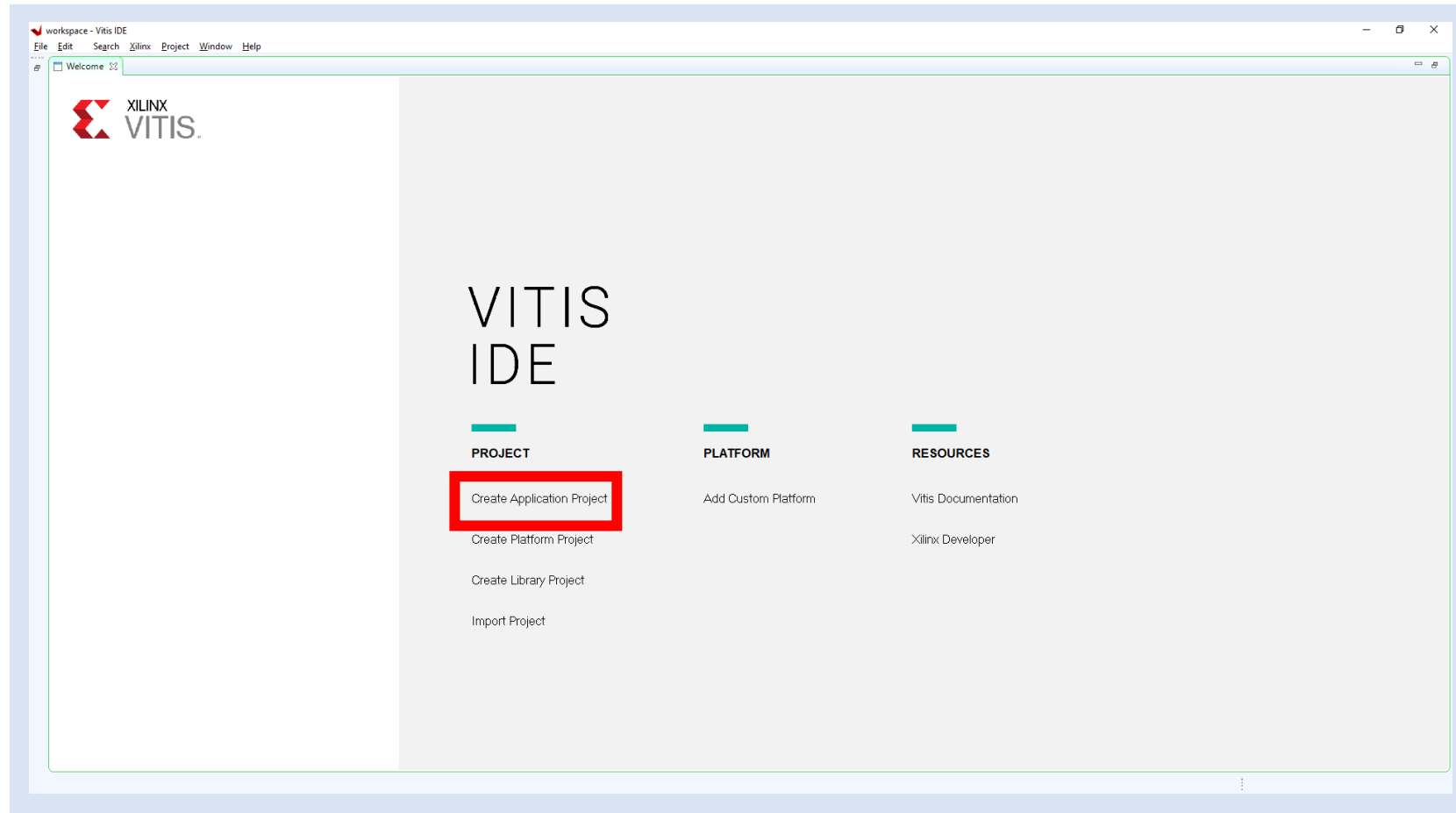
Lab 4: Overview and Introduction to Vivado

Step 44 – Select a workspace – I recommend creating a new directory in the project direcotry



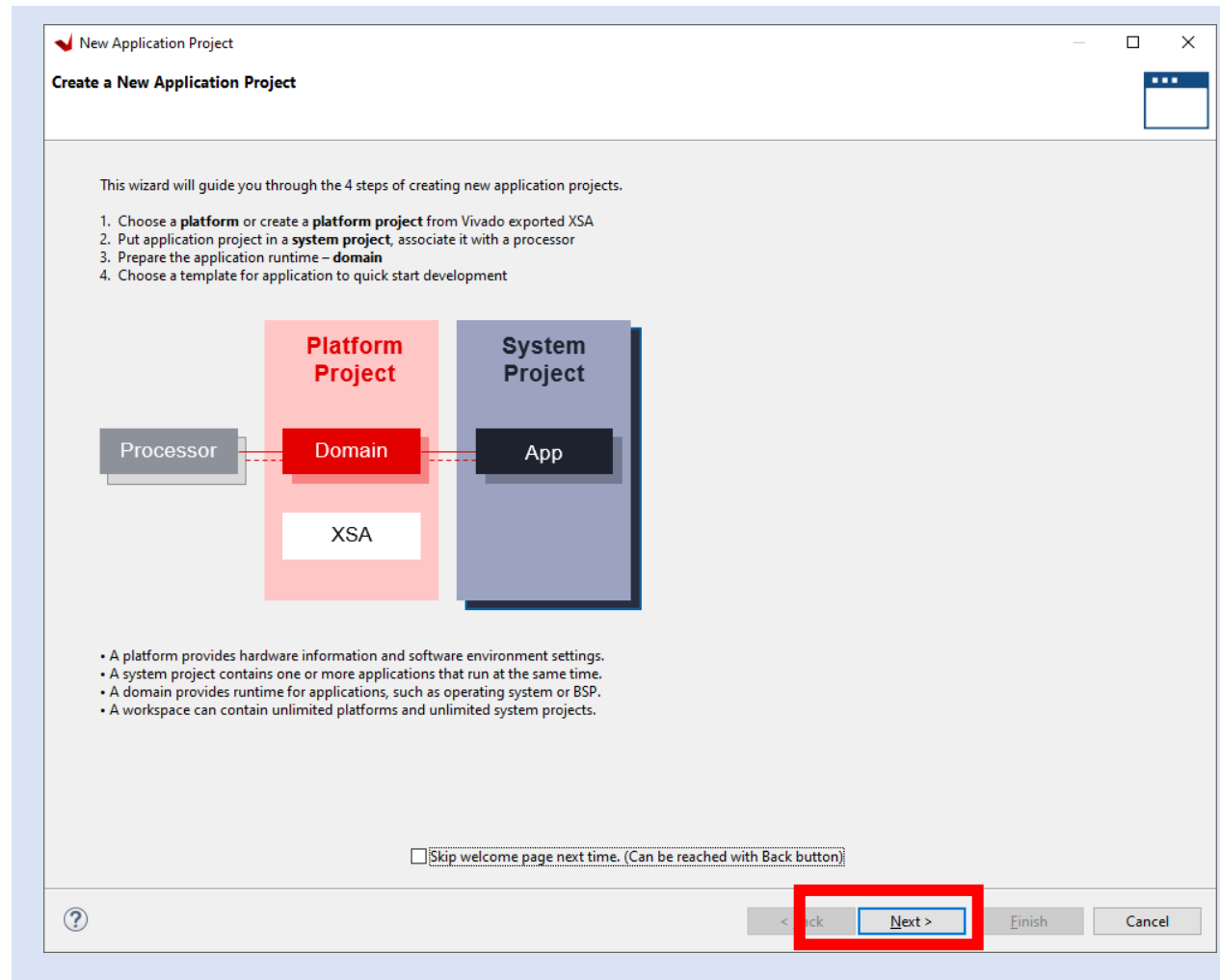
Lab 4: Overview and Introduction to Vivado

Step 45 – When Vitis opens create a new application project



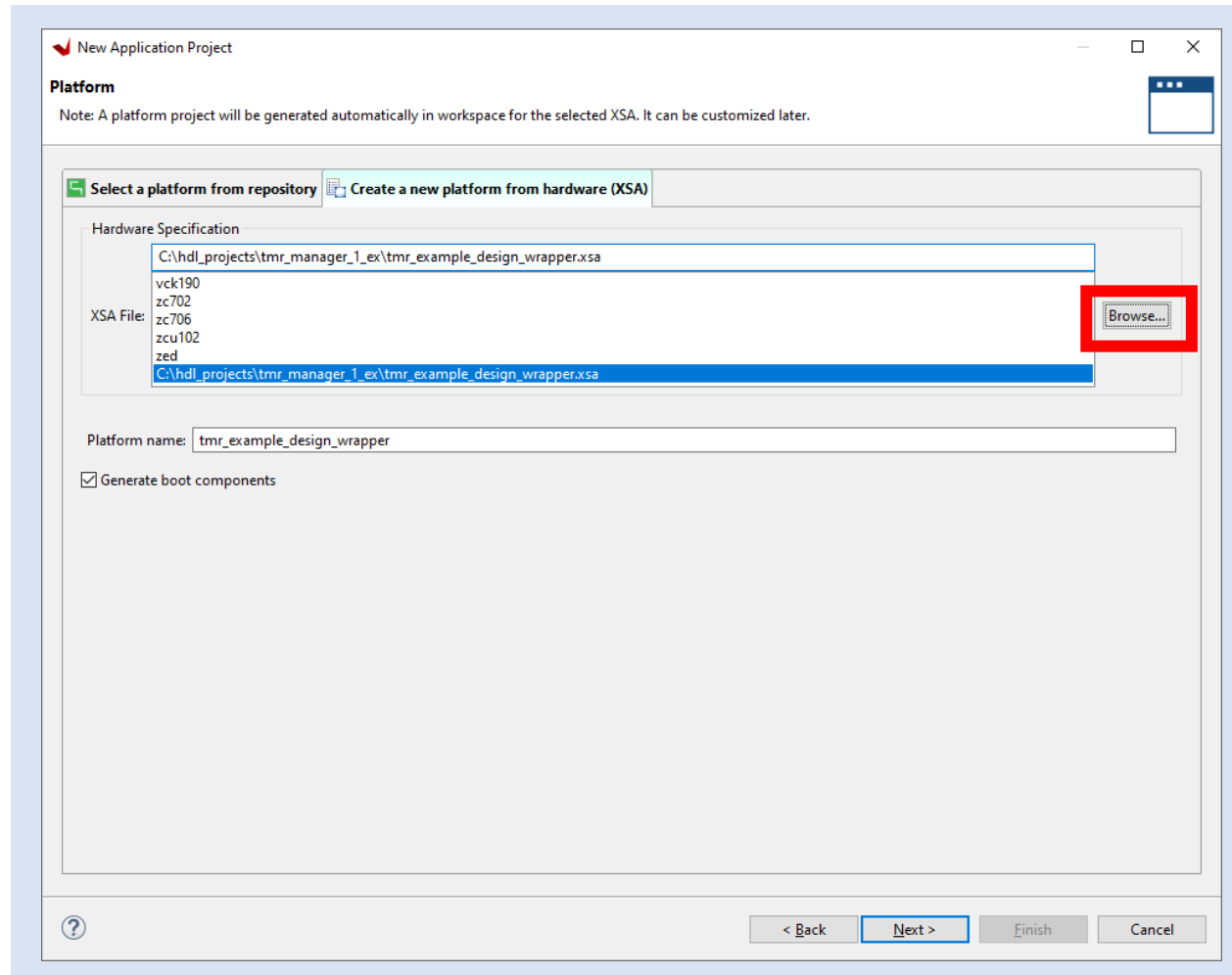
Lab 4: Overview and Introduction to Vivado

Step 46 – Click Next



Lab 4: Overview and Introduction to Vivado

Step 47 – Select Create platform using XSA and navigate the to XSA file just exported



Lab 4: Overview and Introduction to Vivado

Step 48 – Give the project a name and leave the target processor a MB1

New Application Project

Application Project Details
Specify the application project name and its system project properties

Application project name:

System Project
Create a new system project for the application or select an existing one from the workspace

Select a system project
[+ Create new...](#)

System project details
System project name:

Target processor
Select target processor for the Application project.

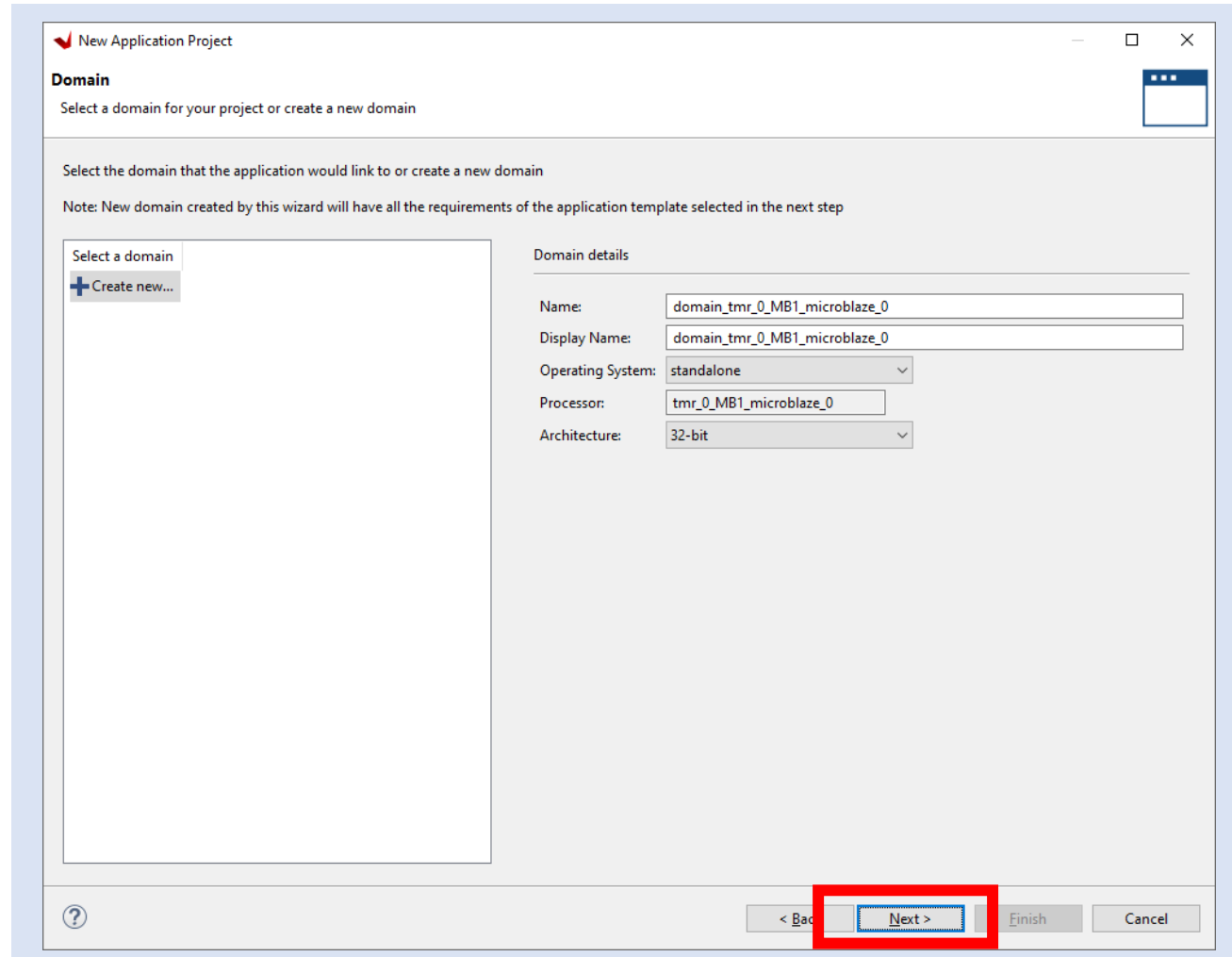
Processor	Associated applications
tmr_0_MB1_microblaze_0	hw_sw
tmr_0_MB2_microblaze_0	
tmr_0_MB3_microblaze_0	

Show all processors in the hardware specification ☒

< Back Next > Finish Cancel

Lab 4: Overview and Introduction to Vivado

Step 49 – Click on next



The screenshot shows the 'New Application Project' wizard in Vivado, specifically the 'Domain' step. The window title is 'New Application Project'. The 'Domain' section has the instruction 'Select a domain for your project or create a new domain'. Below this, it says 'Select the domain that the application would link to or create a new domain' and includes a note: 'Note: New domain created by this wizard will have all the requirements of the application template selected in the next step'.

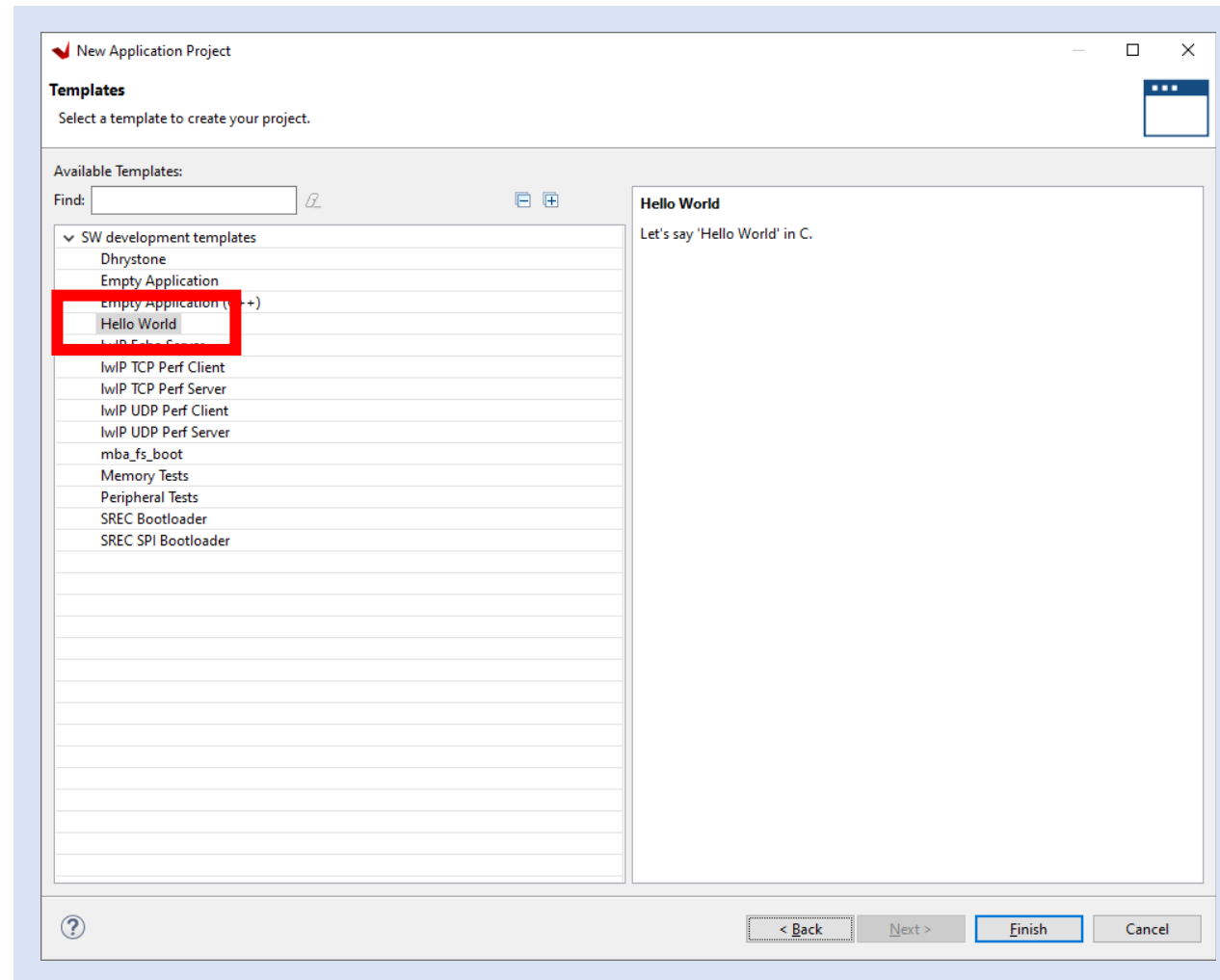
On the left, there is a 'Select a domain' area with a '+ Create new...' button. On the right, the 'Domain details' section contains the following fields:

- Name: domain_tmr_0_MB1_microblaze_0
- Display Name: domain_tmr_0_MB1_microblaze_0
- Operating System: standalone (dropdown)
- Processor: tmr_0_MB1_microblaze_0
- Architecture: 32-bit (dropdown)

At the bottom of the wizard, there are four buttons: '< Back', 'Next >', 'Finish', and 'Cancel'. The 'Next >' button is highlighted with a red rectangle.

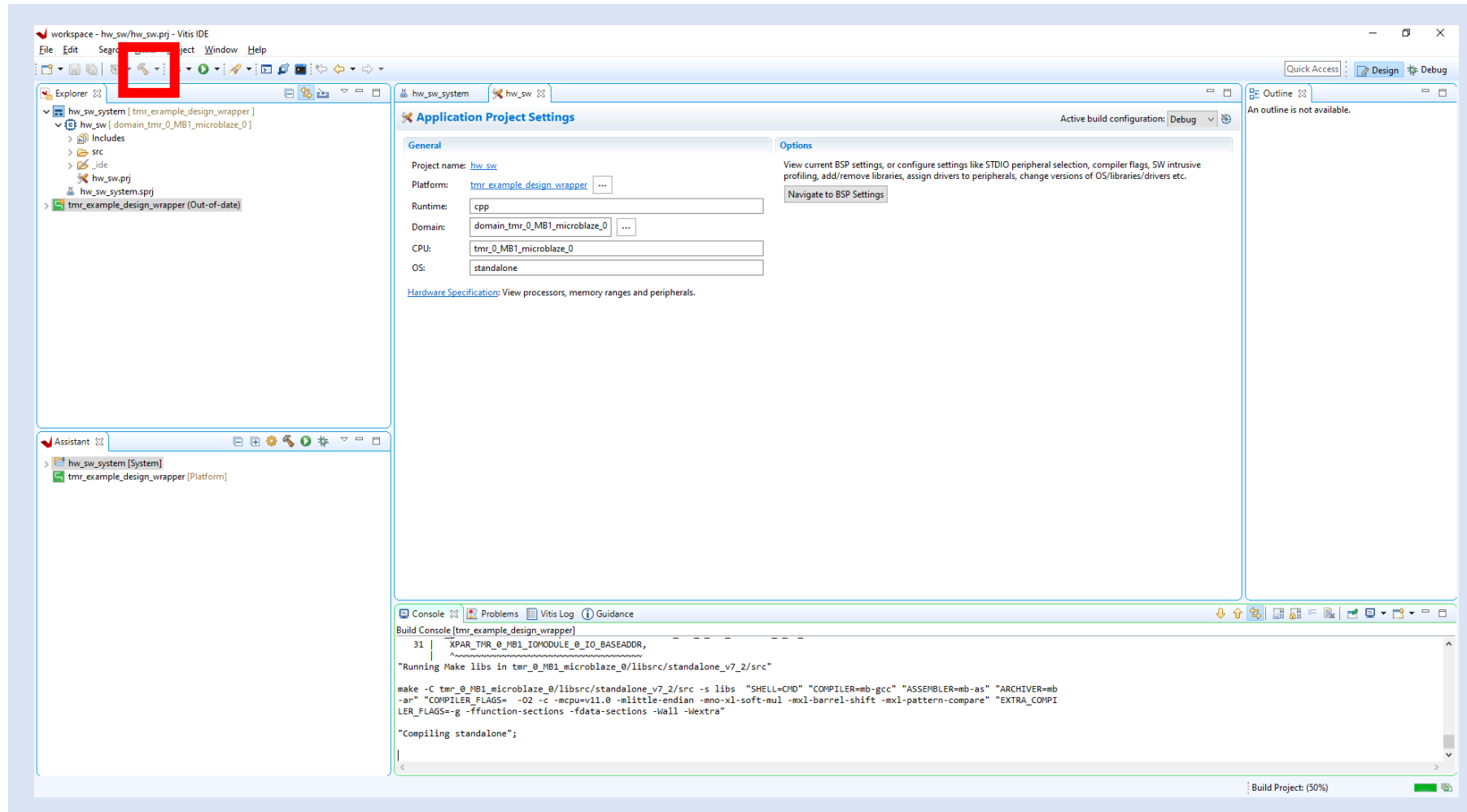
Lab 4: Overview and Introduction to Vivado

Step 50 – Select the hello world application



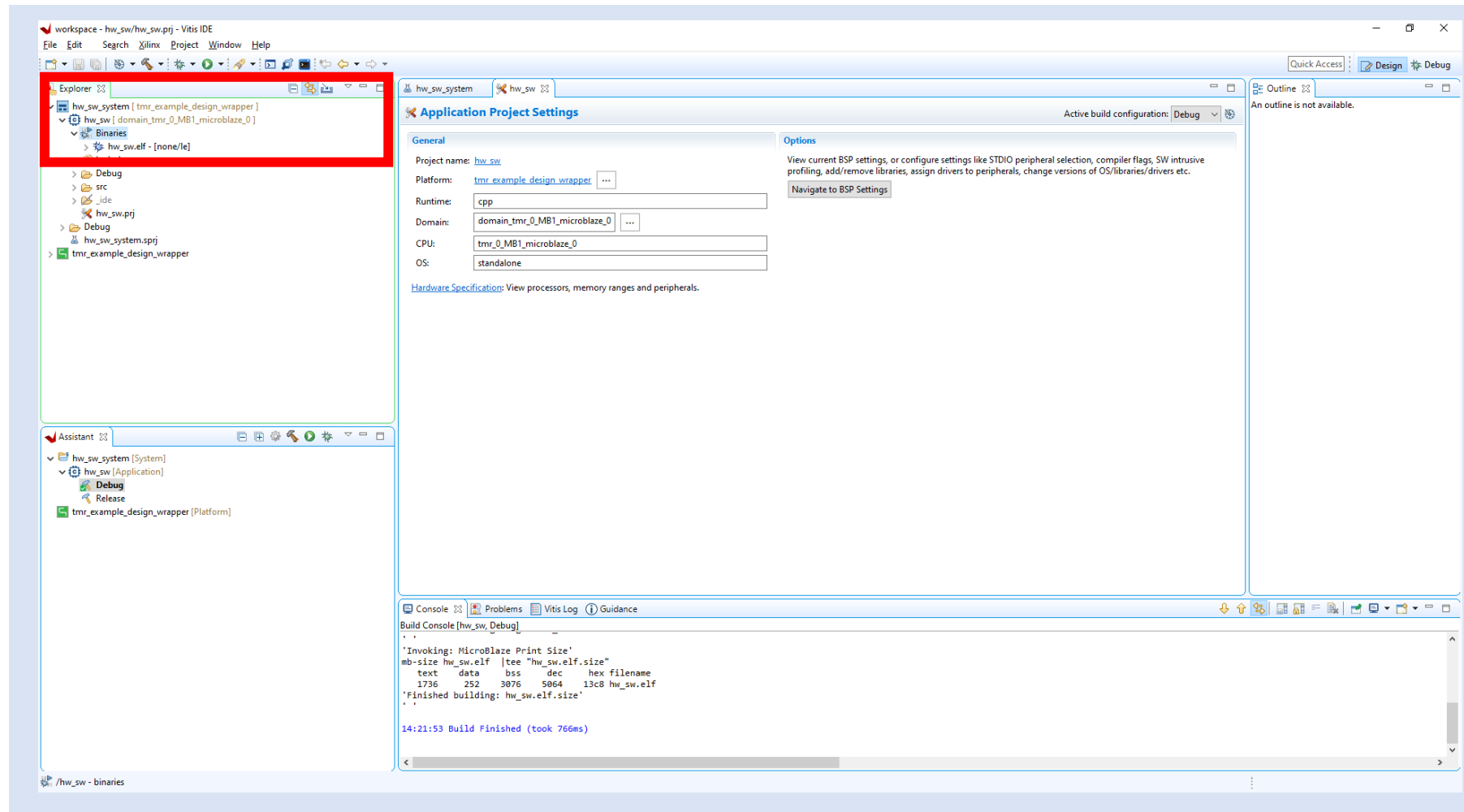
Lab 4: Overview and Introduction to Vivado

Step 51 – Select the platform and build compile the BSP, do the same for the application



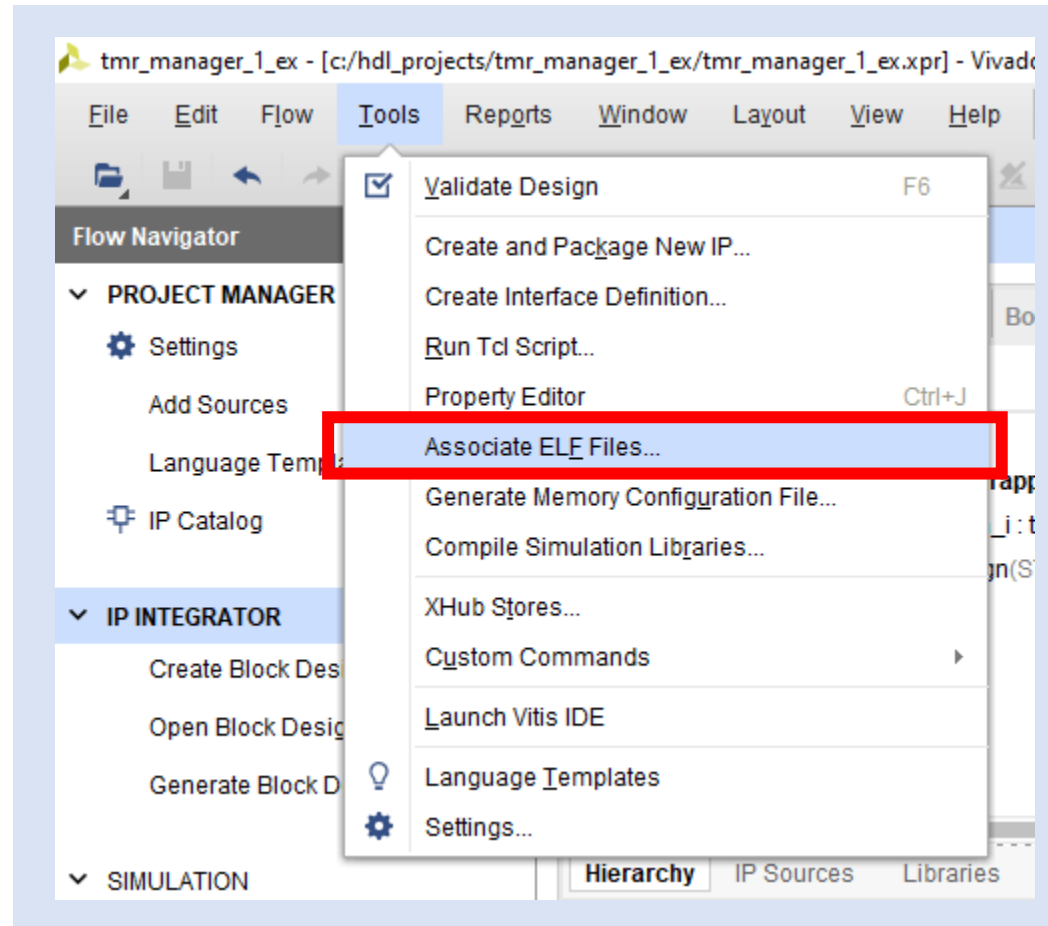
Lab 4: Overview and Introduction to Vivado

Step 52 – Expand the Hw project to double check there is an ELF



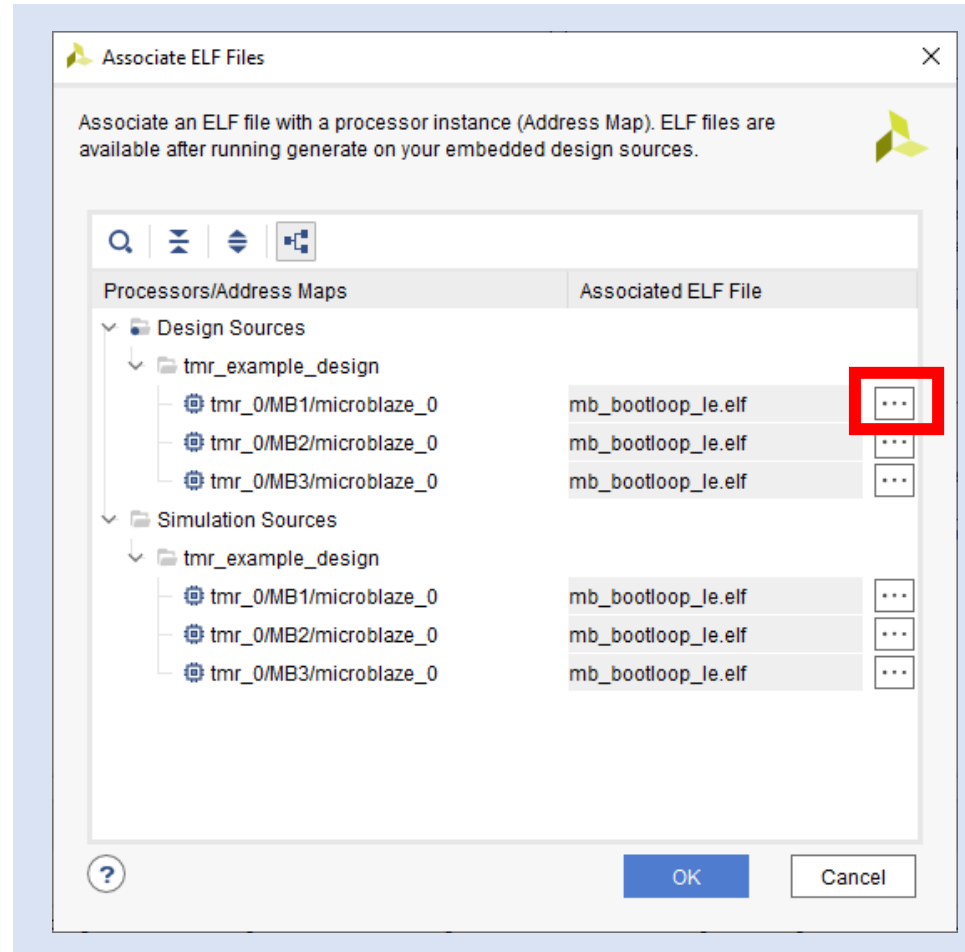
Lab 4: Overview and Introduction to Vivado

Step 53 – Switch Application back to Vivado and Select Associate ELF files from the Tools menu



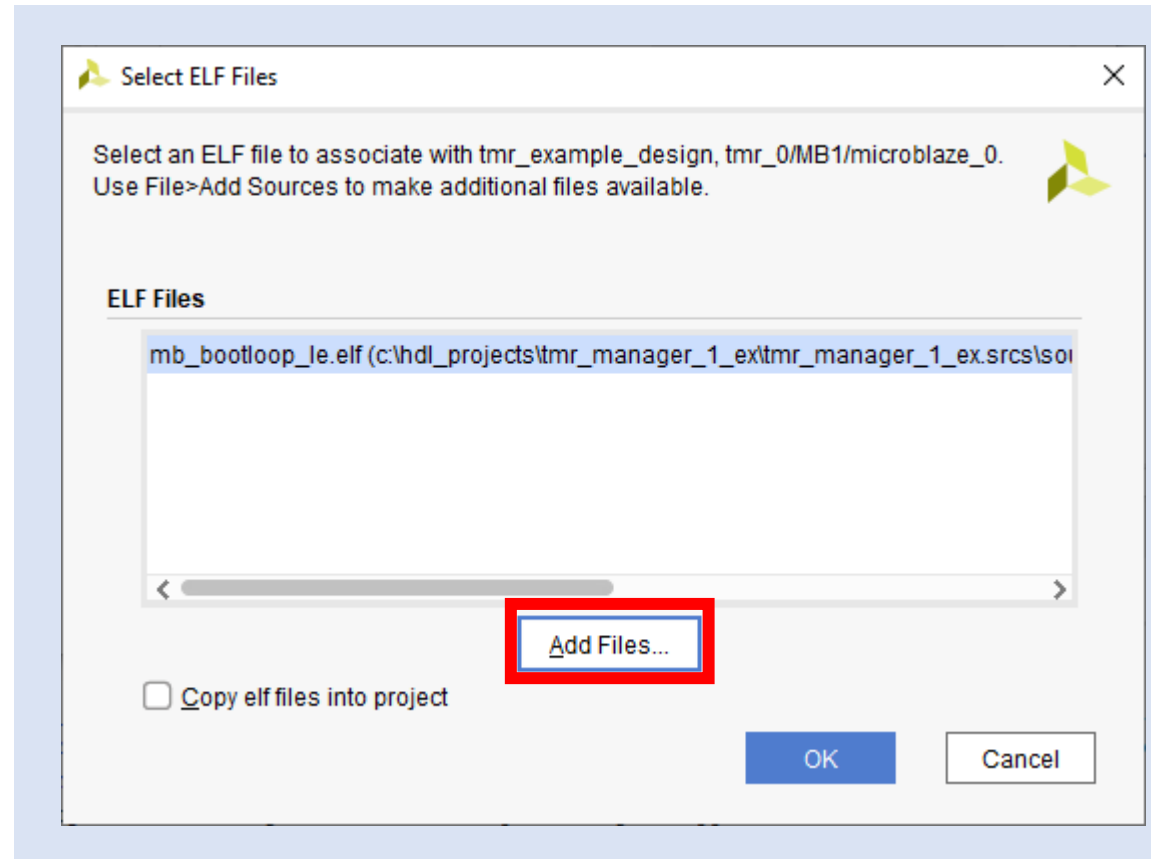
Lab 4: Overview and Introduction to Vivado

Step 54 – Click on the ... icon to a a new ELF file



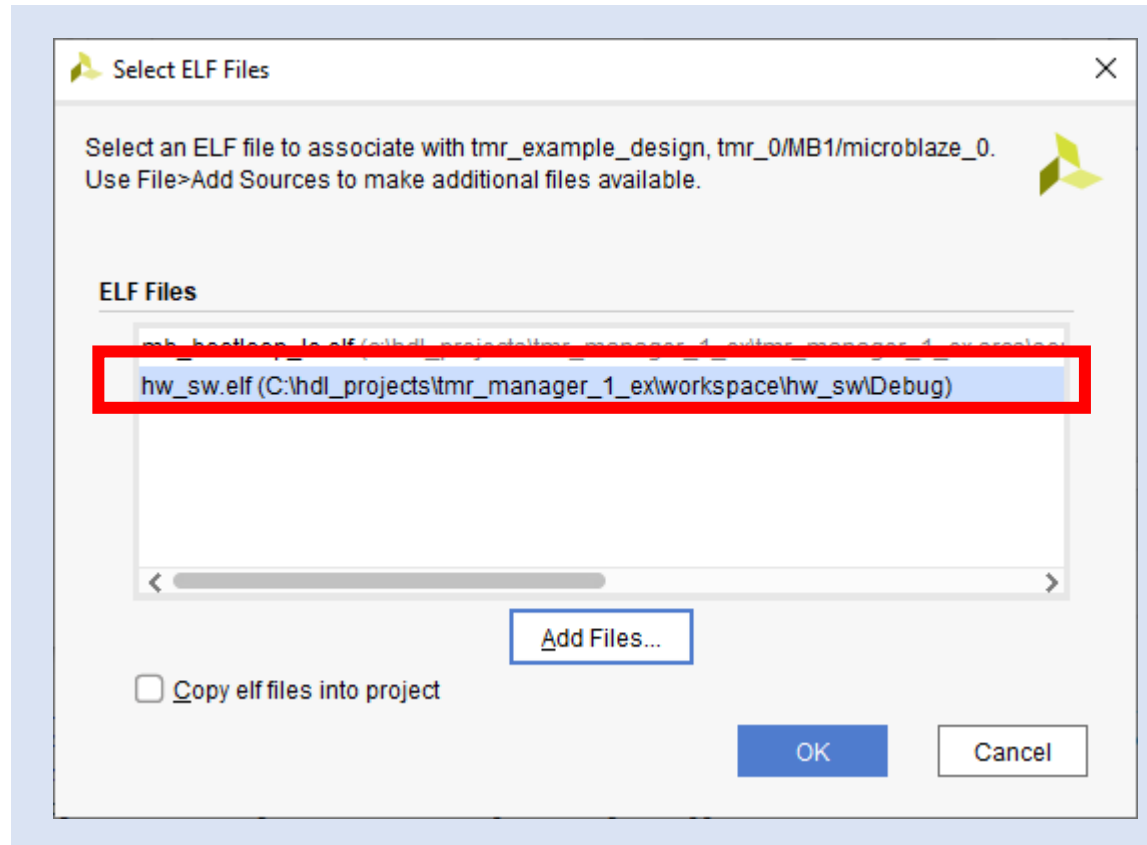
Lab 4: Overview and Introduction to Vivado

Step 55 – In the resulting dialog select Add Files



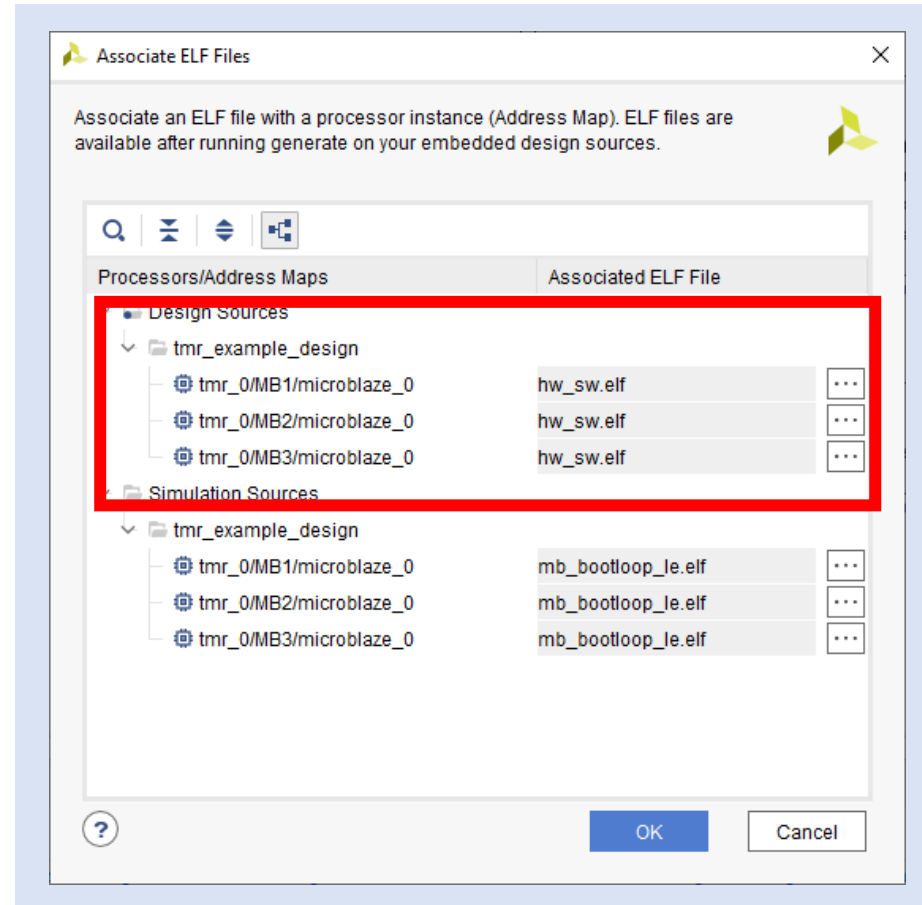
Lab 4: Overview and Introduction to Vivado

Step 56 – Navigate to the ELF file created by Vitis and add in the file and select OK



Lab 4: Overview and Introduction to Vivado

Step 57 – Select the same ELF for all three MicroProcessors



Lab 4: Overview and Introduction to Vivado

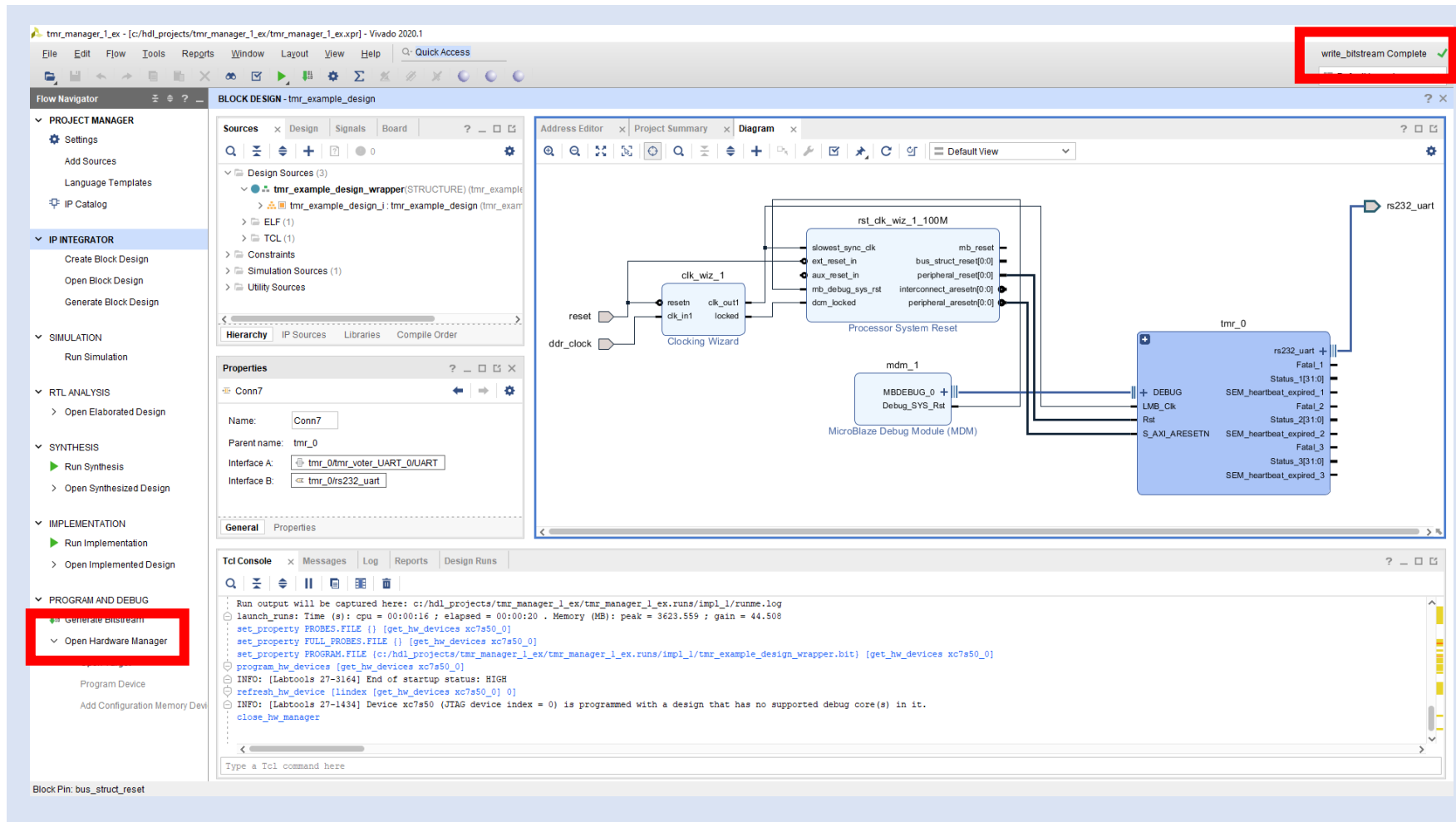
Step 58 – Generate the bit file with the SW included

The screenshot displays the Vivado 2020.1 IDE interface during the 'Generate Bitstream' step. The 'Flow Navigator' on the left shows the project workflow, with 'Generate Bitstream' highlighted under the 'IMPLEMENTATION' section. The 'BLOCK DESIGN' window shows a block diagram of the design, including components like 'clk_wiz_1' (Clocking Wizard), 'rst_clk_wiz_1_100M' (Processor System Reset), 'mdm_1' (MicroBlaze Debug Module), and 'tmr_0' (Timer Manager). The 'Tcl Console' at the bottom shows the execution of the 'write_bitstream' and 'launch_runs' commands, indicating the successful generation of the bitstream.

```
write_hw_platform: Time (s): cpu = 00:00:06 ; elapsed = 00:00:05 , Memory (MB): peak = 2200.176 ; gain = 31.105
add_files -norecurse C:/hdl_projects/tmr_manager_1_ex/workspace/hw_sw/Debug/hw_sw.elf
set_property used_in_simulation 0 [get_files C:/hdl_projects/tmr_manager_1_ex/workspace/hw_sw/Debug/hw_sw.elf]
set_property SCOPED_TO_REF tmr_example_design [get_files -all -of_objects [get_fileset sources_1] [C:/hdl_projects/tmr_manager_1_ex/workspace/hw_sw/Debug/hw_sw.elf]]
set_property SCOPED_TO_CELLS { tmr_0/MB1/microblaze_0 tmr_0/MB2/microblaze_0 tmr_0/MB3/microblaze_0 } [get_files -all -of_objects [get_fileset sources_1] [C:/hdl_projects/tmr_manager_1_ex/workspace/hw_sw/Debug/hw_sw.elf]]
launch_runs impl_1 -to_step write_bitstream -jobs 8
[Thu Oct 8 14:25:41 2020] Launched synth_1...
Run output will be captured here: c:/hdl_projects/tmr_manager_1_ex/tmr_manager_1_ex.runs/synth_1/runme.log
[Thu Oct 8 14:25:41 2020] Launched impl_1...
Run output will be captured here: c:/hdl_projects/tmr_manager_1_ex/tmr_manager_1_ex.runs/impl_1/runme.log
```

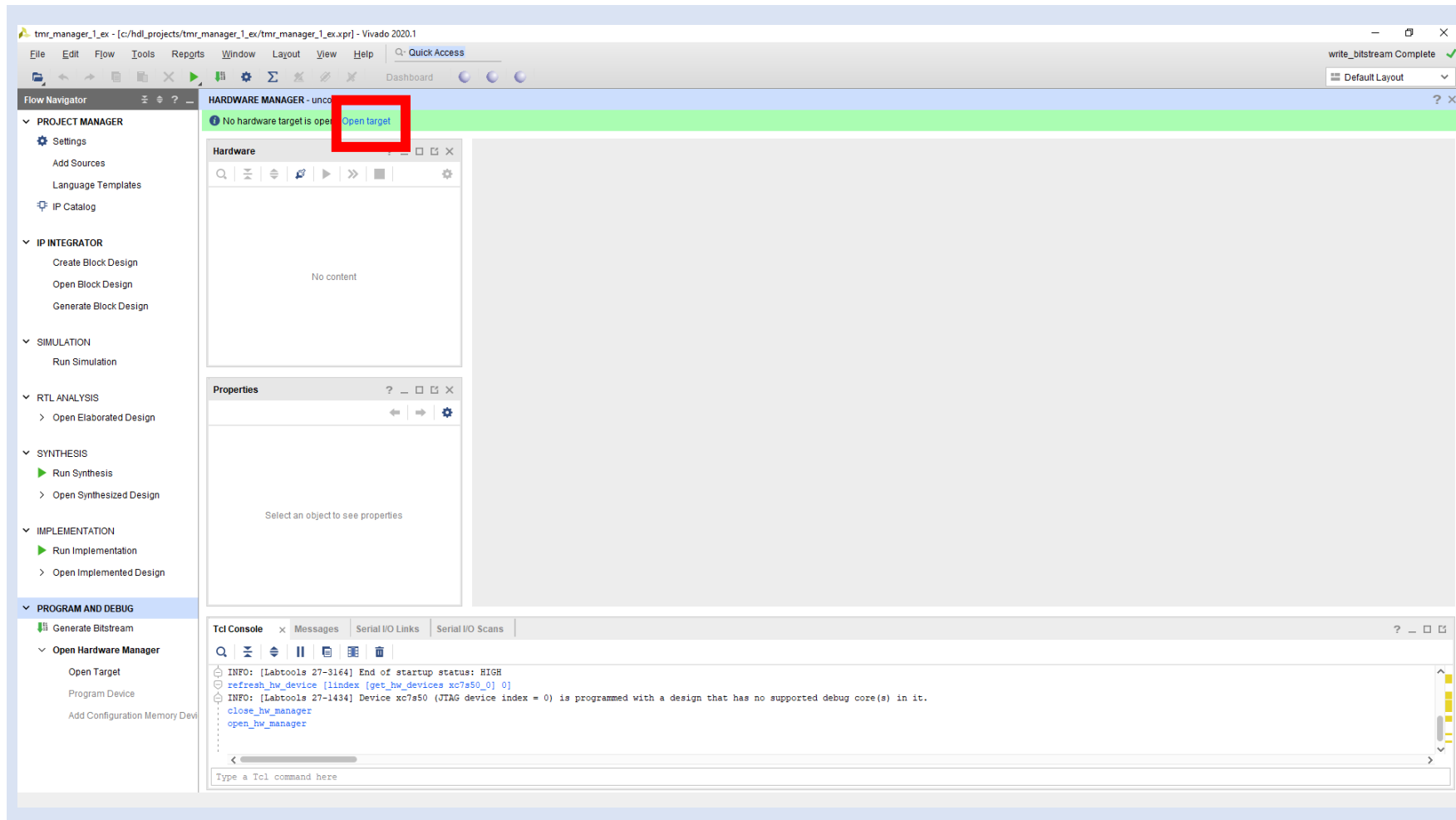
Lab 4: Overview and Introduction to Vivado

Step 59 – Wait for the bitfile to be completed, Open the hardware manager



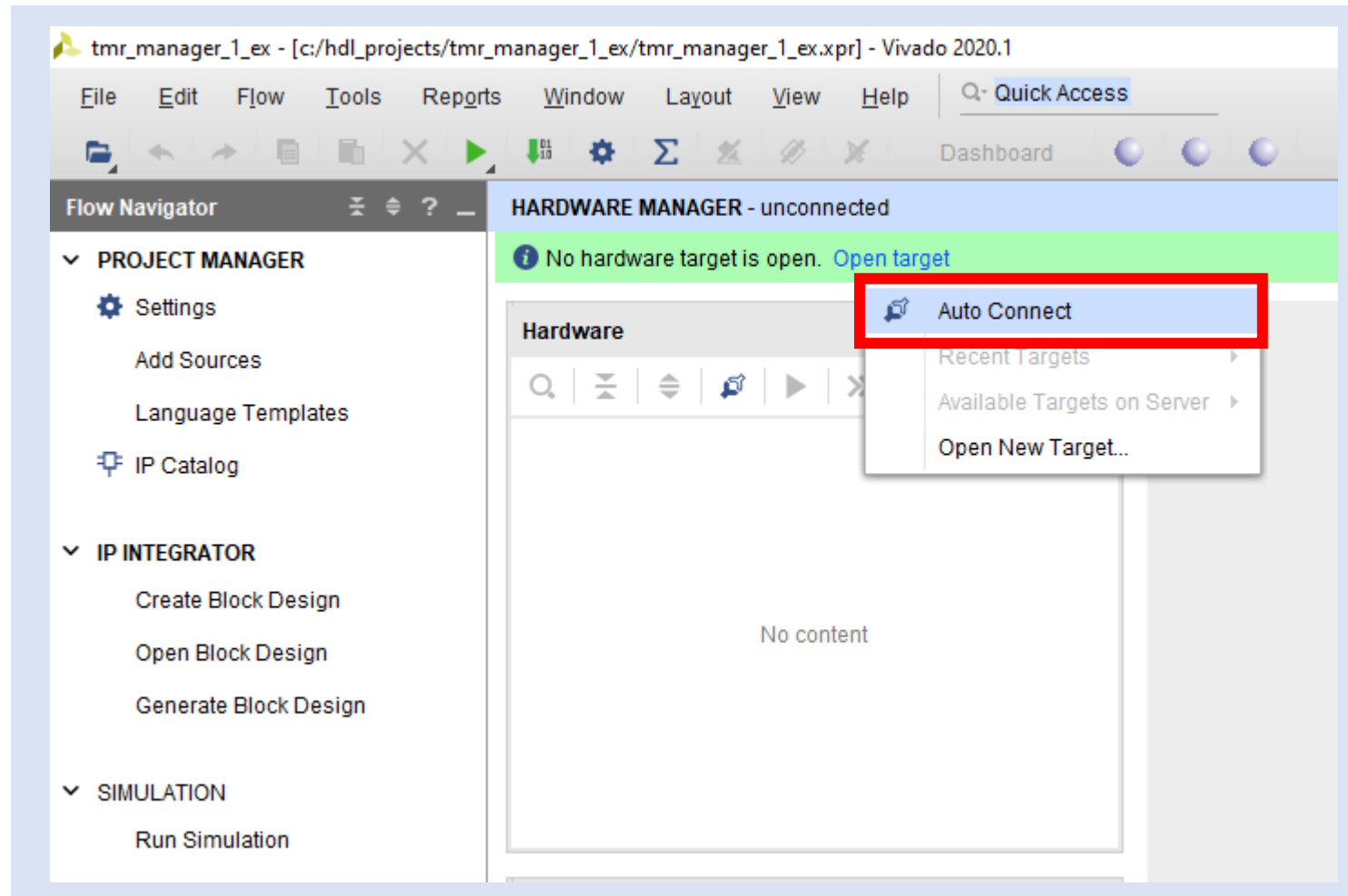
Lab 4: Overview and Introduction to Vivado

Step 60 – Select Open Target



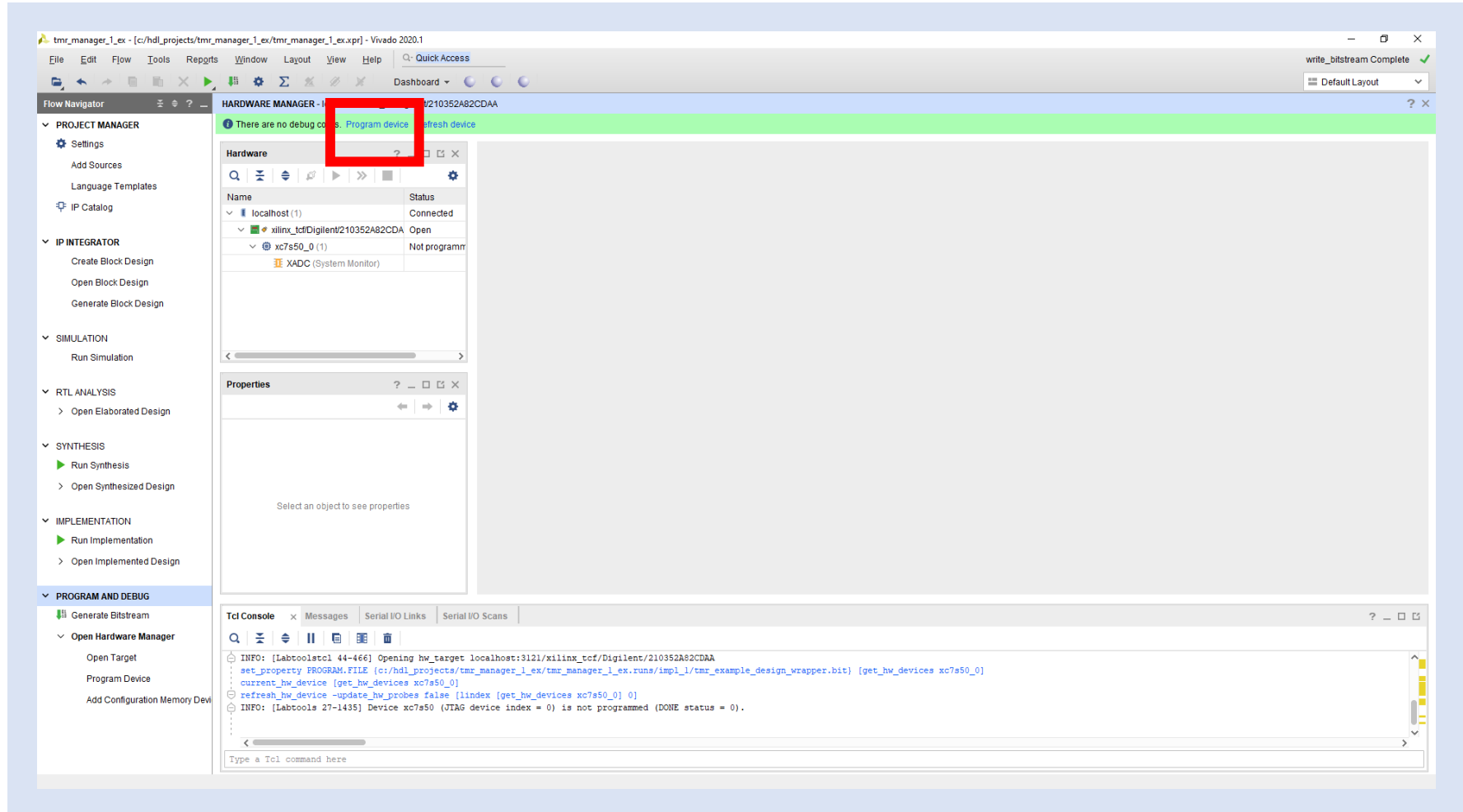
Lab 4: Overview and Introduction to Vivado

Step 61 – Select Auto Connect



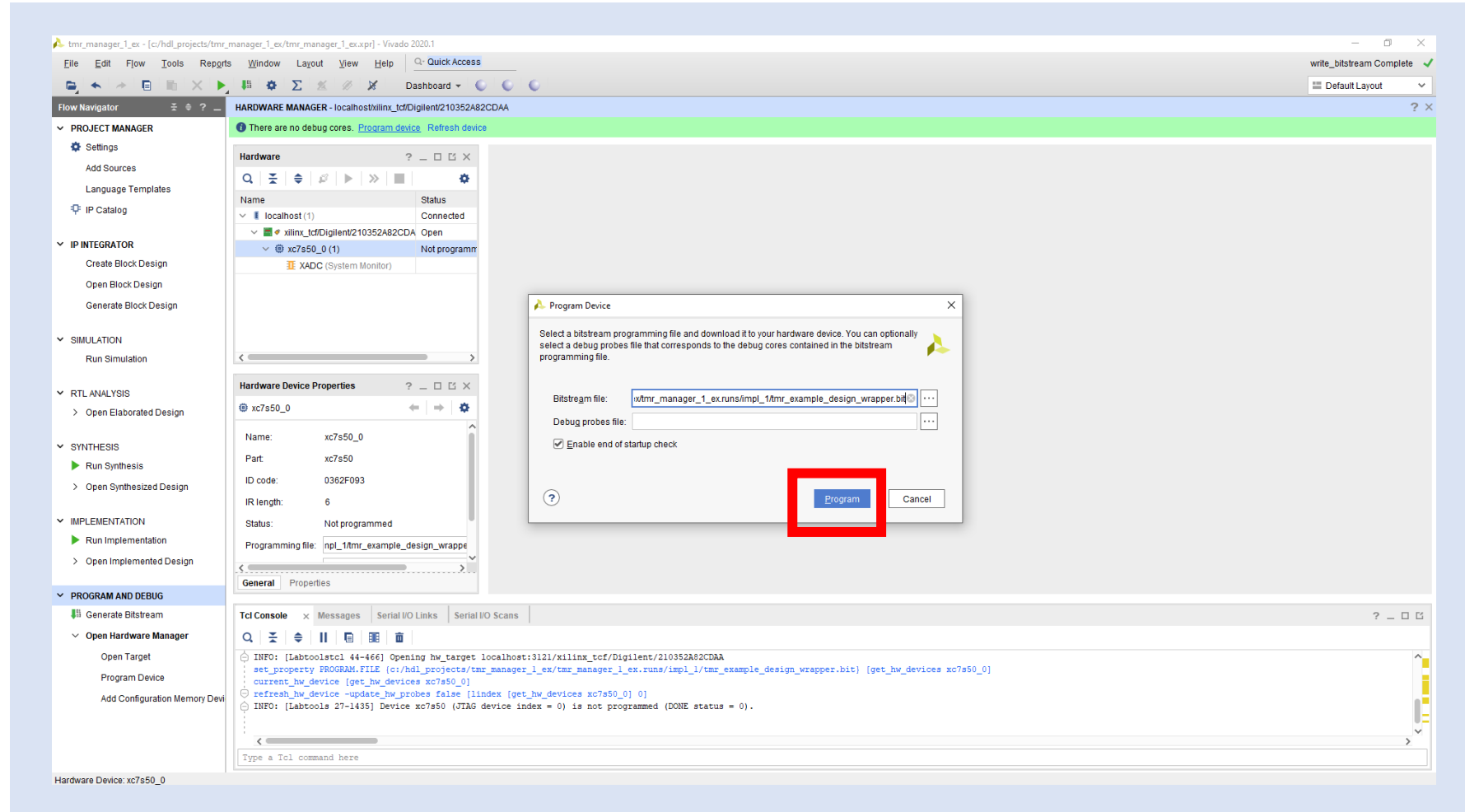
Lab 4: Overview and Introduction to Vivado

Step 62 – Select Program Device



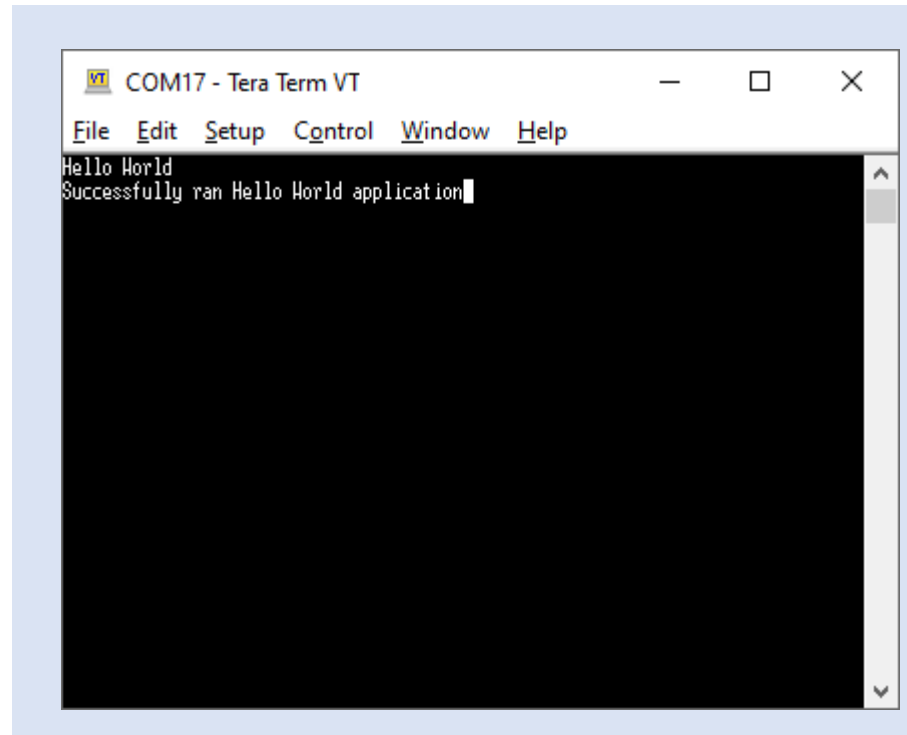
Lab 4: Overview and Introduction to Vivado

Step 63 – Connect the Arty S7 and ensure a terminal window is open, click program



Lab 4: Overview and Introduction to Vivado

Step 64 – In the terminal window you should see the output from the application



Lab 4 Summary

The concludes lab 4, throughout this lab we have demonstrated

1. How to quickly and easily create a TMR MicroBlaze System
2. How to re-target it to any Seven Series device
3. How to create a new Vitis project for the new application
4. How to merge the ELF from Vitis to all three processors
5. How to program the design onto the target board